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MAGAZINE

April 2009

WALL-E

Just another
sophisticated
toy ... or a
hack-worthy
experimenter's
platform?

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Dynamixel AX-12+
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Protocol

Allegro Micro A3950
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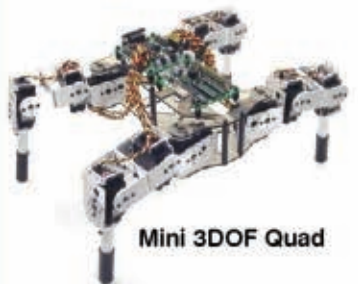
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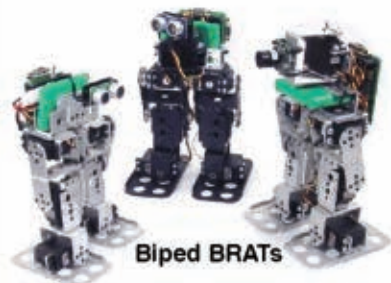
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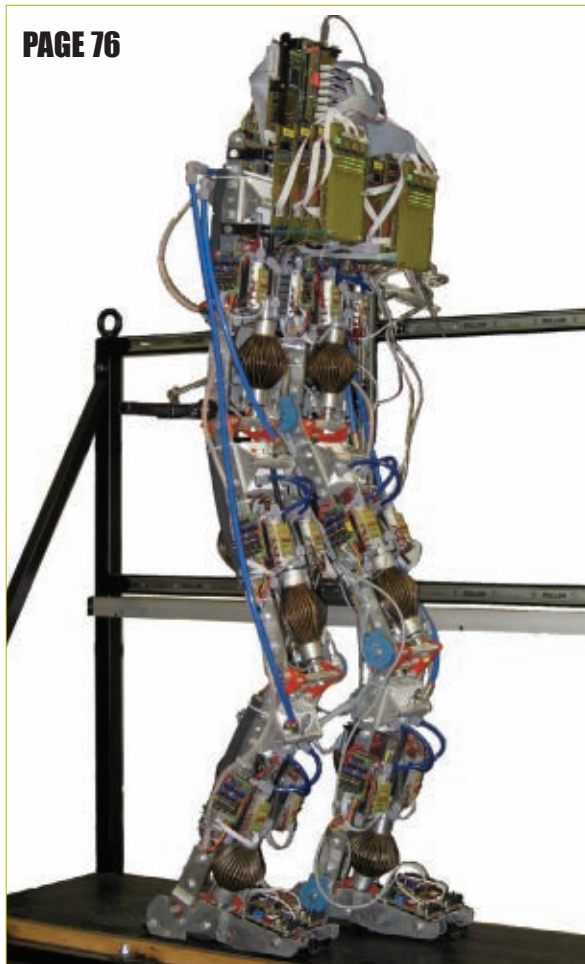
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MAGAZINE

PAGE 76



PAGE 58



Columns

08 Robytes

by Jeff Eckert
Stimulating Robot Tidbits

10 GeerHead

by David Geer
The Ultimate WALL-E Robot Toy

13 Ask Mr. Roboto

by Dennis Clark
Your Problems Solved Here

58 Twin Tweaks

by Bryce and Evan Woolley
Bug Sport

68 Robotics Resources

by Gordon McComb
Organizing Your Robotics Workbench

72 Beginner Electronics

by William Smith
Basic Atom & Robotics

76 Then and Now

by Tom Carroll
European Robots

Departments

06 Mind/Iron

07 Bio-Feedback

16 Events Calendar

18 New Products

21 Showcase

64 SERVO Webstore

81 Robo-Links

81 Advertiser's Index

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PAGE 46

Features & Projects

30 Unwinding the AX-12+ Communication Protocol

by Fred Eady

Build up some special hardware and firmware to drive the Dynamixel AX-12+ robot actuator.

38 Computer Control and Data Acquisition

by David A. Ward

Part 2 introduces National Instrument's most affordable computer interfacing hardware and how to use its digital features.

42 Robot Vision for Everyone

by John Blankenship and Samuel Mishal

See how easy it is to implement visual capabilities with RobotBASIC.

46 Navigation and Tree Measurement

by Jaakko Jutila

Learn how the Forestrix Project implements sensor technology for data gathering.

50 A3950 DC Motor Controller Tips and Tricks

by Jose Quinones

Allegro Micro's A3950 provides a versatile and economical solution for small to medium sized DC motor control applications.

The Combat Zone...



Features

- 22 BUILD REPORT:**
T6 — Evolution of a Full Body Spinner
- 28 PARTS IS PARTS:**
Custom Colson Wheel Hubs

Events

- 25 Results and Upcoming Competitions**
- 25 Event Report: 2009 Chattanooga Robot Battles**

Robot Profile

- 29 Upheaval**



Mind / Iron



by Bryan Bergeron, Editor

Stimulus Package for Robotics

If you've been tracking the economic developments in robotics, you know that only real business opportunities have been primarily in the military and entertainment industries. Several of my friends in consumer robotics joke about how their niche desperately needs a 'stimulus package' to get things moving again. They take the position that nothing revolutionary has happened in every day use of robotics for decades.

Although I think that significant progress in consumer-level robotics has been made, the relative stasis in robotics was brought home to me on a recent visit to an MIT museum that featured prototype planetary crawlers built for NASA in the 1970s. Although today it's possible to replicate the functionality of those crawlers with off-the-shelf components, they are just as compact, expertly machined, and functional as anything on the consumer market today. So, what happened to the innovation and rapid evolution characteristic of robotics decades ago, and why hasn't it percolated down to the consumer level?

Economics is obviously an issue. Most small robotics companies that don't have the good fortune to have contracts with the government have no choice but to innovate in order to make due with less. Because as enthusiasts we're in the same practical situation, our innovation tends to be in the realm of

duplicating technology as inexpensively as possible. While this is certainly a worthy and practical exercise, it doesn't necessarily advance the field of robotics. But it can.

Furthermore, it doesn't take a team of scientists to innovate. Consider the developers of the 'killer apps' that eventually made the personal computer a consumer commodity. Most of these applications were created by one or two innovators, working with little or no capital, while holding down a regular day job.

Clearly, the equivalent of the electronic spreadsheet has yet to be developed in practical robotics for the consumer market. And without the killer app in consumer robotics, it will be decades before the family pooch is replaced by a robot capable of fetching the newspaper — assuming newspapers are still around.

All is not lost, however. Another way to frame the apparent lack of innovation is to argue that the pressure to innovate in a way that advances "the field" isn't there. Robotics, for many enthusiasts, is a means to an end. Students considering a career in engineering use robotics to gain practical, hands-on experience at problem-solving electrical and mechanical systems.

Engineers in training employ robotics to demonstrate concepts they've mastered, and computer scientists use robotics as a platform to explore artificial intelligence and machine learning concepts. For these

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Mind/Iron Continued →

innovators, the current state of readily available, somewhat standardized robotics platforms allows them to innovate in their areas of interest.

So, perhaps there isn't really a need for a stimulus package for robotics after all. It's just adjusting expectations that's in order. While robotic technology can help you park your car today, it will likely be decades before Wal-Mart offers personal care robots that will ferry your slippers and paper-thin view screen from one room to the next, check on your medications, and take out the trash. I have to admit that such a future would be nice.

For now, my CrustCrawler arm, fleet of Parallax carpet rovers, and homemade robots provide stable, useful platforms for my AI experimentation and, hopefully, innovation. **SV**

BIO-FEEDBACK

Dear *SERVO*:

I read Zac O'Donnell's article in the Feb. 09 Combat Zone on running the LiFePO₂ (A123) batteries in series, with the bottom two cells powering the motor and the entire pack powering the weapon. I am a professional (BSEE/MSCS) engineer with 25 years design experience. I have to tell you, the way he configured these batteries is a very bad idea. At the very least, he will severely limit the life of these very expensive cells, and while they are resistant to fire, overheating leading to venting is not outside the realm of possibility.

Lithium cells — even the "safe" LiFePO₂ (A123) cells — need to be protected against overcurrent, overvoltage, and undervoltage. The

way Zac has these cells connected is a problem waiting to happen. While the batteries may have enough capacity to survive a normal three minute run in the ring without "reversing," any number of things could happen to drive the lower two cells under their minimum voltage point of 2-2.6V. This could occur if the ring was especially "sticky" (hot) or rough, causing the drive motors to draw more than their usual current. Or, battle damage could cause a wheel to drag, increasing current. Or, the weapon motor could stall or become heavily loaded, chewing into another bot. If the lower cells are driven below 100% DOD (depth of discharge) of about 2V, they will be ruined. If they are driven in reverse, they may overheat and vent dangerous

continued on page 55



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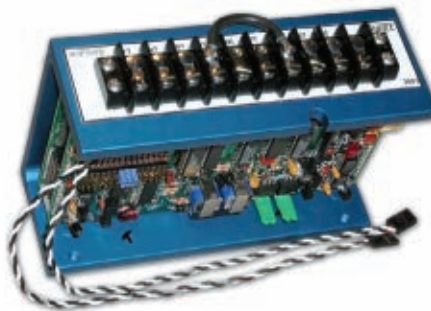
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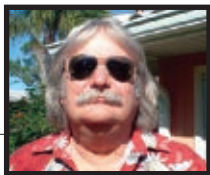
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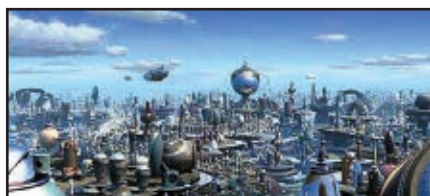
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Robot Theme Park on Track



Artist's conception of what Korea's Incheon robot park certainly won't look like.

A few weeks ago, the South Korean government authorized construction of the "world's first robot theme park," emphasizing the country's view of the robotics industry as a prime area for economic growth. The Ministry of Knowledge Economy has set aside a 767,286 m² (about 8.3 million ft²) development area in the Incheon Free Economic Zone for the park, which is budgeted at 784.5 billion won (\$562.3 million).

Proving that the Korean government is a lot smarter than ours, almost three-fourths of the investment capital (680.5 billion won) will come from private sources, with the remainder chipped in by the central and local governments. This, and a second park to be built later in the South Gyeongsang province, are expected to create 18,000 new jobs and generate 2.8 trillion won in industrial output.

The parks won't be all fun and games, though. Sure, they will feature amusements, exhibition halls, water parks, and stadiums for bot competitions. But there will also be R&D and education centers. The Incheon park is not scheduled for completion until 2014, so there's no need to book your hotel room yet.

Steep Terrain Rover Unveiled

Axel Rover, developed by engineers at NASA's Jet Propulsion



The Axel rover from NASA JPL and Caltech. Courtesy of NASA/JPL.

Lab and students from the California Institute of Technology (www.caltech.edu), sounds a bit like a rock star, and in a way it is. It's designed to rappel off cliffs, travel over steep and rocky terrain, and explore deep craters.

Based on the idea that simplicity is a good thing, it is basically just a symmetrical two-wheeled rover with a trailing link. The machine uses only three motors: one for each wheel and a third to control a lever. The lever is fitted with a scoop to gather lunar or planetary material, and it also adjusts the robot's two stereo cameras, which can tilt 360 degrees. The tether attached to its midriff can be unreeled, allowing it to descend from a larger lander, rover, or other anchor point. Axel can be fitted with wheel types ranging from large, foldable units to inflatable ones, which help it deal with hard landings and rocky terrain.

"Axel extends our ability to explore terrains that we haven't been able to explore in the past, such as deep craters with vertically-sloped promontories," said Axel's principal investigator, Issa A.D. Nesnas. "Also, because Axel is relatively low mass, a mission may carry a number of Axel rovers. That would give us the opportunity to be more aggressive with the terrain we would explore, while keeping

the overall risk manageable."

For a video of an Axel prototype at work, see www.jpl.nasa.gov/video/index.cfm?id=806.

Avoiding the Sand Trap

A recent study published in the Proceedings of the National Academy of Sciences offered what is possibly the first precise look at "the problem of robot locomotion on granular surfaces." Translated into standard English, that means "why they get bogged down in dirt, sand, and piles of leaves."

The answer might seem obvious to anyone who has ever managed to get his car stuck at the beach, but



The Georgia Tech sandbot in a poppy seed-filled trackway. Image courtesy of Daniel Goldman.



The Georgia Tech sandbot tries to plow its way through a pit of poppy seeds. Image courtesy of Daniel Goldman.

maybe the Georgia Tech guys don't get out much. In any event, according to Daniel Goldman, assistant professor in the School of Physics (www.physics.gatech.edu), "Sand is a uniquely challenging terrain because it can shift quite easily from solid to fluid to solid and requires different locomotion strategies."

Experiments were conducted by placing a device with six C-shaped legs on an eight-foot trackway filled with poppy seeds, which simulates conditions that might be found in deserts, extraterrestrial surfaces, and a number of agricultural operations in Afghanistan.

The result? "We have discovered that when a robot rotates its legs too fast or the sand is packed loosely enough, the robot transitions from a rapid walking motion to a much slower swimming motion."

The researchers discovered that the bot can easily traverse the grit if it maintains a constant rpm and some parameters are properly adjusted: the durations of the slow and fast phases and the angle at which each limb changes from slow to fast. But in a pinch, you can just pour some water on the sand.

Robot Darwinism?

In a curiously abstruse press release, researchers at Aberdeen, Scotland's Robert Gordon University (www.rgu.ac.uk) claim to have "caused a stir in the world of engineering by taking the first steps in developing a robot that has the ability to evolve in the same way as animals." They say that the technique "offers the potential to make machines which can interact with their environment and perform useful tasks in difficult or dangerous circumstances — or even around the home.

It does this by gradually



Skippy the mudskipper.
Courtesy of the Aquarium of the Pacific; photo by Hugh Ryono.

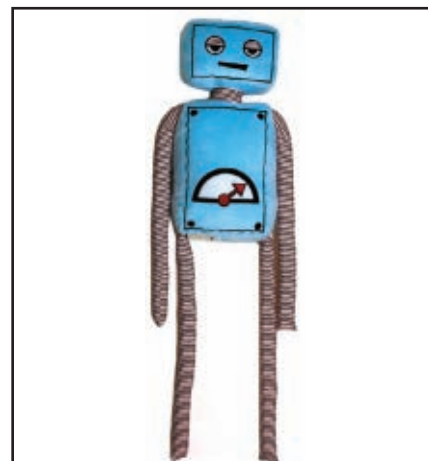
developing the robot's body and environment from simple to complex, while at the same time growing its brain (a special control circuit called an 'artificial neural network') by adding new parts — one on top of the other — to end up with a structure rather like the layers of an onion.

Using this system researchers have produced a complex robot. The robot started off pulling itself along in a primitive way — rather like a 'robotic mudskipper' — and then went through a series of developing body plans, actuators, sensors, and environments until it had evolved into a walking quadruped, able to react to visual stimuli, avoid obstacles, and react to predefined objects as 'predators' or 'prey.'

Unfortunately, no details were offered as to how it accomplishes such evolutionary upgrades, and there were no accompanying photos that might give us a clue. But the announcement inadvertently posed a much more interesting question: "What the heck is a mudskipper?" For a fascinating answer, visit www.aquariumofpacific.org/onlinelearningcenter/species/mudskipper. This is evolution at its best.

Morose, Rejected Fauxbots

The way the yarn goes, 100



The Mellowtron stuffed robot in need of Prozac®. Photo by Bill Brown.

Mellowtron robots (not to be confused with the Mellotron tape-loop keyboards employed by the Moody Blues et al. back in the good ol' days) were created and sold as service robots. But the only service they were designed to perform was to dream all day.

When their owners discovered that the Mellowtrons could not cook, clean out the cat box, or otherwise assist around the house, they were all fired, sent back to the factory, and dumped in a warehouse where they are now idle, sad, and confused.

If this somehow brings a tear to your eye, be advised that you can adopt one of the little slackers by logging onto www.stuffedrobot.com and shelling out \$70 (plus S&H). Although they all look exactly the same, each one professes to have its own story and personality. No. 32, for example, has trouble distinguishing between reality and his dreams. No. 47 gets his feelings hurt if everyone doesn't say "hi" to him. And so on.

As of this writing, 43 of them have already found homes, but you might still get one if you hurry. Or, even if you don't hurry. **SV**



GEER HEAD

by David Geer

Contact the author at geercom@windstream.net

The Ultimate WALL-E Robot Toy

On the big screen, WALL-E (the last, functional Waste Allocation Load Lifter-Earth class robot) is a curious, blue-collar working trash compactor robot tasked with cleaning up mountains of consumer garbage from the Earth's surface. WALL-E is completely alone in this mess because human beings have moved off the world on a permanent vacation, due to pollution and the inability of the planet to support life.

To keep this endless, mundane chore interesting, WALL-E collects artifacts and treasures from the glut of consumer by-products, which help him in his search for the meaning of life and his own worth. WALL-E manages to squeeze in a little fun, too, in the brief, shallow respites he takes from his manual labors.

On the toy shelves at Disney, WALL-E is another 'Stor-E' altogether. The child-sized robot (approximately three feet tall) consists of genuine, movable arms, the central body unit, eyes, and mobile tracks for locomotion. The robot

responds and interacts by talking to the user, following the user, dancing, and playing MP3s.

Sensors, Sounds, and Responses

Ten deluxe grade brushed motors manipulate WALL-E's movements based on input from several multi-directional smart sensors. Here's what WALL-E can do, and how he does it!

FollowMode

When WALL-E detects any movement, he will turn around and follow it. When WALL-E detects movement behind him, he will turn around to examine the source of the movement, according to the owner's manual. These features are available when the unit is in FollowMode.

WALL-E has a square FollowMode button on top of his front container area, signified by a red circle. Users can also activate FollowMode via remote through an oval button with a picture of WALL-E following sound waves.

WALL-E will use his three forward microphones to detect and follow you. Simply clap twice and he will follow. Continue clapping twice as you move and WALL-E will continue to follow the sound of your hands.

Using his infrared obstacle detection and avoidance sensors, WALL-E can detect obstacles in his path, stop movement, and change course to avoid them so as not to hang itself up on the landscape.

There are four infrared sensors: three in the front and one in the back. WALL-E emits an infrared light, and an infrared receiver processes the reply when the light bounces



Ultimate WALL-E robot toy.

Processing Head and Body Actions

WALL-E uses one chip to control his head and another to control his body. His roboticists have connected the chips via Serial Peripheral Interface (SPI) in order to communicate with WALL-E's actuators.

"The head motion is triggered by audio motion sensors and pre-coded digital IR signals. The roboticists pre-programmed the head to produce actions for WALL-E's emotive expressions. The head motion is also controlled by secondary 16-bit chips on a board inside the robot," says Albert Chan, Thinkway Toys CEO and President.

"WALL-E uses five motors to direct the head alone. One motor is used to make the head nod up and down; one is used for left to right head motion/turns; still another actuates the head's opening and closing to form the expressions. The final two open and close each of WALL-E's eyes," explains Chan.

WALL-E uses two motors to control the arms, wrists, and hands. "Each motor controls an arm's up and down motion, wrist turns, and hand opening and closing (once the arms are active). The arms are time sharing when commanded by one of the main IC chips boarded on WALL-E," continues Chan.

For the body and track motions, WALL-E uses three motors. One motor controls the right track's turns while another controls the other track. A third motor actuates the body's tilts forward and backward. "The body and track motions can be interactively triggered by the audio motion sensors and pre-coded IR digital signals. The internally pre-programmed actions also generate commands from the main 16-bit chip," Chan concludes.

off an object so WALL-E can determine where and how far an object is.

Three motion sensors in the front and one in the back

does this by isolating the origins of varying sounds. He then uses a Digital Signal Processing (DSP) unit to calculate the abstracted sound's location of origin in order to respond to

(lower, center) trigger WALL-E's programmed "curiosity." "The CdS motion sensor is a special resistor that responds to changes in light in the surrounding environment, which WALL-E can interpret and then react to," says Albert Chan, Thinkway Toys CEO and President.

"WALL-E uses his DSP and motion sensors to locate and follow people by the sounds that they make. WALL-E recognizes a single, short, discrete sound such as a hand clap and determines the sound's location, whether to the right, left, or directly in front of him," continued Chan.

TalkBack

WALL-E's four audio sensors (microphones placed in the front, back, and each side of the robot) enable him to hear voices and sounds. When WALL-E recognizes voice patterns, he can respond verbally using his TalkBack feature. WALL-E



Children with Ultimate WALL-E shows actual robot size.



to the DSP, which also does analog-to-digital signal conversions.

ExploreMode

WALL-E can explore his environment using his DSP, IR obstacle sensors, and a pre-assigned road map, according to company technicians. This road map actually consists of three individual routes that WALL-E can follow.

Users activate ExploreMode by selecting the green triangle button on top of WALL-E's yellow body. Users can also select ExploreMode via an oval button on the remote control that looks like an asterisk with WALL-E moving in front of it. WALL-E does not scale stairs or navigate steep slopes. Users should avoid these when selecting an area for WALL-E to roam.

DanceMode and Playing MP3s

WALL-E has a square, yellow DanceMode button on his body and an oval button with a picture of two notes and a staff on the remote control. Through a small, active, pre-programmed sequence of steps and synchronization, WALL-E synchronizes his dance moves with the rhythm of the beat. WALL-E comes with a compartment for MP3 players, which users plug into WALL-E so he can play back music from the device and put on a light show.

The Ultimate Programmable Remote Control

The white, black, and yellow Ultimate Programmable Remote Control console is a wireless infrared device that WALL-E operators can use to program and navigate the robot from up to 25 feet away.

The remote will send action commands instantly, or program and store action sequences for WALL-E, with up to 1,000 different combinations of actions possible. "WALL-E will record up to 64 programmed actions in a sequence sent from the remote control. With the press of a single button, WALL-E will play out the actions in the sequence," explains Chan.

Conclusion

WALL-E has already become a popular robot for experimenters. Check out the posted modifications and ways to build your own WALL-E robot in the Resources sidebar. **SV**

that audio input. WALL-E's TalkBack feature is on by default. To turn it off, select the TalkBack on/off button which will beep twice as the feature is turned off. The button's graphic looks like sound coming out of a speaker. Sound sensors and the TalkBack feature resume when the user selects the button again. The user will hear a single beep for "on." WALL-E uses his 16-bit DSP to play any number of customized sounds.

If you are to one side of WALL-E, you can get his attention and get him to turn and face you by clapping loudly at him twice. When he turns his head, clap again twice or three times so he can hear you. WALL-E should then turn to face you.

In order to ensure that WALL-E can hear you, recognize your speech, and give the appropriate response, you do the following: Stand one foot in front of WALL-E while he is in TalkBack mode. Speak loudly with long sentences so WALL-E receives plenty of input to make a decision about the sounds he hears. If he can hear your voice, you will see an indicator light flashing on his chest.

WALL-E will respond by saying his name as in the movie or with sounds, lights, and movement. His response will be different based on the volume of your voice. Loud sounds evoke a frightened response (a combination of responses that make him appear frightened). WALL-E's unique sound capabilities are the result of the DSP which provides dual channel voice mix output.

WALL-E designers have connected his many sensors and motors via I/O control pins

Resources

Search for the Ultimate WALL-E on <http://disneyshopping.com>

WALL-E movie overview
www.imdb.com/title/tt0910970

PIXAR animation, films
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Links to build your own WALL-E
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by
Dennis Clark

This column marks one year for me answering your questions and hoping that I am helping you along with your aspirations towards building useful or entertaining robots. The fact that you are still reading the column makes me believe that I'm doing something right. Please keep those questions coming; they inspire me to keep researching and writing about what I have learned to help you with your projects. Life is all about constantly learning. One of my favorite authors, Robert A. Heinlein, once had a character utter these memorable words: "When you stop learning, you start dying." With this in mind, I hope to live forever — and I'll do everything in my power to help you live forever, too!

The current economic times are less than ideal for most of us to be spending money on non-essentials, let's face it. There are however lots of things that get thrown away that could net some really good robotic components. Does anyone out there have any questions about how to salvage good stuff from technological marvels that we no longer want to keep using for their original purpose? I'd love to hear them and offer you suggestions to get the most from what you have on hand already. Let's move on to your questions — or rather question — there was only one this month, but it was a good one.

Q I am an old *SERVO* reader and I really enjoy your column. I have learned very much from it and it answered almost all my questions, except this one.

First, a little introduction is needed. I've been using continuous rotation servos to propel my robots, but I want to move on to using DC motors with encoders. I have read the articles in *SERVO* about PID control but that does not teach

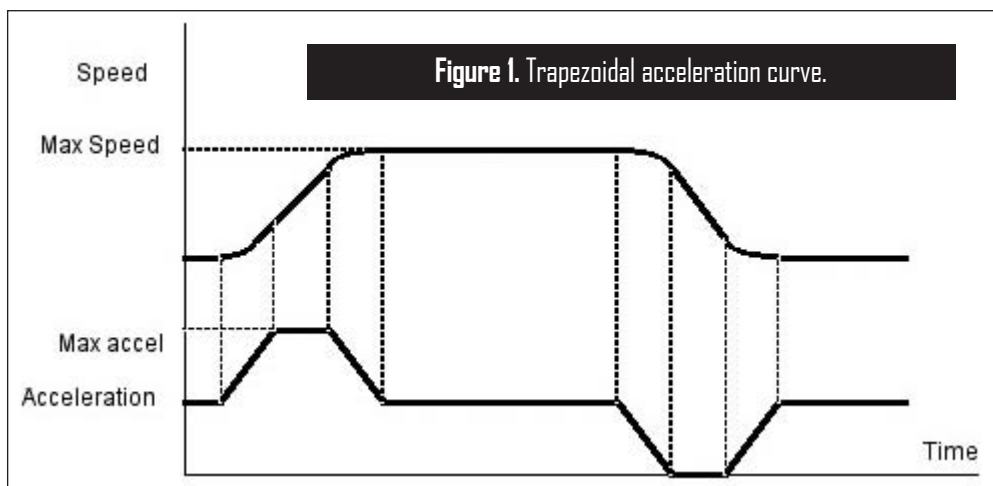
one how to make the robot move smoothly — ramping up to the maximum speed and slowing down just before reaching the destination point. Looking up on Google, I have found that a trapezoidal or even a more advanced S-curve type control is needed.

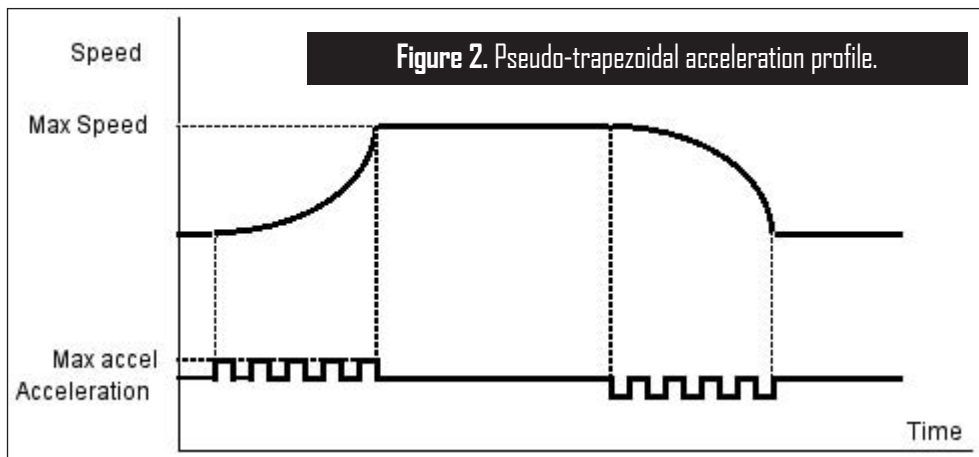
My question is: Can you show me how to implement (using pseudo code) a trapezoidal motion control? And for a more advanced movement, can it be combined with a PID control?

Thank you very much in advance!

— **Gabriel Petrut,**
Canada

A Thanks for your support! You ask a good question and using a ramped acceleration and deceleration curve will most definitely smooth your robot's direction and speed changes. In fact, if you have ever seen a robot (or any motorized machine, for that matter) that controls its acceleration and deceleration, you will most likely be very impressed. Using a ramped curve will help reduce wear and tear on your gear train and if properly implemented, will eliminate the "wheelies" that a highly





powered drive train can cause in a powerful robot. It's worth the effort!

There are a variety of acceleration curves that you can use; over many years, the trapezoidal acceleration curve has pretty much been considered the best to use because it minimizes the jerk characteristic in a drive train. Without acceleration ramping, acceleration is full-on or full-off which can be a major source of vibration in our robots; you already know this if you use a camera on your robot. The trapezoidal acceleration curve smoothes out acceleration/deceleration changes, which will remove much of the vibration attributed to the drive train. The one drawback to a trapezoidal acceleration curve is that instantaneous changes in speed or direction become impossible.

In a remotely controlled robot, this may make the robot feel sluggish or unresponsive. In a fully autonomous robot, the motion looks more deliberate and controlled. So, pick your poison. If you are driving your machine remotely, you won't want the trapezoidal curve; you'll control your rate of acceleration with your thumbs. If you have a fully autonomous robot, then you can tune your trapezoidal curve to the desired level of responsiveness you'll need. In fact, with a closed-loop system that is always looking for environmental conditions that necessitate a velocity change, you can change the ramp rate based upon the impending collision or other need for action.

In case you are wondering what we're discussing, Figure 1 shows what a trapezoidal acceleration and deceleration curve looks like. Figure 1 also shows that a trapezoidal acceleration profile will give you an S-curve velocity profile; that velocity profile is what makes your robot look so smooth on the go. Note how the ramp in the acceleration causes a curve in the change of velocity. The vertical dashed lines simply delineate the parts of the curve affected by the ramps. You calculus buffs should immediately recognize those first derivative buffs! And you thought you'd never use that math!

As you can imagine, calculating a constantly changing acceleration would require that you know your current acceleration, your maximum acceleration, and your time delta. This is a very complex calculation that requires as much knowledge of your motor power package as a PID

loop does. Maximum acceleration IS NOT your maximum PWM; the maximum PWM is your maximum motor speed. You need to calculate your change in speed divided by your change in time ($\frac{\Delta v}{\Delta t}$). We usually don't have that kind of horsepower in our microcontrollers if we want to do anything else in the program we're writing for our robot. A dedicated processor can handle this alongside any PID algorithm we might want to implement.

This gives us spectacular results, but we might not want to put forth that much effort to get there (some motor control chips have this capability built in (the HP/Agilent HCTL 1000, for instance), but these chips are expensive (\$40 for the HCTL 1000).

You knew it wouldn't be as easy as just asking, didn't you? Well, you'd be correct if you did know that, but we can come reasonably close to a trapezoidal acceleration curve by taking some shortcuts. Way back in November 2008, I wrote about a simple PID algorithm that you could implement without doing any difficult math operations. In that article, I referenced source code `picpid.c` that you can get from *SERVO* website (www.servomagazine.com). In that source code, I used a simplified trapezoidal acceleration curve which I'll call the sawtooth acceleration profile.

Figure 2 shows an approximation of what that acceleration profile would look like and its affect upon the velocity of the robot. After a speed was chosen, a simple formula was applied such that several intermediate speeds were selected for the PID loop to reach. The end result of that formula is that we moderate the acceleration to our desired speed. Figure 2 shows an approximation of the effect this pseudo-trapezoidal acceleration has on our robot's speed. It isn't perfect, but it comes reasonably close to what we want to achieve without taking over our microcontroller. (I'll apologize ahead of time for these graphics; I simply could not find a program that would allow me to do a better job with those curves.)

The simple functionality of this program is to avoid large jumps in speed by introducing many intermediate steps in the PID target speed. My code has a constant interval — the time between PID loop calls — in which an incrementally higher speed is assigned as the PID target speed. This results in a small acceleration since the delta between the current velocity and the new target velocity is rather small. So, we get a short period of acceleration, then a short period of constant velocity at the new speed, then another short period of acceleration, and so on. It isn't a real trapezoidal acceleration profile, but it simulates some of the better aspects of the true profile.

Now about that program `picpid.c`. I'm getting a lot of

mileage from it! The general ramping function is shown in Listing 1. There are three variables that are of interest here: endPos, calcPos, and vAcc. The variable endPos is the eventual terminal velocity, calcPos is the intermediate velocity; that is incremented by vAcc on every pass through the PID loop. This simple construct will limit the acceleration by only accelerating in small increments. The overall effect is to smooth that acceleration by limiting the effect of any instantaneous velocity change. Note that this will prohibit an immediate or sudden direction change, as well! We're not changing the rate of acceleration; we're limiting how much velocity change each acceleration pulse will cause. This will give us a rough approximation of the effect of a trapezoidal acceleration profile.

In main(), there are two keyboard inputs that will change the rate of change of calcPos: the a and A keys will decrease the ramp rate by one and increase the ramp rate, by one respectively. (see Listing 2). Can you see a way to better simulate the trapezoidal profile? How about if we increase vAcc each time it is used? In that way, we have a larger Δv (delta-v) each pass through the loop, which essentially will increase the rate of change each time through the loop. Again, it wouldn't — strictly speaking — be trapezoidal acceleration, but simply a closer (simple) approximation. Another way to come even closer to a true trapezoidal acceleration curve would be to start out with a longer interval between increments to calcPos which will shorten as more passes through the PID loop occur. This would simulate less acceleration at the front of the ramp, and with shorter intervals at the end of the ramp, simulate faster acceleration because of a larger delta between the current speed and the desired target speed. Perhaps we could combine these two improvements and induce smaller, longer deltas at the start of the ramp up (or down), and shorter time deltas with larger velocity deltas at the end of the acceleration ramp.

The source code for picpid.c is available for the November 2008 column at www.servomagazine.com under Mr. Roboto as picpid.zip. Rather than duplicate this file for this article, I'll point you back to the code as it was set up with the earlier article. In the spirit of "green" energy, perhaps this will save us a few electrons.

As I have said before, this isn't a perfect implementation, but it is a serviceable one that won't interfere too badly with the execution of any other code you may have running in your robot. A better implementation would be to implement the PID and trapezoidal acceleration curve in a dedicated processor so that it can concentrate on the job of simply moving the robot by turning the

Listing 1. Pseudo-trapezoidal acceleration ramp code.

```
//Figure the speed ramping value.
if (action == A_CONST)
{
    if(abs(endPos - calcPos) > counter)
        calcPos += (dir*vAcc);
    else // skip if we are close
        calcPos = endPos;
}

Vc = getCorrection(calcPos);
setMotor(Vc);
```

Listing 2. Pseudo-trapezoidal acceleration ramp setup code.

```
case 'a': //Decrease the ramp rate by 1
    vAcc -= 1;
    if (vAcc < 1)
        vAcc = 1;
    counter = vAcc+1;
    break;
case 'A': //Increase the ramp rate by 1
    vAcc += 1;
    counter = vAcc+1;
    break;
```

motors. Well, that's it for another Mr. Roboto column. I hope you've felt that it was time well spent. As usual, I can be reached for questions, comments, and criticisms at roboto@servomagazine.com and I'll be happy to work on it! Until next time, keep on building those robots! **SV**

In a month where the first day is used as an excuse for pranks and the major holiday revolves around rabbits pooping eggs, there may not be a lot we can say here to get your attention.

...GIANT FLAMING TACOS!

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Although... I'm sure the annual Presidential Easter Egg Hunt would be a lot more exciting if the Whitehouse grounds were littered with tacos instead. It'd smell good, too.



EVENTS CALENDAR



Send updates, new listings, corrections, complaints, and suggestions to: steve@ncc.com or FAX 972-404-0269

Know of any robot competitions I've missed? Is your local school or robot group planning a contest? Send an email to steve@ncc.com and tell me about it. Be sure to include the date and location of your contest. If you have a website with contest info, send along the URL as well, so we can tell everyone else about it.

For last-minute updates and changes, you can always find the most recent version of the Robot Competition FAQ at Robots.net: <http://robots.net/rcfaq.html>

— R. Steven Rainwater

April

4-5 Trinity College Fire-Fighting Home Robot Contest

Trinity College, Hartford, CT

The well-known championship event for fire-fighting robots.

www.trincoll.edu/events/robot/

14-16 DTU RoboCup

Technical University of Denmark, Copenhagen, DENMARK

Line-following and wall-following events for autonomous robots.

www.robocup.dtu.dk/

16 Austrian Hexapod Championship

FH Hagenberg, Austria

Events include Hexapod dancing and a Hexapod race.

www.fh-ooe.at/campus-hagenberg/aktuelles/events.html

16-18 FIRST Robotics Competition

Georgia Dome, Atlanta, GA

National Championship for the regional FIRST winners.

www.usfirst.org/

17 Carnegie Mellon Mobot Races

CMU, Pittsburgh, PA

The traditional Mobot slalom and MoboJoust events.

www.cs.cmu.edu/~mobot/

17-18 BlimpDuino Aerial Robotics Competition

Georgia Dome, Atlanta, GA

Autonomous blimps must complete five tasks.

<http://robots.net/article/2739.html>

17-18 National Robotics Challenge

Marion, OH

Student competition designed to complement classroom instruction.

www.nationalroboticschallenge.org/

18 Penn State Abington Fire-Fighting Robot Contest

Penn State Abington, Abington, PA

Based on the Trinity College Fire-Fighting contest. Autonomous robots must locate and extinguish a flame in a scale model of a home.

www.ecsel.psu.edu/~avanzato/robots/contests/

18 RoboRodentia

Mott Gymnasium, California Polytechnic, San Luis Obispo, CA

Autonomous micromouse-like robots must navigate a maze while picking up and moving small balls.

<http://tiedye-srv.csc.calpoly.edu/~jseng/robotics.html>

18 UC Davis Picnic Day MicroMouse Contest

University of California, Davis campus, CA

Standard micromouse contest.

www.ece.ucdavis.edu/umouse/

24 Trenton Computer Festival Robotics Contest

College of New Jersey, Ewing Township, NJ

This is the 34th annual computer festival, so there's plenty to see and enjoy besides the robot events which include Micromouse, robot racing, remote-control robot racing, and a variety of judged awards including most elaborate construction and most humorous appearance.

www.tcf-nj.org/web

25 Historical Electronics Museum Robot Festival

Linthicum, MD

Events include fire-fighting, FIRST, robot Sumo.

EVENTS CALENDAR



www.robotfest.com/

- 25 Istrobot**
Slovak University of Technology, Bratislava, Slovakia, EU
 Multiple events including line-following, IEEE Micromouse, mini Sumo, and free style.
www.robotics.sk/
- 25 Penn State Abington Mini Grand Challenge**
Penn State Abington, Abington, PA
 Autonomous outdoor ground robots must navigate around the campus — both on and off-road — avoiding obstacles.
www.ecsel.psu.edu/~avanzato/robots/contests/outdoor/
- 27-30 Alcabot/HISPABOT**
University of Alcalá, Madrid, Spain
 This year, both these robot competitions will be held simultaneously at the University of Alcalá.
www.depeca.uah.es/alcabot/ or
www.depeca.uah.es/alcabot/hispabot/
- 30 UNI Mini-Sumo Robotics Challenge**
Maucker Student Union, University of Northern Iowa, Cedar Falls, IA
 This is a mini Sumo competition. One difference from usual mini Sumo events is that this one allows "ship-ins." You can ship your mini Sumo robot to them and compete without even being there. The event is broadcast live on the Internet so you can also watch your robot compete.
<http://list.dprg.org/archive/2008-March/031554.html>

Autonomous. Prizes and certificates awarded for each event.

www.dprg.org/competitions

- 9 RoboFest**
Lawrence Technological University, Southfield, MI
 Game Competition — Two autonomous robots work together, Robot Exhibition, RoboFashion Show, Mini Urban Challenge, Fire-fighting, and VEX.
<http://robofest.net>
- 20-24 International Joint Robotics Competition and Workshop**
Suleyman Demirel University, Isparta, Turkey
 This event includes line-following, mini Sumo, Labyrinth, Stock Car, Virtual Soccer. Size is 500-1,500 gm. Cash awards and certificates.
www.ijrcw.org
- 22 NATCAR**
UC Davis Campus, Davis, CA
 Autonomous robot car race. Size of 14 inch max wheelbase, three feet max length.
www.ece.ucdavis.edu/natcar

May

- 9 DPRG RoboRama**
Dallas, TX

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NEW PRODUCTS

MODULES

Web-based, Remote Control, and Monitoring Platform

After completion of a world-wide beta testing, ioBridge Corporation (www.iobridge.com) has released the IO-204 Monitor and Control Module and integrated



web service. Along with a set of online tools, the module allows for easy creation of interactive web-based projects. ioBridge solves many of the hardware and software problems associated with getting most projects online including network configuration, web programming, mass deployment, and security.

The IO-204 module removes the need to run a local web server, track dynamic IP addresses, or even open firewall ports. Once the IO-204 is connected to a network using Ethernet, the module operates over an encrypted connection with ioBridge web services establishing a gateway to handle remote control, monitoring, and interactivity with devices connected to the IO-204.

By itself, the IO-204 module can control digital outputs and monitor both digital and analog inputs. However, more advanced functions are capable through a suite of interface boards that allow for instant project integration. Interface boards are available for relay control, temperature measurement, full servo control, X10 home automation, and RS-232 serial communication.

ioBridge modules tie into integrated web services hosted by **ioBridge.com**, which allow for web-based configuration, control, and real-time monitoring. Anytime access to the module is compatible with mobile devices such as BlackBerry and iPhone and all major web browsers. ioBridge acts as a hub for module-to-module connections, allowing for interconnected projects spanning the globe. Supported web services include (but are not limited to) event-based text and email messaging alerts, Twitter and UberNote integration, and data reporting with Google Charts.

Web widgets used for monitoring inputs or controlling outputs are created using step-by-step wizards to eliminate complex microcontroller and web

programming. ioBridge offers a secure dashboard to access widgets, and copy-and-paste embed codes to drop widgets into external web pages.

The platform features an API that developers can use to extend functionality into their own applications using Python, Perl, PHP, JavaScript, JSON, XML, and Java.

ioBridge is supported by an active community of developers and users. Collectively, they have inspired and created projects featured in *Popular Science*, *Digg*, *Wired*, *Instructables*, *Hack-a-Day*, and *Make*. Projects range from real-time household power monitoring to a Twittering toaster, and from interactive fishcams to Internet-enabled pet products.

For further information, please contact:

ioBridge Corporation

Website: www.iobridge.com

ENTERTAINMENT ROBOTS

KIRK — Knowledgeable, Interactive, Robotic Kiosk

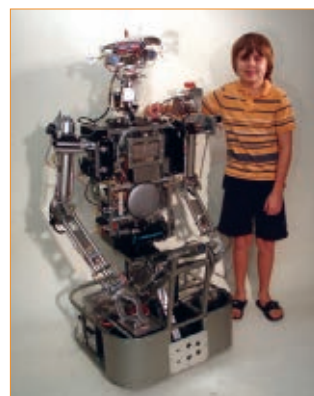
Florida Robotics has completed construction on a new, advanced entertainment robot. The first unit was delivered to Troutville, VA based Separation Technologies LLC and was promptly nicknamed RALF — Robotic Artificial Life Form.

"We love RALF ... he is *beautiful*," says Pat Borders, President of

Separation Technologies LLC. "RALF has been extremely well received and is fairly widely traveled now. We are very proud of him and happy to have him!" The company uses the robot for community awareness events at schools in order to teach children about eco-friendly concrete and the environment (www.visitralf.com/blog).

Florida Robotics has sold several units so far and at the price of approximately \$50,000, KIRK is affordable compared to some of the six figure machines available with similar features and functions. A limited number of units will be built per year and rentals will be available domestically.

While other complex robots require dozens of



buttons and levers to control, one person can operate this robot with one hand via a proprietary remote control and wireless microphone housed in a drinking cup. In the autonomous mode, KIRK can detect observers and deliver pre-recorded messages, give directions, tell jokes, or any other routine desired by the owner.

For further information, please contact:

**Florida
Robotics**

Website: www.floridarobotics.com

MOTORS & CONTROLLERS

TightDrive Motor-Mounted Speed Control

Bison Gear's new TightDrive™ speed control for permanent magnet DC (PMDC) motors was designed for applications that require a convenient location for the control ... on the motor. The majority of PMDC gearmotors and motors used in industry today require speed controls in order to operate. Typically, the control is wired to the motor through a costly, complicated cable system and the motor and control can be many steps away from each other. Bison's voice of the customer research brought forth the need for a convenient "point of use" motor speed control that was robust for industrial environments while offering practical features and exceptional value.

Bison Gear & Engineering's new TightDrive motor-mounted speed control can be easily field mounted on Bison PMDC gearmotors up to 1/6 horsepower (124 watts) and offers a 20:1 speed range with maximum output of 90 volts. The TightDrive is housed in a durable aluminum extrusion which offers NEMA 1 (IP 30) protection and superior heat dissipation. Speed is controlled with a combination on/off switch and speed potentiometer. In addition, three easily accessible adjustable potentiometers provide settings for minimum RPM, maximum RPM, and current limiting. The simple, yet innovative, SCR control architecture also provides much tighter speed regulation than alternative controls.

"Simple solutions are often the best," said Matt Hanson, Bison Gear vice president, portfolio management. "The new TightDrive enables machine builders to put the control, the power, and the gearmotor more conveniently at the point of use, while saving installation time and reducing costs. As a bonus, users can maximize energy savings by easily changing speeds as requirements change."

The TightDrive is designed for 115 volt 50/60 Hz operation and comes complete with a three foot power cord and NEMA 5-15P plug.

Users can mount the drive directly to the motor in 90° increments to optimize position of cord exit and motor leads. The new TightDrive speed controls are

available for immediate shipment from Bison's distribution network or direct from the Bison Gear website.

Like all of Bison's gearmotor products, the TightDrive speed controls are American made to Bison's high quality standards in order to ensure reliable, long-life operation and direct supply chain fulfillment from Bison's St. Charles, IL facilities. In addition, Bison's Innopreneurial™ application and design engineering capabilities to customize or adapt these controls to meet specific OEM needs are readily available.

For further information, please contact:

**Bison Gear &
Engineering Corp**

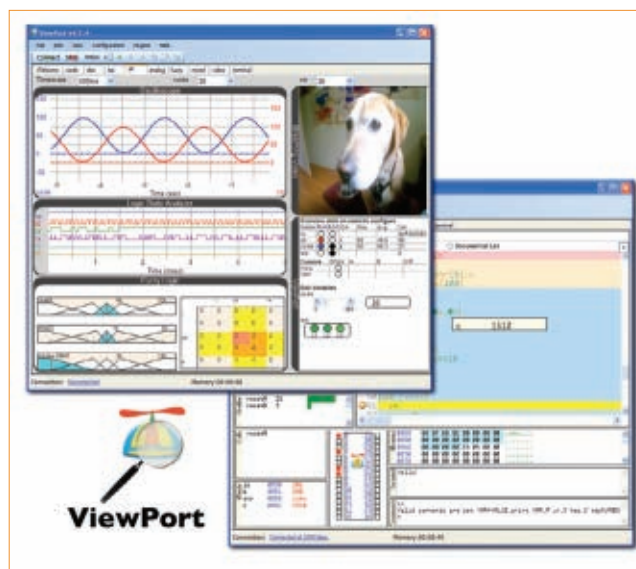
Website: www.BisonGear.com

SOFTWARE

ViewPort Software Version 4.1

Parallax is now selling ViewPort Software (Standard and Ultimate versions). ViewPort, developed by Hanno Sander at **MyDanceBot.com**, is the premier debugging environment for Parallax's eight-cog multiprocessing Propeller microcontroller. The tool combines an integrated debugger with powerful graphics that show you what's going on within the Propeller. Users can monitor variables over time with the built-in oscilloscope or change values while the Propeller is running. You can also solve hardware problems with the logic analyzer at sampling rates up to 80 Msps and add intelligence to programs with the fuzzy logic module or integrate computer vision using the OpenCV library.

"ViewPort v4.1 is solid, stable, and versatile, a very impressive achievement. Now, I am going to incorporate ViewPort in several of my research investigations," says a Parallax customer. Both Standard and Ultimate versions



come equipped with a debugger. The Standard version is low speed (up to 115 kbps) while the Ultimate version is high speed (up to 2 Mbps) and includes OpenCV (state-of-the-art computer vision processing), Development Kit, and Designer (customize the graphic instrumentation via drag and drop).

ViewPort can be integrated into any Spin program. It requires one cog and a single line of code at the start of the program. There are several tutorials, videos, and documentation included. ViewPort is also configurable and extensible so users can customize it to their needs.

The Propeller chip makes it easy to rapidly develop embedded applications. Its eight processors (cogs) can operate simultaneously — either independently or cooperatively — sharing common resources through a central hub. The developer has full control over how and when each cog is employed. There is no compiler-driven or operating system-driven splitting of tasks among multiple cogs. A shared system clock keeps each cog on the same time reference, allowing for true deterministic timing and synchronization. Three programming languages are available: C (via ICC for Propeller); the easy-to-learn high-level Spin (native); and Propeller Assembly (native), which can execute at up to 160 MPS (20 MIPS per cog).

ViewPort is available for purchase from Parallax, Inc., or **MyDanceBot.com**; the Standard version is \$59 and the Ultimate version is \$149.

For further information, please contact:

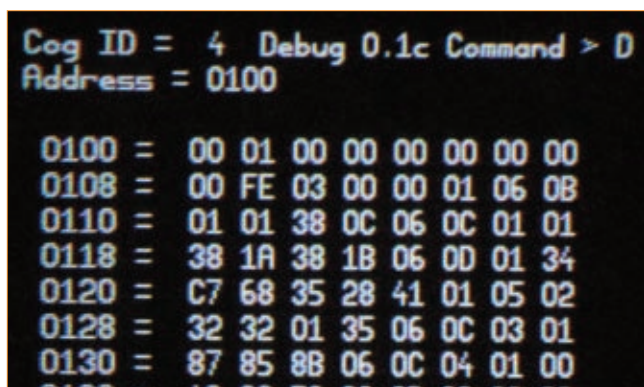
Parallax, Inc.

Web: www.parallax.com

TOOLS & DEBUGGING

Propeller Chip "KISS" Debugger Program

The KISS Debugger is an indispensable tool for writing and debugging software applications available now from Machine Intelligence Technologies.. A debugger saves time and can serve as a test bed to get projects up and running quickly. This debugger is a great source for code to incorporate into your own project.



Many routines that run on a simulator won't run on the target machine, so a simulator can't provide a real time development environment like the KISS Debugger.

This simple debugger is designed to use very little of the RAM in your development cog while providing valuable information on what is actually happening in real time.

Debugger Commands

- A: Display Main RAM address of label — See where a piece of SPIN code or data starts in Hub RAM.
- D: Display Main Memory Block — Display consecutive memory locations in Hub RAM.
- E: Set/Clr DIRA.
- G: Go To Loc.
- H: Output this list of cmds.
- Help command.
- L: List cog RAM contents — List the contents of 32-bit words in cog RAM addresses 0 through 495 (\$1EF).
- M: Display/Change Main Memory Bytes — View and optionally change the contents of memory bytes in Hub RAM.
- O: Set/Clr OUTA.
- R: Dump Regs — List the contents of 32-bit special purpose registers in cog RAM addresses 496 (\$1F0) through 511 (\$1FF).
- S: Start Next Free cog.
- T: Execute Test Code — Insert your test code in this command. Then use this command to execute it.
- X: Select Start cog — Select a specific cog, download your ASM code, and execute in the cog.
- Del Key: Exit This Repeat — Exit from a command that performs a variable number of operations.
- End Key: End Cmds D, E, L, O, M — Exit from instructions that display or modify consecutive locations of memory.
- Esc Key: Exit Debugger.

The KISS Debugger is available for \$20.

The Propeller Servo Control Software

Another indispensable tool from Machine Intelligence Technologies is the Servo Control software. It is used for controlling up to 16 independent servos with no perceptible jitter or servo buzz. Servo Control v1.1 features include:

- No external circuitry needed.
- Connect your servos directly to Propeller I/O pins.
- Controls 16 servos independently.
- Zero jitter.
- No servo buzz.
- Full rail-to-rail servo motion.
- User selectable framerate — adapts easily to any

The Escape Robot's built-in microprocessor enables it to "think" on its own. (KSR4) **\$29.95**



The robot frog moves forward when it detects sound and repeats: start (move forward) -> stop -> left turn -> stop -> right turn -> stop. (KSR2) **\$19.95**



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20 second voice recorder/playback module.

The electret microphone is on the board. One button records, the other button is momentarily pressed to replay the message.

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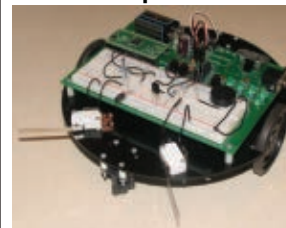
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COMBAT ZONE

Featured This Month:

Features

22 *BUILD REPORT:*
T6 — Evolution of a Full
Body Spinner **by Dennis Beck**

28 *PARTS IS PARTS:*
Custom Colson Wheel
Hubs **by Travis Schmidt**

Events

25 *Jan/Feb 2009 Results and*
Apr/May 2009 Upcoming
Events

25 *EVENT REPORT:*
2009 Chattanooga Robot
Battles **by Thomas Kenney**

ROBOT PROFILE — Top Ranked Robot This Month:

29 *Upheaval* **by Kevin Berry**

BUILD REPORT

T6 — Evolution of a Full Body Spinner

● by Dennis Beck

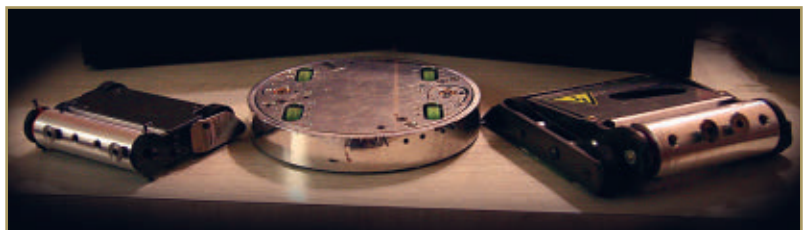
Conception

With so many combat robot builders building so many robots, it is given that there are going to be many robots that resemble each other. Proven and successful designs are a safe way to go. However, when I designed my first combat robot 1.5 years ago, I had only ever seen one other insect sized combat robot in my life. That robot was "Mars Attacks 2" built by Travis Schmidt of Rumble Robotics, a coworker of mine at the time. Before Travis took up employment with the company that I worked for, I had no previous knowledge of combat robotics outside of the late TV shows *BattleBots* and

Robot Wars. But when Travis brought his robot into work for me to see, I knew instantly that I had to build a combat robot. Without knowing about any other robot than Travis', I set out to design mine.

The destructive capabilities of a full body spinner were obvious, but their inability to operate when inverted was a huge turn-off for me. It didn't take long for me to decide to build an invertible full body spinner, or a "ring spinner" as I learned they are called inside the sport. Having the outer perimeter of the robot rotate while keeping the top and bottom plates stationary would allow the

Third revision of T6 (center) with antweight "Metroid" and beetle "Utopia."



Friction drive and ring support bearings from the original T6.

wheels to extend out of the top and bottom of the robot, allowing it to be invertible. I thought it was pretty clever.

Naturally, I wasn't the first person to ever think of this design, but little did I know that my first robot build was something rarely attempted by other roboticists due to the inherent complexity. I had my work cut out for me.

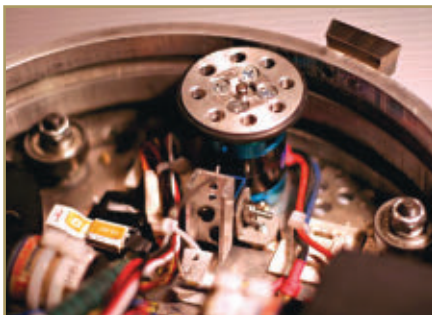
Design Hurdles

The most distinctive and defining feature of a ring spinner is also the biggest challenge to get to work effectively: how to support the ring so that it would spin freely, and transmit power to get it spinning? The idea for supporting the ring came quickly, but getting it to work reliably would be the biggest challenge.

I decided to use a section of 8" OD 0.25" wall 6061-T6 aluminum tube for the ring which I purchased from an online vendor. I used 0.090" thick 6061-T6 for a base plate since I had some leftover scraps in my garage from a previous project. The robot soon became known as "T6," for obvious reasons.

I contracted AJ Machine to turn a 0.125" deep groove into the inner wall of the ring in which four equally spaced skateboard bearings would reside. With the bearings mounted to the aluminum base plate, the ring "should" spin freely on the bearings.

Deciding on a method to transmit rotational power to the ring did not come so easy. I wanted to keep the height of the robot to a minimum for stability and weight reasons. With common brushed DC motors typically being quite long, the weapon motor would have to lay down in order to meet the strict height limitations of T6. It was then that I was introduced to the wonderful world of brushless motors.



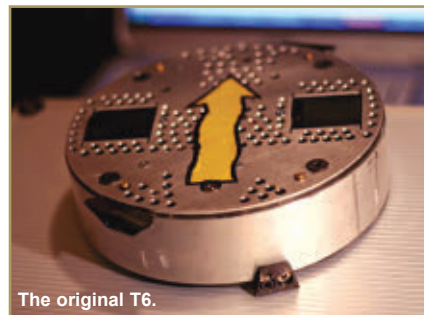
With the large variety of sizes that brushless motors were available in, it was possible for a motor to stand on end and still fit inside of T6. This would make transferring power to the ring via a friction drive much more simple. Friction drive was very desirable to me since it transfers less mechanical shock to the weapon motor under impact compared to a custom gear driven alternative, and prevents the weapon motor from stalling in the event that the ring becomes jammed.

Design Changes Throughout the Life of T6

Revision 1:

The original T6's drive system consisted of a Dimension Engineering Sabertooth 2X5 motor controller wired to a pair of B62 gearmotors. The tires were Lynxmotion 2.5" OD Sumo tires. A friend of mine worked for a local hobby store and gave me a good price on a Himax brushless outrunner, Castle Creations Phoenix 25 speed controller, and a Thunder Power 1,320 mAh three cell LiPoly battery. (And to think that I had intentions of keeping this robot as inexpensive as possible!)

In the first event that T6 competed at, it went 1-2, with the only win being against the robot that inspired the creation of T6 — Mars Attacks 2! While somewhat disappointed with the 1-2 record, I was very impressed with the huge beating T6 took from Team



The original T6.

Acme's "AMP," and drum spinners who's drum makes up two-thirds of the robot's weight!

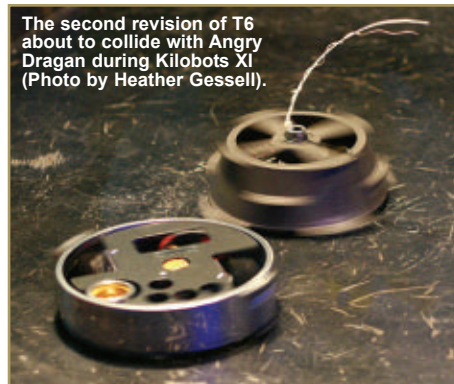
There were definitely improvements to be made. The B62's were powerful but slow. The weapon motor got the ring up to a decent speed, but took a while to do so. The Sumo tires were slippery, causing the inside of the robot to counter-rotate the direction of the ring during spin-up. The chunks of tool steel bolted to the ring were perfect for drum spinners to catch on. With all of these deficiencies, T6's future was bleak.

Revision 2:

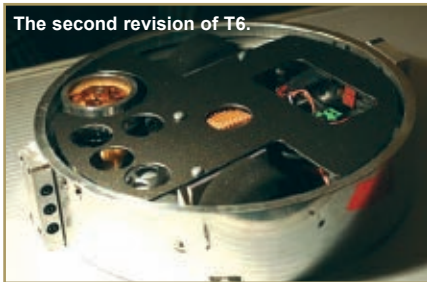
It was 11 pm the night before the Saskatoon Combat Robotics Club's (SCRC) Kilobots XI event when I decided to attempt to completely rebuild T6. Could T6 be rebuilt in nine hours? That was the true test.

Reusing only the ring and 2X5 motor controller, the rest of the robot would have to be built from scratch. I mounted the new aluminum base plate and garolite top cover inside of the ring for a

The second revision of T6 about to collide with Angry Dragan during Kilobots XI (Photo by Heather Gessell).



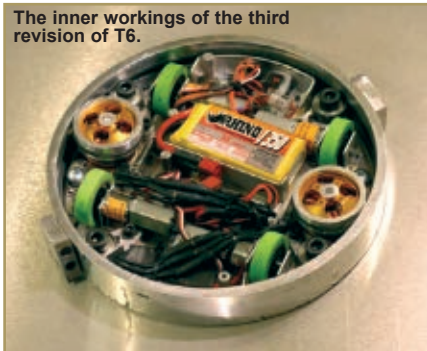
The second revision of T6.



Self-tensioning friction drive used in the second revision of T6.



The inner workings of the third revision of T6.



lower profile and for better protection. The overall height of the robot shrunk to 1.875". Lynxmotion 595 rpm gearmotors were used to make for a faster robot. The wheels were staggered in a skid-steer fashion to lessen the counter-rotation of the chassis under spin-up. A much larger brushless outrunner was used to power the ring. I fabricated a new self-tensioning system for the outrunner. The more power that is sent to the weapon motor, the harder the friction drive pushes into the ring. A more effective tooth design was also employed.

A flexing baseplate resulted in the bearings coming out of the groove in the ring under impact, and the drive motors jammed up, resulting in a 0-2 showing at KBXI.

Maybe it's time to give up on this design ...

Revision 3, the latest revision:

I don't know what keeps drawing me back to this design. Perhaps it's the challenge.

While taking a break from my other robots, I decided to see if it would be possible to fit four drive motors and wheels inside of T6's ring. It looked like it would fit ... barely! The catch was that I'd have to use 1-3/8" diameter 0.4" wide Banebot wheels. That meant that the ring and weapon teeth would have to be cut down to 1-1/8" tall in order for the wheels to extend out of the top and bottom of the ring.

Finding a weapon motor to fit in such a cramped space would be tough! Luckily **Hobbycity.com** has a huge selection of brushless motors, and I found a 37 mm OD outrunner that would fit. Since T6's layout was going to be a mirror image side-to-side, there would be room for two weapon motors!

Excited with my findings, I quickly worked towards getting the robot built as quickly as possible. Fitting the larger 1,350 mAh 30C Lipoly battery and the wiring for two brushless speed controllers into the super low profile robot was proving to be another big challenge.

Shiny finishes make good looking bots.



Keeping the wiring from contacting the inside of the ring was also a problem. I fabricated inner walls out of more aluminum to keep the wiring and components away from the moving parts.

It was soon obvious that I had run out of room for the 2X5 motor controller. Fortunately, I had a pair of Fingertech Tiny ESCs laying around from another robot. But would they be up to the task of running all four Banebot 24:1, 16 mm gearmotors? Yup! In a weaponed robot, they worked just fine.

The self-tensioning rocker arm design from revision 2 was used again, but had to be dramatically reduced in size to fit in the very cramped quarters.

Completed the night before a SCRC Demo event, I was anxious to try out the new version of T6! To my surprise, it worked great! Spin-up was almost instant, and the weapon was very effective for smashing up the chassis' of old mantisweight (6 lb) robots! T6 was even able to remove an internal combustion engine from its 1/8" garolite base plate! Halfway through the demo, a brushless motor came apart ejecting motor parts throughout the arena, but T6 kept working! Having two weapon motors was not such a silly idea after all!

The latest revision has not yet seen a true combat event, but another rebuild with a host of improvements is due to happen in the near future. A new heavier ring with angled sides milled from a billet of Fortal aluminum, a carbon fiber base plate for weight savings and strength, and four weapon teeth are on the list of improvements.

Taking the road less travelled, while being more of a challenge, has turned out to be quite rewarding. Employing a proven design may be easier to be successful with, but the challenge of making an unproven design work well can be much more rewarding. Well, at least for me it is! **SV**

Blunt tool steel teeth as used on the second and third revisions of T6.



EVENTS

Event Results and Upcoming Events

Event Results for January 10 to February 9, 2009

RoboWars 6 was held by RoboWars in Oakleigh, Melbourne, Australia on January 16th and 17th. Twenty-three bots were registered.



NW Model Hobby Expo 2009 was held by Western Alliance Robotics in Monroe, WA on February 7th. Fourteen bots were registered.



Upcoming Events for April-May 2009

(Note: All registration quantities are as of press time.)

Seatle Bot Battle 7 will be presented by Western Allied Robotics in Seattle, WA on April 12th. Twenty-six robots are

registered. Go to www.westernalliedrobotics.com for more information.



BattleBots 2009 High School, Collegiate, and Pro Championships will be presented by BattleBots in Vallejo, CA on April 21st. Thirteen robots are registered. Go to www.battlebots.com for more information.



BotsIQ 2009 National Competition will be presented by BotsIQ in Vallejo, CA on April 22nd. Fifteen robots are registered. Go to www.botsiq.org for more information.



HORD Spring 2009 will be presented by the Ohio Robot Club in Brecksville, OH on May 9th.



Eighteen robots are registered. Go to www.ohiorobotclub.com for more information.

CCR May Mayhem will be presented by Carolina Combat Robots in Greensboro, NC on May 23rd. Eight robots are registered. Go to www.carolinacombatrobots.com for more information.



The Tijdens Antweight World Series will be held in the Netherlands on April 10th. Go to www.dutchrobotgames.nl for more information.



dutchrobotgames.nl for more information. **SV**

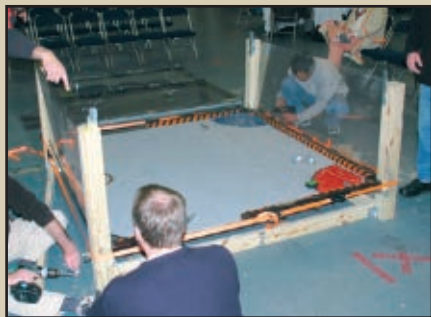
EVENT REPORT: 2009 Chattanooga Robot Battles

● by Thomas Kenney

The 2009 Chattanooga Robot Battles were held on Saturday, January 24th at the Chattanooga Choo Choo Holiday Inn. This

was the second annual Robot Battles event in Chattanooga, the first of which saw the revival of Tennessee's only robot fighting

competition. It also marked Robot Battles' 36th event since the competition began at Dragon*Con in 1991, making it the second



The arena is mostly reconstructed with new materials (Photo courtesy of Kelly Lockhart).

oldest competition still active.

The tournaments are much more relaxed than many RFL sanctioned events with a loose schedule, no judge's decisions, and an MC that fills the gaps between fights by poking fun at robots, builders, and occasionally members of the audience. The arena used is six feet square with polycarbonate walls on all sides and two 18" or so pit openings. It also includes a revamped arena hazard after the previous one was sheared off of its hub at the Dragon*Con robot battles.

The new arena hazard was just like the previous in appearance — a large rubber caster wheel covered in rough sandpaper, but this new version spins at speeds of up to 30,000 rpm, greatly increasing the destruction involved. Unfortunately, like many betas, it ran into some unexpected flaws and its use had to be cut short in the middle of the competition.

Lastly, like all Robot Battles competitions, the Chattanooga Robot Battles event was held at a

Hunter Wood and Randy Farmer from Team "We're Gonna Die" with beetleweight Dr. Feelgood (Photo courtesy of Kelly Lockhart).



MC Kelly Lockhart displays the award plaques for the antweight and beetleweight champions (Photo courtesy of Chattanooga).

science fiction convention, in this case, Chattanooga.

Set-up was scheduled to begin at 10:00 am, and the fights at 12:00. Unfortunately though, like nearly all things related to the sport, nothing went as planned. The event started out rough when the arena's lights and steel side posts were left in Atlanta by mistake. The absence of the lights could be easily dealt with, but the side posts, not so. The way the arena is constructed, the posts not only raise its height, but also hold the arena floor, the polycarbonate side walls, and the ceiling, or in short, the entire arena all together.

Through a quick run to the local Home Depot, the arena's owners, Rob Dillard and Brandon Davis of Team Found Object Robots, managed to find some suitable lights and wood beams to temporarily replace the steel posts. For the next hour, all of the competitors worked together in an attempt to hack together something resembling the arena's frame. Finally, through the use of copious amounts of wood, cardboard, and

duct tape, the arena was completely fit for combat.

Six teams, all Robot Battles regulars, brought a total of five one pound antweights and seven three pound beetleweights, including several new and heavily upgraded robots. In the antweights, Found Object Robots (FOBOT) added a titanium servo powered lifting arm to their joke antweight entry "Tigger," in an attempt to increase its competitiveness. Jason Brown from Evil Robots, Inc., increased his antweight "Rippy's" weapon speed, to reach an estimated 30 joules of stored energy. Finally, my team, MH Robotics, completely rebuilt their antweight "Gilbert," one of last year's winners, adding a razor sharp wedge and giving it enough drive power to be the zippiest robot in the competition.

The antweight double elimination tournament kicked off with a match between Dragon*Con robot battles champion "Misdirected Pedestrian" and the dreaded Tigger. The fight went through several rounds, ending with Tigger's armor being completely ripped off by the arena hazard.

The antweight fights continued with several anticipated matchups. Rippy, driven by Jason Brown of Evil Robots, Inc., was paired up with "Segs," an innovative, eight-wheeled, flexible bot. The fight ended with Segs driving out of the arena after a massive hit by Rippy tossed it through the air like a rag doll. The tournament continued with matches between Segs and

Misdirected Pedestrian, and Rippy and Gilbert, both of which the latter robots won.

Finally, the antweight finals turned out to be teammates Gilbert and Misdirected Pedestrian. The match looked like it could go either way at first, with Gilbert slamming Pedestrian into the wall repeatedly and Pedestrian getting several



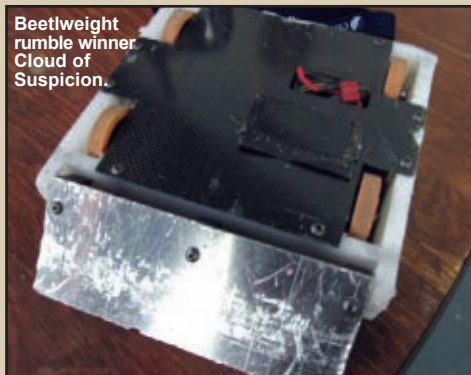
Beetleweight champion Tipan.

good bites in with its lifter. Eventually though, Gilbert ended up driving out of the arena and into one of the pits. In true double elimination format, the two robots were required to fight again. The same result followed, giving Misdirected Pedestrian the title of antweight champion.

Moving on to the three pound beetleweights, Jason Brown brought "Family Joules," a new four-wheeled robot with a 1.25 pound drum spinning at over 15,000 rpm. In the same spirit, my team, MH Robotics brought our creation, "Misdirected Aggression," a horizontal spinner with a 13 ounce steel bar spinning at 10,000 rpm. MC Kelley Lockhart temporarily amended the Robot Battles rules to allow Randy Farmer to enter "Dr. Feelgood," a wheeled platform with a small mobile turret mounted on top that could rapidly shoot airsoft pellets.

Among the other robots present were the pyramid shaped wedge "Pyramid of Death," the solidly built eggbeater robot "Drumbeat," the massively overpowered ramming brick "Cloud of Suspicion," and the simply huge dustpan sporting robot, "Tipan."

Several exhilarating matches ensued. The ramming brick Cloud of Suspicion fought drum spinner, Family Joules. Cloud of Suspicion was able to bully Family Joules around the arena, actually flipping the drum spinner several times by slamming it into the walls, while likewise, Family Joules managed to get in several good hits with its drum. Eventually, just like Gilbert, Cloud of Suspicion proved to be at a point, too uncontrollable. It often could not drive in a straight line, often slamming into the walls during its intended attacks. It eventually missed its target in an unlucky box rush and flew out of the arena. Cloud of Suspicion won a fight in the losers bracket against



Beetleweight rumble winner Cloud of Suspicion.



The Beetleweight bar spinner Misdirected Aggression.

Drumbeat, but lost its next and final match against Pyramid of Death when it once again slammed into a wall and bounced out of the arena.

Misdirected Aggression first fought Pyramid of Death. Its weapon proved to be just as deadly as it appeared, but equally unreliable. It loosened itself from the shaft after dishing out several impressive hits. Pyramid of Death had the fight won at that point, but unfortunately, like many others, it drove itself out of the arena and Misdirected Aggression took the win.

Misdirected Aggression's next fight was against Family Joules. Both robots spun up and cautiously approached each other at first. Then in one huge collision, Family Joules was knocked back and Misdirected Aggression was sent flying through the air, its weapon no longer secured to its shaft. Although Family Joules had lost half of its drive motors in the hit, lacking any way to attack or defend itself, Misdirected Aggression tapped out.

The beetleweight finals ended up being fought between Family Joules and Tipan, who had advanced through the loser's bracket.

Although it lacked the ability to drive properly, Family Joules was still able to take several decent sized chunks out of Tipan's large plastic dustpan, though it was eventually pitted, giving Tipan the beetleweight championship.

After the award plaques were presented, all of the antweights present were placed in the arena for one last untimed free-for-all. Rippy was the first to go out of the arena, and was eventually followed by Misdirected Pedestrian. Finally, Segs and Gilbert went out of the arena, giving the robot least expected to win — Tigger — the victory.

Event Results

ANTWEIGHTS

- 1st — Misdirected Pedestrian, MH Robotics
- 2nd — Runner Up: Gilbert, MH Robotics
- 3rd — Rumble Winner: Tigger, Found Object Robots

BEETLEWEIGHTS

- 1st — Tipan, Found Object Robots
- 2nd — Runner Up: Family Joules, Evil Robots, Inc.
- 3rd — Rumble Winner: Cloud of Suspicion, MH Robotics



Jason Brown from Evil Robots, Inc., with antweight Rippy and beetleweight runner-up Family Joules (Photo courtesy of Kelly Lockhart).



Antweight champion Misdirected Pedestrian pushes rumble winner Tigger towards the arena hazard (Photo courtesy of Chattacon).

Links

- **robotbattles.com** — Robot Battles website with a full listing of upcoming events, photo galleries of past events, and more.
- **chattacon.net** — Chattacon, the host of this Robot Battles event.
- **youtube.com/kenneyth** — Several fight videos from the event.

In the beetleweight rumble, Tipan and Family Joules teamed up when the builders placed the now immobile drum spinner inside Tipan's large dustpan beforehand, though once it began, Family Joules was quickly knocked loose by Cloud of Suspicion. Throughout the rest of the rumble, Dr. Feelgood continually shot airsoft pellets at random all through the arena, eventually causing the new arena light to break loose from the

ceiling. By the end of the rumble, Cloud of Suspicion was the only robot left moving, with Drumbeat and Family Joules still left in the arena, though sessile.

After the event was over, Brandon Davis from Found Object Robots taught a small class to some audience members on how to build a combat robot, answering questions with the help of other builders. The full event results are listed in the sidebar. **SV**

PARTS IS PARTS: Custom Colson Wheel Hubs

● by Travis Schmidt

1. Use a hacksaw to cut 2" aluminum round bar into several 1/4" thick blanks. I highly recommend a bandsaw for this job:



2. This hub is designed to be used with standard drill motors. I used a 21/64" drill bit to drill the center hole:



3. Tap the center hole for 3/8-24 threads. The holes to mount the hub to wheel are also drilled:



4. I machined the core of the 4x2" Colson wheel out so that the hub would pop into place so that the overall width of my bot would be shorter:



5. The drill motor threads onto the hub and is secured on the other side with a 3/8-24 locknut. I attached the hub to the wheel with a pair of 10-32 screws:



6. It is very important to make sure your motors are mounted as firmly as your wheels are. **SV**



ROBOT PROFILE

TOP RANKED ROBOT THIS MONTH

● by Kevin Berry

BotRank.com

Ranking Your World.

Top Ranked Combat Bots

History Score			Ranking		
Weight Class	Bot	Win/Loss	Weight Class	Bot	Win/Loss
150 rams	VD	26/7	150 rams	The Joke	4/0
1 pound	Dark Pounder	44/5	1 pound	Black Death	6/0
1 k	Roadbug	27/10	1 k	Roadbug	7/4
3 pounds	3pd	48/21	3 pounds	Yeti	8/0
6 pounds	G.I.R.	17/2	6 pounds	G.I.R.	8/0
12 pounds	Solaris	42/12	12 pounds	Tourinho	11/0
15 pounds	Humdinger 2	29/2	15 pounds	Humdinger 2	29/2
30 pounds	Helios	31/6	30 pounds	Touro Feather	8/2
30 (sport)	Bounty Hunter	9/1	30 (sport)	Upheaval	8/4
60 pounds	Wedge of Doom	43/5	60 pounds	K2	6/0
120 pounds	Devil's Plunger	53/15	120 pounds	Touro	18/3
220 pounds	BioHazard	35/5	220 pounds	Original Sin	10/2
340 pounds	SHOVELHEAD	39/15	340 pounds	Ziggy	5/2
390 pounds	MidEvil	28/9			
History Score is calculated by performance at all events known to BotRank			Current Ranking is calculated by performance at all known events, using data from the last 18 months		
Data as of February 10, 2009					

Upheaval – Currently Ranked #1



Class: 30 pound Sportsman Hobbyweight
Team: Mad Scientist
Builder(s): Alan Young
Location: Freehold, NJ

BotRank Data	Total Fights	Wins	Losses
Lifetime History	16	10	6
Current Record	12	8	4
Events	5		

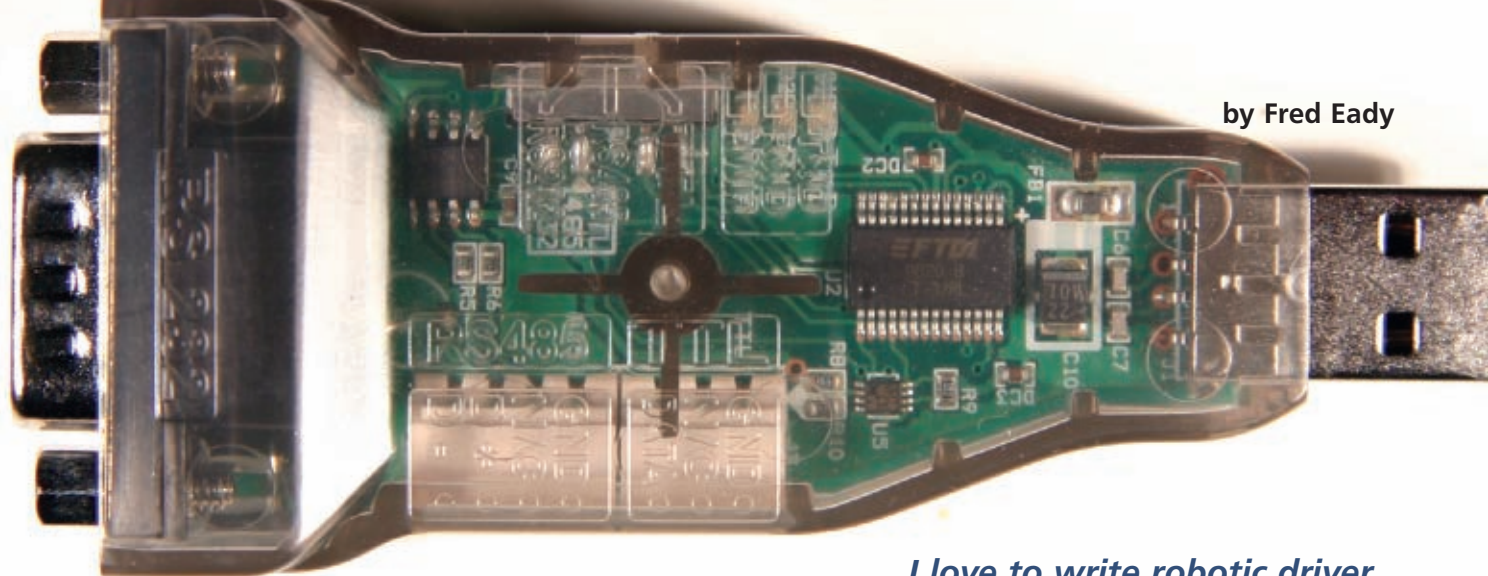
Upheaval has competed in: Franklin Institute Robot Weekend, Motorama 2008, Franklin Institute Robot Conflict, Motorama 2007, and House of NERC 2006. By press time, Upheaval will also have fought at Motorama 2009. Details are:

- **Configuration:** Pneumatic flipper.
- **Frame:** Welded 6061 aluminum and garolite.
- **Drive:** 2WD Banebots 36 mm planetary gearmotors.

- **Wheels:** 3" Colsons.
- **Drive ESC:** Victor 883.
- **Drive batteries:** Thunderpower four cell 2,300 mAh LiPo.
- **Weapon type:** Pneumatic launcher.
- **Weapon power:** 200 psi CO₂.
- **Armor:** UHMW plastic and duct tape.
- **Radio system:** Spektrum DX6.

- **Future plans:** Currently on the third rebuild ... I think I'll stick with this version for awhile.
- **Design philosophy:** Pack in as much flipping power as possible, while retaining reliability.
- **Builders bragging opportunity:** Record for most opponent flips in NERC event history. **SV**

Photo is courtesy of Eric Scott and information is courtesy of Alan Young. All fight statistics are courtesy of BotRank (www.botrank.com) as of February 10, 2009. Event attendance data is courtesy of BotRank and The Builder's Database (www.buildersdb.com).



UNWINDING THE AX-12+ COMMUNICATION PROTOCOL

In the discussion and hardware build-up that follows, our collective programming and hardware design/assembly efforts will be focused on the Microchip PIC18F2620, which will be coded to drive a Dynamixel AX-12+ robot actuator. There are several Dynamixel robot actuators in addition to the AX-12+. This month, we will center exclusively on instructing the F2620 to drive a Dynamixel

AX-12+ robot actuator. We've got some specialized AX-12+ hardware to design and assemble before we can begin to code the driver firmware. So, let's get started.

The Dynamixel AX-12+

I can easily describe the Dynamixel AX-12+ robot actuator you see in Photo 1 with a single word: SuperServo. The AX-12+ can do everything a standard hobby servo can and better. For instance, to obtain continuous rotation you don't have to disassemble and intentionally "break" it. You simply command it to perform an endless turn. Need to know where the servo shaft is? Don't ask a hobby servo, because unless it's one of the new digital models, it can't tell you. A Dynamixel can not only tell you where its shaft is, it can also tell you if it's moving.

In that you're reading this magazine, odds are that you have some prior exposure to hobby servos. For those experienced readers, you know that hobby servos move at their own mechanical pace with a set amount of torque. The Dynamixel can be



PHOTO 1. The daisy-chained one-wire RS-232 link I am referring to is actually a DATA line and a GROUND line. The AX-12+ also distributes power on a third daisy-chained line.

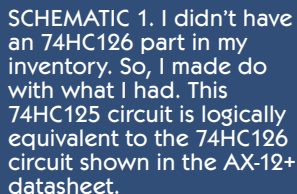
Standard hobby servos use a variable duty cycle pulse train to control their shaft's angular velocity and position. The duty cycle of the servo control pulse determines the servo shaft's rotational position while the angular velocity of the servo shaft is dictated by the speed of the duty cycle modulation. Thus, the slower the duty cycle change, the slower the angular velocity. A pulse width of 1.0 ms will move a standard hobby servo shaft to the extreme left, while a 2.0 ms pulse will move the shaft to the far right. Centering the shaft requires a pulse with a width of 1.5 ms.

The Dynamixel robot actuators don't depend on pulse widths for their position information. Instead, a half-duplex, one-wire, RS-232 protocol-based TTL communications link transfers command and status information between a host controller and the robot actuators. The TTL-level status and command messages are called digital packets. The term half-duplex means that devices attached to a common communications link are only allowed to talk when no other device is talking. In the case of the Dynamixel robot actuators, all of the robot actuators that are daisy-chained on the one-wire link spend most of their time listening and only speak after being spoken to.

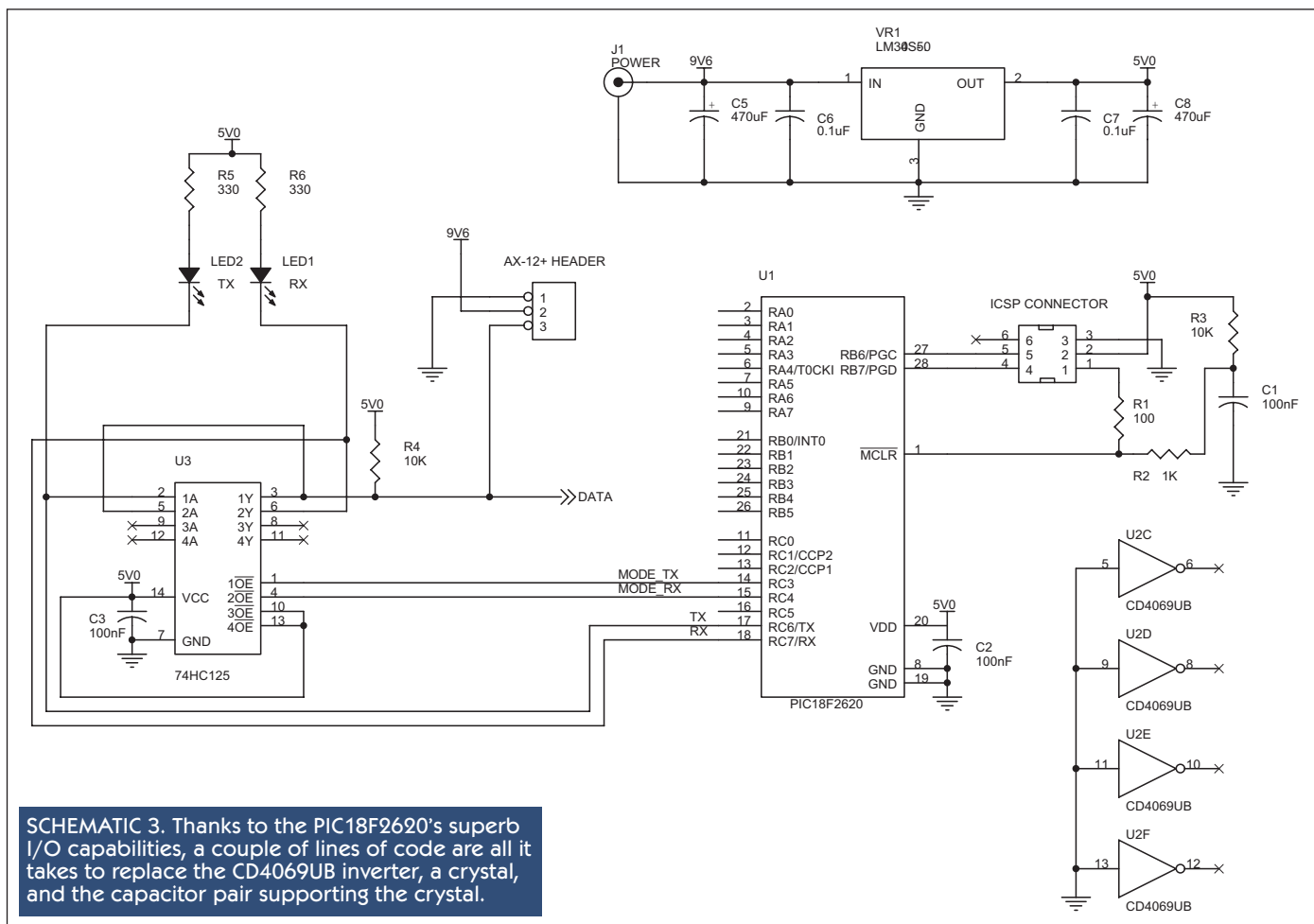
the daisy chain to listen and obey only. All of the Dynamixel robot actuators on the link are able to hear every message that is transmitted on the wire. However, each actuator that participates on the half-duplex one-wire TTL link is assigned a unique address between 0 and 253 decimal. If an actuator hears a message that does not contain its assigned address, the message is ignored. The only way to get the attention of every AX-12+ on the link at the same time is to send a digital packet using the broadcast address, which is 254 decimal (0xFE).

I could expound on the virtues of the AX-12+ all day. However, you're not here to listen to me talk. You're here to get the skinny on how to put an AX-12+ to work under the control of a PIC18F2620. With that, let's determine what we need in a hardware way to get the PIC and an AX-12+ to communicate with each other.

Behold Schematic 1. I've used a pair of CD4069 inverter gates to mirror the half-duplex transmit/receive logic that is set forth by the AX-12+ datasheet. A TTL high applied to the *MODE_SW* inverter input enables U3A, the transmit buffer, and tristates the output of U3B, the receive buffer. Conversely, a TTL low at the *MODE_SW* input tristates the transmit buffer's output and enables the receive buffer output of U3B. This simple circuit is the key to the implementation of the one-wire half-duplex TTL link required by the AX-12+. The AX-12+ datasheet presents the

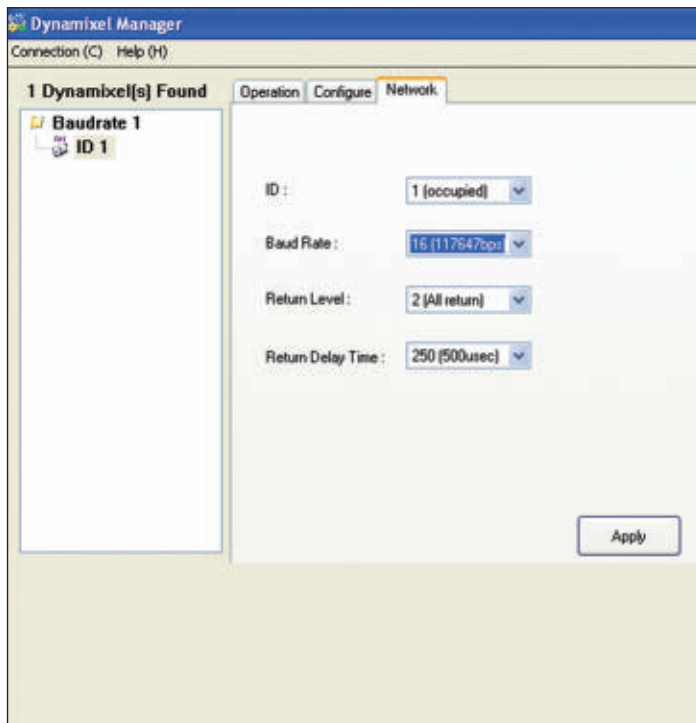


1. POWER FOR CD4069 AND 74HC125 -- PIN 14 = +5.0 - PIN 7 = GND.
2. ALL UNUSED CD4069 INPUTS TIED TO GROUND.
3. ALL UNUSED 74HC125 OUTPUT ENABLE PINS TIED TO +5.0.



There's no rocket science in the power supply or the

SCREENSHOT 1. I recommend adding this little utility to your programming arsenal. Although it looks like the original site is gone, search the web using PicMultiCalc and you'll find archives that will allow you to download the executable.



ICSP portal. Strip away the 74HC125 and CD4069 circuitry and this becomes a baseline PIC design.

Driving the AX-12+ with a PIC18F2620

The AX-12+ comes from the factory addressed as 0x01 and ready to communicate at its maximum speed of 1 Mbps. If you don't believe the PIC18F2620's EUSART can run with the big dogs at 1 Mbps, swing your eyes over to Screenshot 1. The math is courtesy of PicMultiCalc. This PIC utility runs on a PC and is the brainchild of Mister E Engineering. The absence of the PicMultiCalc download on the Mister E website seems to indicate that Mister E is out of the country at the moment. However, I did find a downloadable copy of the utility in the microEngineering Labs

SCREENSHOT 2. Exploring the functionality of this software tool with a USB2Dynamixel and AX-12+ attached to your PC's USB port is a quick and easy way to get acquainted with the Dynamixel robot actuator system.

PICBASIC forum. Here's what our EUSART initialization code looks like after incorporating the PicMultiCalc numbers:

```

//*****
/*      Init EUSART Function
//*****
void init_EUSART(void)
{
    SPBRG = 7; //7 = 1 Mbps with 32MHZ clock
    //SPBRG = 68; //68 = 115200 bps with 32MHZ clock
    TRISC7 = 1; //receive pin
    TRISC6 = 0; //transmit pin
    BRG16 = 1;
    TXSTA = 0x04; //high speed baud rate set BRGH = 1
    RCSTA = 0x80; //enable serial port and pins
    EUSART_RxTail = 0x00; //flush Rx buffer: Head=Tail
    EUSART_RxHead = 0x00;
    EUSART_TxTail = 0x00; //flush Tx buffer: Head = Tail
    EUSART_TxHead = 0x00;
    RCIP = 1; //receive interrupt = high priority
    TXIP = 1; //transmit interrupt = high priority
    RCIE = 1; //enable receive interrupt
    PEIE = 1; //enable all unmasked peripheral irq's
    GIE = 1; //enable all unmasked interrupts
    CREN = 1; //enable EUSART1 receiver
    TXIE = 0; //disable EUSART1 transmit interrupt
    TXEN = 1; //transmitter enabled
}

```

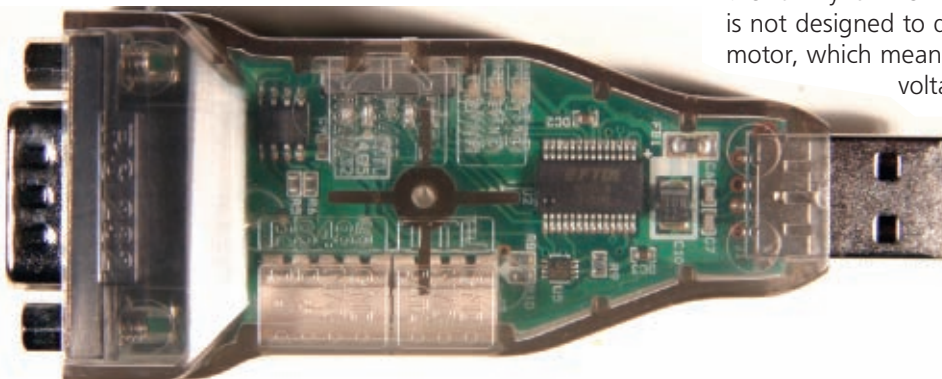
Note that I added a commented-out *SPBRG* statement for 115200 bps. The reasoning behind this is that I initially tested the AX-12+ driver at 115200 thinking that 1 Mbps was not a realistic baud rate for the PIC18F2620. If you want to experiment with other baud rates, you'll need to preload the out-of-the-box AX-12+ with your desired baud rate. The easiest way to do this is to use a Robotis USB2Dynamixel dongle like the one smiling at you in Photo 3.

The USB2Dynamixel is based on the FTDI FT232R USB UART IC. With the flick of a slide switch, the USB2Dynamixel can transform a PC's USB data stream into RS-232, RS-485, or TTL voltage levels suitable for use with the full Dynamixel robot actuator line. The USB2Dynamixel is not designed to drive the robot actuator electronics and motor, which means that we have to supply the bulk motor

voltage if we wish to exercise the AX-12+ using the USB2Dynamixel. The USB2Dynamixel is supported by a number of PC-based control and test programs, which can be downloaded from the Robotis website.

The AX-12+-USB2Dynamixel hardware hookup was simple. I crimped a pair of Molex female terminals (Molex part number

PHOTO 3. If you've been keeping up with my RS-232 to USB conversion projects in *Nuts & Volts*, you already know quite a bit about what's going on inside of the USB2Dynamixel. The USB2Dynamixel is designed around the FTDI FT232R USB UART IC.



16-02-1125) onto the opposite end of a couple of wires that I ultimately connected to a +9.6 volt power source. The AX-12+ comes with a three-wire jumper that I used to connect it to the USB2Dynamixel. I plugged the USB2Dynamixel into my laptop serial port, powered up the +9.6 volt supply, and kicked off the Dynamixel Manager application. As you can see in Screenshot 2 — with the help of the USB2Dynamixel and a home-made power cable — I used the Network portion of the Dynamixel Manager to preload the 115200 baud rate into the AX-12+. Naturally, I used the same process to return the AX-12+ to its original 1 Mbps baud setting. We don't need to run the PIC18F2620's internal oscillator at full blast to pump 1 Mbps out of its EUSART. According to PicMultiCalc, we can run with an 8 MHz clock and still achieve 1 Mbps throughput at the EUSART. Let's settle on running at full blast, which is 32 MHz. The clock coding for 32 MHz goes like this:

```
//*****
//*      INITIALIZE CLOCK AND I/O PORTS
//*****
OSCON = 0x70; //set for 32 MHz operation
PLLEN = 1;    //enable PLL for 32 MHz
TRISA = 0b01111111;
TRISB = 0b11111111;
TRISC = 0b10000000;
```

All of the I/O that relates to the AX-12+ is currently taking place on PORT C of the PIC18F2620. To get things moving as quickly as possible, I like to set the I/O port directions as soon as I can in the code.

To send a digital packet, we must know how the data is laid out within each packet. So, let's look at a digital packet from the viewpoint of a programmer using the C programming language. We'll stash our digital packets into a 128-byte array called *xmit_buff* until we're ready to send them.

Every digital packet begins with a pair of 0xFF sync characters:

```
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
```

Since our EUSART firmware engine uses circular buffers to hold its transmit and receive data, the pair of 0xFF sync characters will always stand as digital packet demarcation points. The byte that immediately follows the sync characters is the AX-12+ ID, which ranges from 0 to 253 decimal (0x00 to 0xFE). There will never be a trio of consecutive 0xFF characters as the ID can never be greater than 0xFE. So, we're still safe triggering on a pair of 0xFF characters to denote the beginning bytes of a digital packet. Here's what our digital packet looks like so far:

```
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = id;   //unique ID 0-253
```

The next byte in a digital packet holds a number representing the length of the digital packet, which is computed as the Number of Parameters + 2:

```
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = id;   //unique ID
xmit_buff[3] = parm_len + 2; //PARMS+INSTR+CHECKSUM
```

The additional two bytes added to the parameter length include the instruction and the digital packet checksum value in the packet length calculation. The parameters — if there are any — are squeezed in between the instruction and checksum bytes:

```
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = id;   //unique ID
xmit_buff[3] = parm_len + 2; //PARMS-INSTR-CHECKSUM
xmit_buff[4] = inst;  //instruction
xmit_buff[p] = parms[0],parms[1] //any number of params
xmit_buff[c] = packet_checksum //packet checksum byte
```

Here's how we define the seven AX-12+ instructions in our firmware:

```
//*****
//*      INSTRUCTIONS
//*****
#define iPING      0x01 //obtain a status packet
#define iREAD_DATA 0x02 //read Control Table values
#define iWRITE_DATA 0x03 //write Control Table values
#define iREG_WRITE 0x04 //write and wait for ACTION
#define iACTION    0x05 //triggers REG_WRITE
#define iRESET     0x06 //set factory defaults
#define iSYNC_WRITE 0x83 //control mult. actuators
```

The instruction parameters are kept in their own 128-byte array, which we call *parms*. Let's dry-run some example digital packets to show you how the *parms* array works with the *xmit_buff* array. We'll begin with building a PING digital packet, which has no parameters. The AX-12+ ID will be 0x01 in all of our examples:

```
//PING DIGITAL PACKET
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = 0x01; //unique ID
xmit_buff[3] = 0x02; //number of PARMS+
//INSTRUCTION+CHECKSUM
xmit_buff[4] = iPING; //instruction
xmit_buff[5] = 0xFB; //checksum
```

The only byte you probably can't figure out right now is held in *xmit_buff[5]*. The digital packet checksum is simply the bitwise inversion (logical NOT) of the sum of the ID byte, length byte, parameter bytes, and instruction byte.

Address	Symbol Name	Value
100	EUSART_RxBuf	
100	[0]	0x00
101	[1]	0xFF
102	[2]	0xFF
103	[3]	0x01
104	[4]	0x02
105	[5]	0x00
106	[6]	0xFC

Any bits that roll out of the checksum's least significant byte are ignored.

Let's dry-run with an instruction that requires some parameters. Let's manually code up a READ_DATA digital packet that will retrieve the AX-12+'s ID from the Control Table. The Control Table is simply a chunk of EEPROM and RAM that holds the AX-12+'s configuration and feedback data. The first 23 Control Table entries are nonvolatile. There are 49 Control Table memory slots:

```
//*****
//*      CONTROL TABLE ADDRESSES
//*****
enum{
    MODEL_NUMBER_L,           // 0x00
    MODEL_NUMBER_H,           // 0x01
    VERSION,                   // 0x02
    ID,                         // 0x03
    BAUD_RATE,                 // 0x04
    RETURN_DELAY_TIME,         // 0x05
    CW_ANGLE_LIMIT_L,          // 0x06
    CW_ANGLE_LIMIT_H,          // 0x07
    CCW_ANGLE_LIMIT_L,         // 0x08
    CCW_ANGLE_LIMIT_H,         // 0x09
    RESERVED1,                 // 0x0A
    LIMIT_TEMPERATURE,         // 0x0B
    DOWN_LIMIT_VOLTAGE,        // 0x0C
    UP_LIMIT_VOLTAGE,          // 0x0D
    MAX_TORQUE_L,              // 0x0E
    MAX_TORQUE_H,              // 0x0F
    STATUS_RETURN_LEVEL,       // 0x10
    ALARM_LED,                 // 0x11
```

SCREENSHOT 3. This is a capture of the PIC18F2620's EUSART receive buffer. A value of 0x00 in the ERROR byte is a very good thing as bits that are set indicate errors.

```
ALARM_SHUTDOWN,             // 0x12
RESERVED2,                   // 0x13
DOWN_CALIBRATION_L,          // 0x14
DOWN_CALIBRATION_H,          // 0x15
UP_CALIBRATION_L,            // 0x16
UP_CALIBRATION_H,            // 0x17
TORQUE_ENABLE,               // 0x18
LED,                         // 0x19
CW_COMPLIANCE_MARGIN,        // 0x1A
CCW_COMPLIANCE_MARGIN,       // 0x1B
CW_COMPLIANCE_SLOPE,         // 0x1C
CCW_COMPLIANCE_SLOPE,        // 0x1D
GOAL_POSITION_L,             // 0x1E
GOAL_POSITION_H,             // 0x1F
MOVING_SPEED_L,              // 0x20
MOVING_SPEED_H,              // 0x21
TORQUE_LIMIT_L,              // 0x22
TORQUE_LIMIT_H,              // 0x23
PRESENT_POSITION_L,          // 0x24
PRESENT_POSITION_H,          // 0x25
PRESENT_SPEED_L,             // 0x26
PRESENT_SPEED_H,             // 0x27
PRESENT_LOAD_L,              // 0x28
PRESENT_LOAD_H,              // 0x29
PRESENT_VOLTAGE,             // 0x2A
PRESENT_TEMPERATURE,         // 0x2B
REGISTERED_INSTRUCTION,      // 0x2C
RESERVE3,                    // 0x2D
MOVING,                      // 0x2E
LOCK,                        // 0x2F
PUNCH_L,                     // 0x30
PUNCH_H                      // 0x31
```

```
};
```

To access the AX-12+ ID byte, we need to build a READ_DATA digital packet to retrieve the fourth Control Table byte. Since the READ_DATA instruction requires parameters, the first step involves defining the parameter table in the *parms* array:

```
parms[0] = ID; //starting read address
parms[1] = 0x01; //number of bytes to read from
                // starting read address
```

Later on, we will write a function to load the *parm*

Sources

- **ROBOTIS — www.robotis.com**
USB2Dynamixel; AX-12+ Robot Actuator; Dynamixel Manager
- **Microchip — www.microchip.com**
PIC18F2620; MPLAB IDE; MPLAB REAL ICE
- **HI-TECH Software — www.htsoft.com**
HI-TECH PICC-18 C Compiler

The AX-12+ firmware driver was compiled with HI-TECH PICC-18 PRO.

A Microchip MPLAB REAL ICE was used as the debugging device.

SCREENSHOT 4. The payload data (AX-12+ ID) is loaded at offset 6 in the EUSART's receive buffer.

Address	Symbol Name	Value
100	EUSART_RxBuf	
100	[0]	0x00
101	[1]	0xFF
102	[2]	0xFF
103	[3]	0x01
104	[4]	0x03
105	[5]	0x00
106	[6]	0x01
107	[7]	0xFA

array entries into the correct memory slots of a digital packet. However, for now, let's insert the parameters manually:

```
//READ ID DIGITAL PACKET
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = 0x01; //unique ID
xmit_buff[3] = 0x04; //# of PARMS+INSTRUCTION+CHECKSUM
xmit_buff[4] = iREAD_DATA; //instruction
xmit_buff[5] = ID; //parameter 1
xmit_buff[6] = 0x01; //parameter 2
xmit_buff[7] = 0xF4; //checksum
```

Both of the digital packets we assembled will trigger a response from the AX-12+. In the case of the PING digital packet, we should receive a status message containing the AX-12+'s ID, an ERROR byte, and a checksum byte. If the PING operation is successful, here's what a returned status digital packet looks like from a programmer's point of view:

```
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = 0x01; //payload value = ID of AX-12+
xmit_buff[3] = 0x02; //# of PARMS+INSTRUCTION+CHECKSUM
xmit_buff[4] = 0x00; //error byte - 0x00 = none
xmit_buff[5] = 0xFC; //checksum
```

This status message contents are confirmed in Screenshot 3, which is the PING response data I received from an AX-12+ with an ID of 0x01. I also took the liberty to capture the response for the READ_DATA digital packet in Screenshot 4. Here's the programmer view of the status message data captured in Screenshot 4:

```
xmit_buff[0] = 0xFF; //sync character
xmit_buff[1] = 0xFF; //sync character
xmit_buff[2] = 0x01; //unique ID
xmit_buff[3] = 0x03; //# of PARMS+INSTRUCTION+CHECKSUM
xmit_buff[4] = 0x00; //ERROR byte - no errors
xmit_buff[5] = 0x01; //parameter 1
xmit_buff[6] = 0xFA; //checksum
```

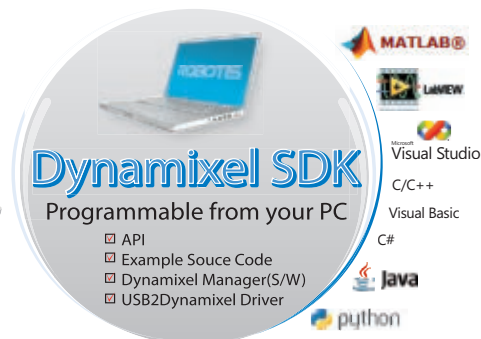
More To Come

I think you've got the idea. So, next time we'll add some meat to our firmware potatoes and code up a number of functions that will give us dominion over the AX-12+ Dynamixel robot actuator. In the meantime, I'll post the preliminary AX-12+ PIC18F2620 firmware I used here to communicate with an AX-12+ as a download package on the *SERVO* website (www.servomagazine.com). Be sure to have your AX-12+ controller hardware ready to roll as next time we're going to concentrate on the firmware. **SV**

Fred Eady can be reached via email at fred@edtp.com

Why DYNAMIXEL?

High-performance networked actuators for robots



AX-12+



- Engineering plastic gear
- Control network:TTL
- Gear Ratio : 254
- Postion Sensor : POT

RX-28 / DX-117



- Full metallic gear
- Control network:RS-485
- Gear Ratio : 193
- Postion Sensor : POT

RX- 64



- Full metallic gear
- Control network:RS-485
- Gear Ratio : 200
- Postion Sensor : POT

EX-106



NEW

- Full metallic gear
- Control network:RS-485
- Gear Ratio : 164
- Postion Sensor : Encode

AX-12+	at 7V	at 9.6V
Holding torque(Kgfc.m)	12.5	17.1
Speed(sec/60deg)	0.269	0.196
Weight(g)	53.5	

	RX-28		DX-117	
	at 12V	at 16V	at 12V	at 16V
Holding torque(Kgfc.m)	28.3	37.7	28.3	37.7
Speed(sec/60deg)	0.167	0.126	0.167	0.126
Weight(g)	72		67	

	at 15V	at 18V
Holding torque(Kgfc.m)	64.4	77.2
Speed(sec/60deg)	0.188	0.157
Weight(g)	125	

EX-106	at 14.8V	at 18.5V
Holding torque(Kgfc.m)	84	106
Speed(sec/60deg)	0.182	0.143
Weight(g)	155	

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COMPUTER CONTROL and DATA ACQUISITION

by David A. Ward

Part 2: An Introduction to National Instruments USB-6008 Data Acquisition Hardware

Part 1 introduced National Instruments LabVIEW software and had the reader build a very simple VI (virtual instrument). This article will introduce National Instrument's most affordable computer interfacing hardware — the USB-6008(9) data acquisition units — and show how to use the digital features of the units.

On the computer interfacing hardware side, NI offers many options, however, at the entry level the most affordable are the USB-6008(9) units. The USB-6008 student version sells for \$170 and the USB-6009 sells for \$280. These two units look identical on the outside (see Photo 1), however, the 6009 unit has several enhanced features that the 6008 does not have. First, the 6009 has a faster analog sampling rate — 48 ks/s versus 10 ks/s (thousands of samples per second). Secondly, the 6009 has a higher resolution analog-to-digital converter (A-to-D); 14 bits versus 12 bits. Thirdly, the USB-6009 digital outputs can be set up as either push-pull or open-drain; the USB-6008 digital outputs can only operate as open-drain.

I have worked with both units and believe that the less expensive USB-6008 will handle any task that an entry level user would require. Therefore, all of the examples given in this article and those that follow will be with the USB-6008, although everything should work fine on the USB-6009.

Let's take a closer look at all of the features available on the units. Each is divided between an analog side and a digital side. On the digital side, they have two digital I/O ports that are CMOS, TTL, and LVTTTL compatible. Port0 has eight digital terminals; P0.0 through P0.7. Port1 has four digital terminals; P1.0 through P1.3. The 12 digital I/O terminals can be individually configured as inputs or outputs, and as was mentioned earlier there is a difference between the types of outputs each unit can provide. There is another terminal, PFIO, that is connected to a 32-bit counter that can count



FIGURE 1



PHOTO 1

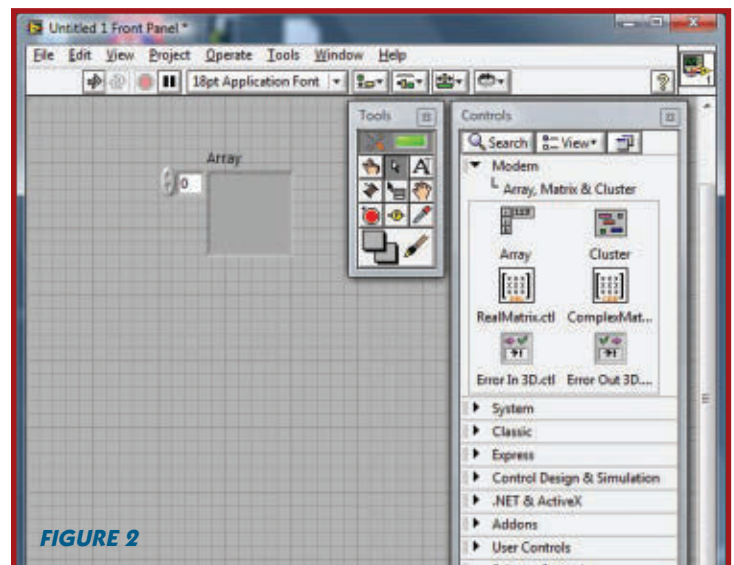


FIGURE 2

signals up to 5 MHz and can also be configured as an external trigger.

Also on the digital side is a +2.5 VDC at 1 mA terminal, a +5 VDC at 200 mA, and a ground terminal. The +2.5 VDC 1 mA terminal is for calibrating the A-to-D channels. On the analog side of the devices are 16 terminals, as well. There are four analog input ports, AI0 through AI3, each with a positive, negative, and ground terminal. When these ports are configured as differential inputs, they have 12 or 14 bit resolution (depending on the model). When they are configured as single-ended inputs, a total of eight individual analog voltages can be converted at one bit less in resolution. There are also two analog output channels, AO0 and AO1; these have a 12-bit resolution and can output an analog voltage from 0V up to 5V at a maximum of 150 Hz. (The analog features of the unit will not be covered until Part 4 of this series.)

Before you connect the unit to your PC, you must first install the NI-DAQmx drivers from the two CDs included with the unit. After the drivers are installed, a pop-up menu should appear when you plug the unit into a USB port on your computer (see Figure 1). From this menu, you can test the USB device, as well as several other choices. If you are having problems with your unit later on, you may try to resolve these problems through this menu; for now, we will close this pop-up menu and begin building a VI for the USB hardware.

Build A Framework

Let's make a simple VI that can be used to output four digital signals to control four LEDs connected to Port0. Begin by opening LabVIEW and selecting Blank VI. If you are having trouble starting a new VI, refer to Part 1. On the

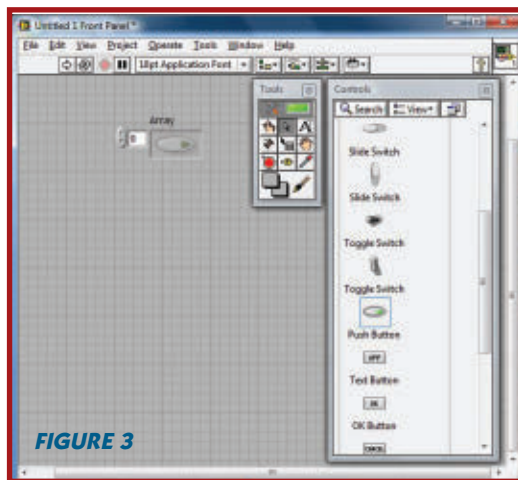


FIGURE 3

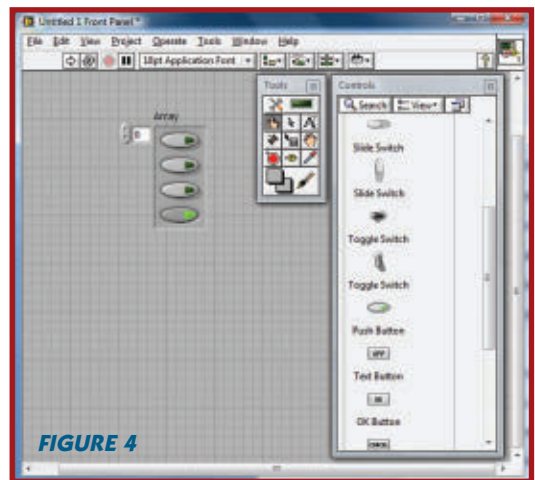


FIGURE 4

front panel, place an array by selecting: Modern>Array, Matrix & Cluster>Array from the controls palette (see Figure 2). Next, place a toggle switch in the box to the right of the array by selecting: Express>Buttons and Switches>Toggle Switch (see Figure 3). Now, stretch the array with the toggle switch in it down until four toggle switches are visible, as shown in Figure 4.

Using the pointer finger cursor, select this from the tools palette; click on the bottom toggle switch in the array. If you do this correctly, the toggle switches will change from a grayed-out color to a more solid color. Add a stop button to the front panel at this time by selecting: Express>Buttons and Switches>Stop. Now, you will need to work in the block diagram window to complete the VI.

Write Some Code

In the block diagram window, click and drag out a while loop by selecting: Programming>Structures>While loop from the functions palette. Be sure to place both the stop button and the array icons inside the while loop. Wire the stop button to the stop icon in the lower right corner of the while loop. You will now need to use the DAQ Assistant to set up the USB unit for use by selecting: Express>Output>DAQ Assistant.

When you place the DAQ Assistant inside the while loop, it will open up a series of pop-up windows to step

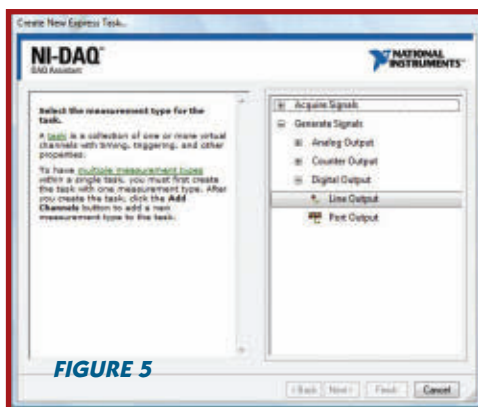


FIGURE 5

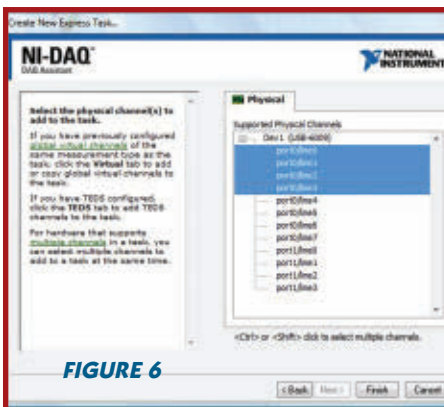
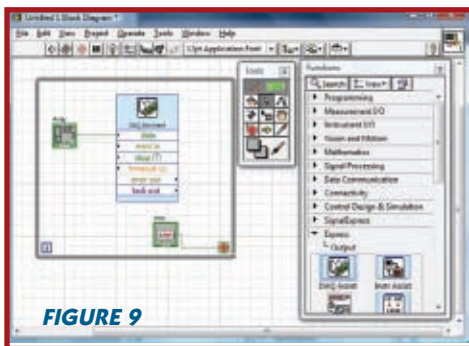
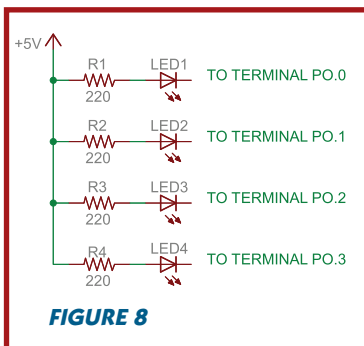


FIGURE 6



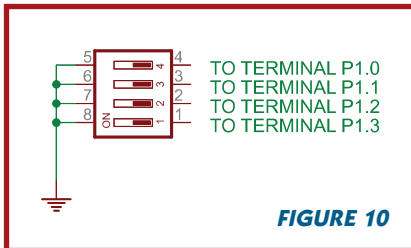
FIGURE 7



for now; press OK and you should be back in the block diagram window.

Create Some I/O

The array of four toggle switches now needs to be wired to the DAQ Assistant, but you'll notice that the DAQ Assistant has several wiring points to connect to. As you bring the wiring tool near each terminal, you should see pop-ups showing each terminal's name; connect the array output to the data



you through the set-up process. From the DAQ Assistant opening window select: Generate Signals>Digital Output>Line Output, as shown in Figure

input of the DAQ Assistant.

You will also notice arrows on the bottom of the DAQ Assistant icon; you can drag the bottom edge of the DAQ Assistant down and all of the terminal names will be visible to you (see Figure 9). Your VI should be ready to run. As you press each toggle switch, the appropriate LED connected to the USB-6008 terminals should turn ON and OFF, and pressing the stop button should put you back into the editing mode.

Now, let's add four digital inputs to the VI. We can connect a four position DIP switch to PORT1 lines P1.0 through P1.3 (see Figure 10). The DIP switches do not require external pull-ups because 4.7K resistors are built into the USB-6008(9)'s circuitry. The switches merely need to connect the terminals to ground when they are closed.

Open another DAQ Assistant by selecting: Express>Input>DAQ Assistant from the functions palette of the block diagram window and place it inside the while loop. From the DAQ Assistant windows that appear, select: Acquire Signals>Digital Input>Line Input, then select Port1/Line0 through Port1/Line3.

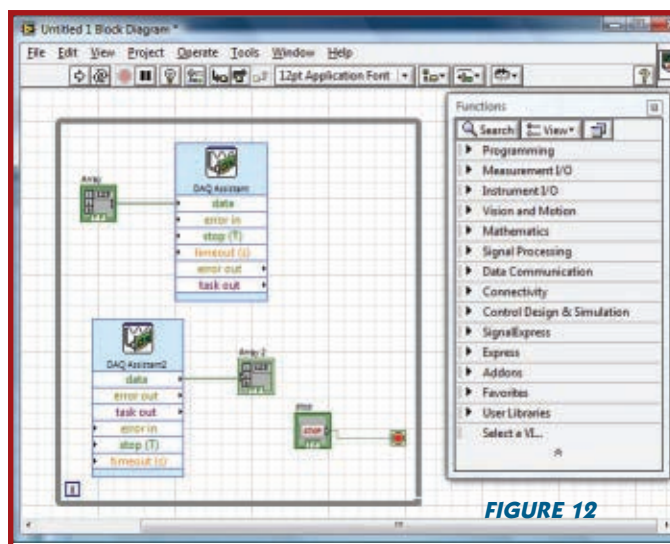
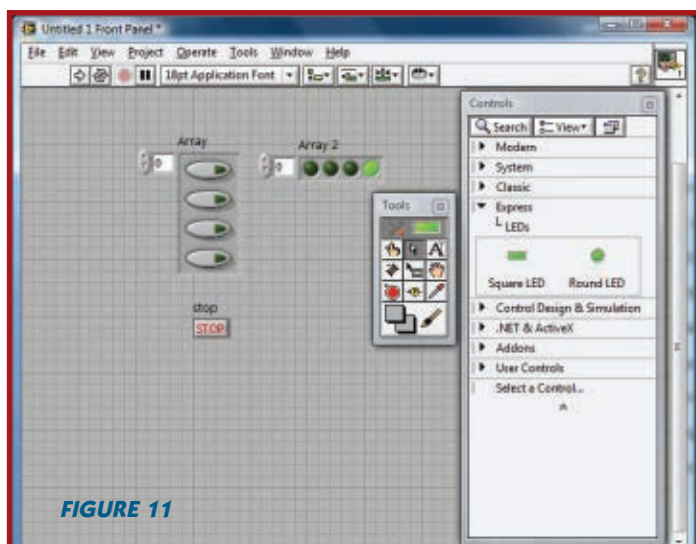
You can test these inputs just like the outputs from the DAQ Assistant window and you can also invert these lines as well, so that a +5V on a terminal pin coming from an external switch is seen as a "0" rather than a "1."

Notice on this DAQ Assistant the data terminal is on the right side of the icon and is now a terminal that will

5. From the next window, select Port0 /Line0 though Port0/Line3 by holding down the control key and left-clicking on each line (see Figure 6), then click on the finish button.

In the final window of the DAQ Assistant, you can test each output at the top of the window and you can also invert one or more lines if desired (see Figure 7). Since the USB-6008 digital outputs are open-drain only, they cannot supply +5V to the terminals but only take them to ground. Therefore, the LED cathodes should be connected to the USB terminals as shown in Figure 8. If we do not invert the lines, the LEDs will be OFF when the toggle switches are pushed and ON when they are released – which is not what the user would normally expect.

To invert each of the four lines, check the invert line box for each one separately or select all four lines by holding down the control key, then clicking on each line and then clicking in the invert line box to invert all four at the same time. We should be done with the DAQ Assistant



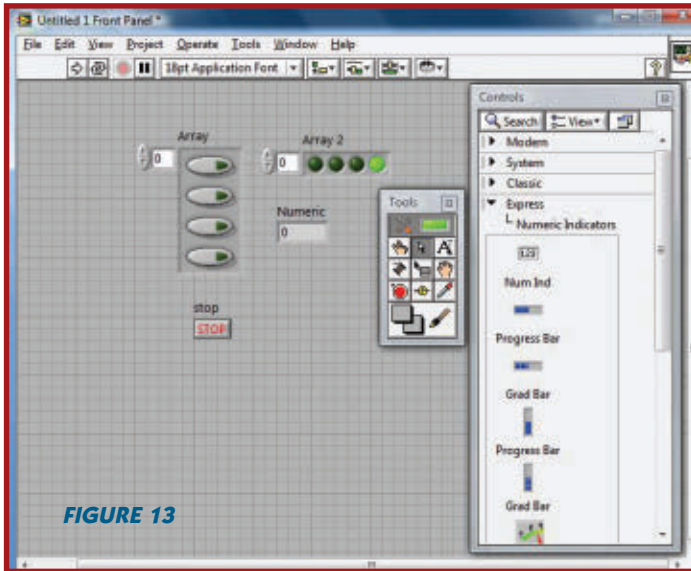


FIGURE 13

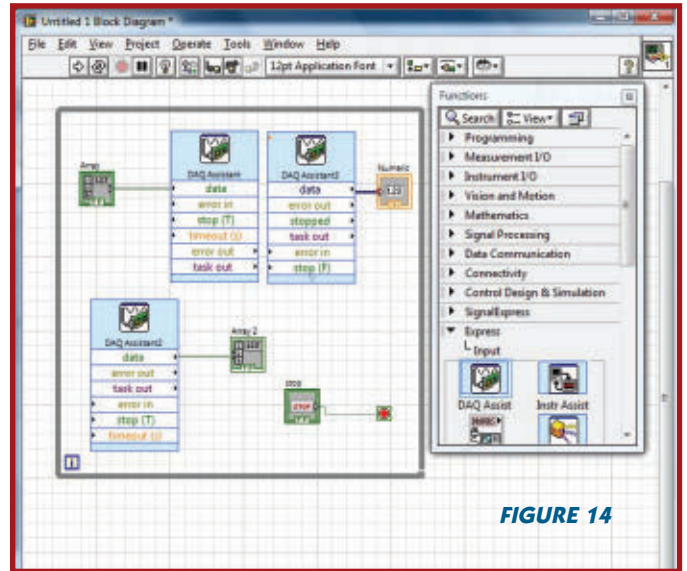


FIGURE 14

output data to an indicator.

Let's add four LEDs to the front panel to display the status of the external DIP switches connected to the USB-6008 unit. On the front panel, add an array again by selecting: Modern>Array Matrix...>Array from the controls palette. Place an LED in the box on the right side of the array by selecting: Express>LEDs>and either a square or round LED. Now, stretch the array either down or to the right until four LEDs are visible.

Now, click on the last LED with the pointer finger cursor to set the size of the array to four; remember that the LEDs will change from a grayed-out color to a more solid color when this is done correctly. Go back into the block diagram window and wire the output of the second DAQ Assistant to the input of the LED array (see Figures 11 and 12). When you run the VI, you should see the LEDs turn ON and OFF as you toggle your external DIP switches.

Next Steps

We can now control external devices and acquire data from digital inputs, however, these must all be TTL types of signals at this point; that is, +5 VDC and ground. Part 3 will introduce devices that can step these signals up to higher voltage AC and DC levels, such as 120 VAC.

The easiest way to demonstrate the 32-bit digital counter input will be to loop back one of the digital output terminals (such as P0.0) to the PF10 input; each time the P0.0 toggle button is pressed or cycled On and OFF, the

counter should increment by one. On the front panel, place another numeric indicator by selecting: Express>Num Inds>Num Ind (see Figure 13). Now, in the block diagram place a third DAQ Assistant by selecting: Express>Input>DAQ Assist. From the DAQ Assistant windows that appear, select: Acquire Signals>Counter Input>Edge Count. You'll also see other types of counter options such as frequency, period, etc.; however, the USB-6008(9) units only support the edge counter option. Next, wire the data output terminal of this third DAQ Assistant to the input of the numerical indicator, as shown in Figure 14.

To test your counter, you can wire the P0.0 output terminal back into the PF10 terminal; you will see the numeric indicator increment each time that output is turned ON and then OFF again. You can also feed a TTL compatible (+5V and ground) function generator output to the PF10 terminal on the USB-6008. You will see the number of cycles completed as you run the VI. With a 32-bit counter, the device can count up to 2^{32} or 4.29×10^9 pulses.

And Coming Up...

As mentioned earlier, the next part in this series will take what was done here and add additional external hardware to boost the outgoing digital signals and condition the incoming digital signals to real-world voltages such as 12 VDC and even 120 VAC. **SV**



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ROBOT VISION FOR EVERYONE

by John Blankenship and Samuel Mishal

If you have ever tried to add vision to one of your robotic projects, you probably appreciate why you seldom see articles exploring the subject at the hobbyist level. The ability to experiment with vision – especially in the past – has generally only been accessible to university researchers or the rare hobbyists with advanced tools and capabilities.

SOME INSIGHTFUL THOUGHTS

Let's start by examining why it is hard to develop vision applications. First, even though there are numerous low-cost cameras and web cams available, it can be difficult to capture images from these devices into your programming environment, especially if you want your programs to control image acquisition in real time. It can be done, of course, but generally, your programming language must allow you to communicate with the camera's driver. Often this means you must be able to utilize advanced programming constructs such as DLLs, the Windows API, and so forth.

After acquiring the image, you will need commands for manipulating individual pixels in the image so that appropriate vision algorithms can perform the desired analysis. You may need, for example, routines to alter the contrast and brightness of the acquired image. Depending on the application, you may also want the ability to alter the resolution of the image, convert it to black and white or gray scale, or detect the edges of objects in the image. It would also be nice to have routines that can detect motion by comparing two consecutively captured images, as well as the ability to track an object by being able to locate specific colors in the

image. Such routines can be complex and difficult to develop without advanced tools and programming skills.

With the release of RobotBASIC Version 3.2 (a free robot control and simulation language), all of the above capabilities are within the reach of every robot enthusiast. This article will explore some of these capabilities and how they can be used to give rudimentary vision to the puppet discussed in the November '08 issue of *SERVO Magazine* (see page 36). The goal of this article is to provide a basic framework that will encourage you to experiment with robotic vision. After demonstrating how easily the above operations can be implemented with RobotBASIC, we will examine other aspects of vision with which you can experiment further on your own.

DOING A DEMO

In order to give vision capabilities to the puppet mentioned above, a Creative Lab's web cam was added as shown in Figure 1. It resembles a miner's light and mounting it required only a rubber band strategically positioned behind the puppet's ears. A better setup would be to integrate the camera into the eye mechanism, but this simple approach is adequate for experimentation.

The goal of this demonstration is as follows: The puppet should be able to detect one of three different objects; we used a red vegetable brush, a yellow fish



FIGURE 1. A webcam provides vision for a humanoid puppet.

food container, and a green Christmas tree cookie cutter as shown in Figure 2. When one of the objects is detected, the puppet will track the object with its eye and head movements. Once the object has been centered in the puppet's field of view, the puppet will verbally announce the object that it sees (for example: "I see a red veggie brush").

In order to give the puppet more personality and to provide feedback for the human operator, the puppet will use its eyebrows to indicate the current visual status. If no object is seen, the eyebrows will lower, giving the impression that the puppet is agitated. When an object is spotted, the eyebrows will be raised to imitate an expression of interest. When the object has been tracked and centered in the field of vision, the eyebrows will assume a neutral position.

The actions described above are best appreciated if seen in real time, so we have created a video demonstrating the puppet responding to various objects. The video is posted on YouTube at the following link: www.youtube.com/watch?v=LwvspYFXJMM.

In order to perform the desired operations, the robot's control program needs to locate a specific color within the images captured by the camera on its head. Let's assume that we have a routine that divides an image into a 5x5 grid and gives us the **x,y** coordinates (each of which range from 0 to 4) of the sector within the image that contains the specified color. Once we have the values for **x** and **y**, we can easily determine which way the robot should turn so as to track a desired object (color) using a statement such as:

if x<2 then gosub LookLeft

The **LookLeft** subroutine can perform any actions you want. In our case, it raises the puppet's eyebrows, turns the eyes left first, then the head left to creating a lifelike movement. Our puppet's movements are controlled by servomotors as described in the previous article. RobotBASIC provides commands for accessing serial, parallel, USB, and Bluetooth ports, though, so you can move your robot using any motor or actuator you wish.

If we add a **LookRight** subroutine that is executed whenever **x>2**, we have the basis for a tracking system. Placing these two **if** statements within a loop will make the puppet continue to move its head (and the camera) to the left or right based on where the color is seen.

Similarly, if we add **LookUp** and **LookDown** subroutines that are executed based on the value of **y**, our robot can track an object (color) both horizontally and vertically. Another **if** statement can determine when the object has been centered in the field of view (when both **x** and **y** are equal to 2) and initiate the desired actions.

In our application, we have the robot continuously look for all three of the previously mentioned colors and track the one it finds. When the color is centered in the field of view, the puppet verbally announces the object that it sees

(based on the color that it has been tracking). The puppet can recognize any number of objects, as long as they have distinct colors. If, for example, the specified color is flesh tone the puppet would be able to track a face.

COMMAND PERFORMANCES

RobotBASIC makes all of the above easy to implement by providing a function to capture pictures called **CaptureImage()**, and a command that can track colors called **BmpFindClr**. The parameters for **BmpFindClr** provide tremendous flexibility when specifying how the search color should be located (all of RobotBASIC's image processing commands provide a similar flexibility). Let's examine the command and its parameters. The parameters in *italic* are optional.

BmpFindClr *FileName, Color, Var1, Var2, ClrTol, GridTol, GridSize, ArrayVar*

Parameter definitions:

- **FileName**: This is the name (and path) of the BMP image file. If the filename is an empty string, then the file is assumed to be on the Window's clipboard. Use RobotBASIC's **Capture** functions to acquire the image.
- **Color**: This is the RGB color to be searched for. If you prefer, RobotBASIC has functions that allow you to specify colors using levels of their red, green, and blue components.
- **Var1**: This variable will be set by the command to the

FIGURE 2. The puppet can recognize and track these items by their color.





FIGURE 3. Motion can be detected by comparing two images.

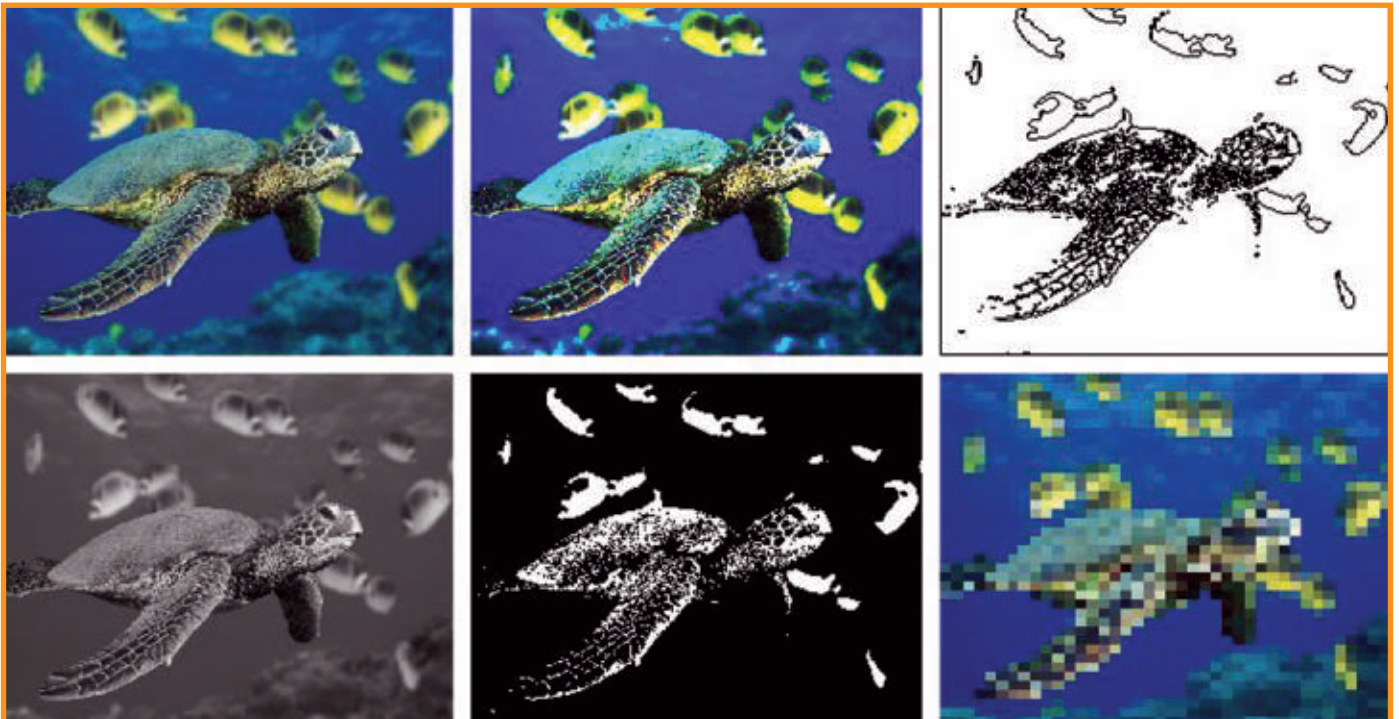
sector number (0-24 for a 5x5 grid) that contains the maximum number of pixels that match (see **ClrTol**) the specified **Color** (or -1 if no sectors contain the color). This sector number can be converted to the **x** and **y** parameters mentioned above as follows (for a 5x5 grid):
 $x = \text{Var1} \# 5, y = \text{Var1} / 5$.

- **Var2**: This variable will be set by the command to the number of sectors that contain a significant number of pixels (see **GridTol**) matching the specified **Color**. Your program can use this information as a validity indicator to decide if different tolerances should be used to acquire more valid information.
- **ClrTol**: This expression should be a number from 0 to 1 that sets the tolerance for deciding if two colors match. For example, if **ClrTol** is 0.1 then each of the RGB

components of a pixel's color must be within $\pm 10\%$ of the components of **Color** to be counted as a pixel that matches **Color**.

- **GridTol**: This expression should be a number from 0 to 1 that indicates the tolerance to be used when determining if a sector contains enough of the specified **Color** to be included in the sector count (**Var2**). For example, if **GridTol** is 0.1, then a sector must have 90% as many matching pixels as the sector with the most matching pixels (reported in **Var1**) if it is to be included in the sector count (**Var2**).
- **GridSize**: This expression should be a number from 3 to 20 that specifies the size of the grid matrix (a value of 5 indicates a 5x5 matrix).
- **ArrayVar**: This is a two-dimensional array (based on the selected **GridSize**) that provides information to facilitate a more detailed analysis. Each of the elements in the

FIGURE 4. RobotBASIC provides many commands to process your images.



array will be populated with the number of pixels within the corresponding sector that match the specified **Color**.

The parameters for **BmpFindClr** allow objects to be tracked in a wide variety of situations using only a few lines of code, as described earlier. As your applications get more complex though, additional commands might be needed. Let's look at some examples to illustrate this point.

GETTING COMPLICATED

When the lighting in the room where a robot is operating changes, the brightness and contrast of the captured images will change. RobotBASIC provides commands that copy pixel information from an image into an array and commands that calculate the standard deviation and average of the data in an array. These statistics provide an indication of the average contrast and brightness, enabling you to create more robust algorithms that allow your robot to work in a changing environment. For example, if your program determines and saves the average contrast and brightness of the image it expects to process, then in the future when an image is captured that is significantly different in brightness or contrast, it can be processed (using the RobotBASIC commands **BmpRGB** and **BmpContrast**) to make it a better match.

For another example, let's assume you want to track a moving object of an unknown color. The RobotBASIC command **BmpCompare** allows you to compare two images and create a new black and white image with the white areas indicating where the images were different (see Figure 3). The **BmpFindClr** command discussed earlier can then be used (searching for WHITE) to find the grid sectors representing the movement. If you want to experiment with more complex vision algorithms, then RobotBASIC can still assist you. There are

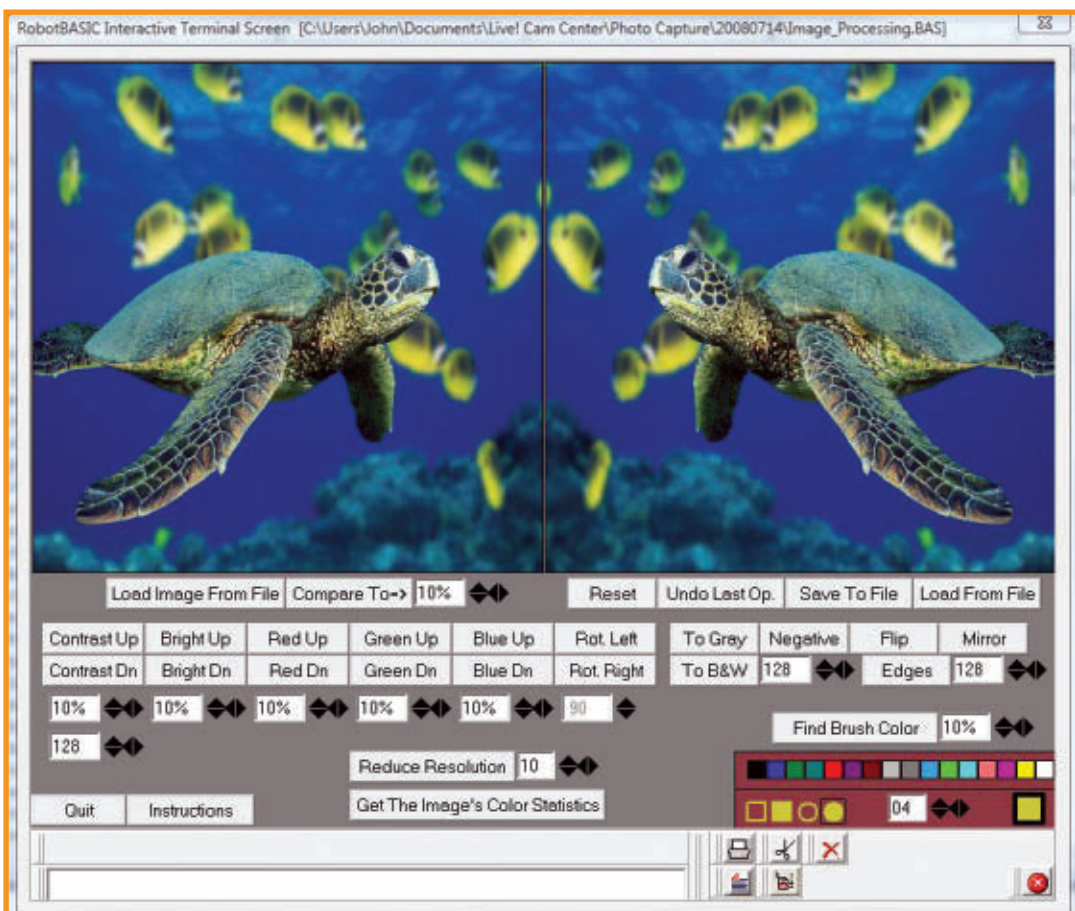
commands, for example, that alter contrast and brightness, reduce resolution, find the edges in an image, and convert it to black and white or a gray scale (see Figure 4). If you want to create your own image processing algorithms, RobotBASIC has commands to create a disk file containing the pixel information generated by any of its internal commands. You can then spawn a specialized program that you have written in the language of your choice to process the data as you wish before passing appropriate information back to RobotBASIC via a disk file.

FINAL INSIGHTS

In conclusion, RobotBASIC provides a platform for allowing the average hobbyist to add vision to their robotic applications. Since most hobbyists have limited experience with vision, we have written a demo program (see Figure 5) that allows you to perform most of the image processing operations discussed in this article on any BMP files, allowing you to try different approaches to see what works before you start coding. You can download the demo program and the vision program demonstrated in the YouTube video (along with your copy of RobotBASIC) from www.RobotBASIC.com.

The new version of RobotBASIC provides tools that allow hobbyists at any level of experience to readily and easily experiment with robotic vision. **SV**

FIGURE 5. This program demonstrates many of the vision and image processing commands available in the latest version of RobotBASIC.



Navigation & Tree measurement

by Jaakko Jutila

Finland — the land of a thousand lakes and forests — is located in northern Europe. Water covers 10% of the total area of 338,000 square kilometers, and forests cover up to three fourths of the area. The country is the most forested one in Europe. The population totals approx. 5.3 million and in every square kilometer there lives approx. 15.5 inhabitants in average (1). In the US, for example, there are approx. 30.6 inhabitants per square kilometer (2).

The terrain is fairly flat, which is a good thing for forestry and forest machines. The mean temperatures go down to -5.7°C in February and in July $+17^{\circ}\text{C}$. This climate is good for the four main tree species in Finland. Nearly 50% of the timber consists of pine, the other three common species are spruce, downy birch, and silver birch (3).

Forestry has long roots in the history of Finland and wood and paper products have the third biggest export share with about 20.3% of the total export. Forest research is also at a high level in Finland and, for example,

the soil type of most of the forests is known. (1)

The Forestrix Project

As forest industry is moving towards more automated machines and efficient production control, there is a growing interest on data gathering from inside the forests. The traditional way of manual or aerial measurements has strict limitations on what kind of data can be gathered and how accurately. So, what does this have to do with robots?

The motivation for this project is twofold and both of the sides are heavily related to the application of sensor technology. The first motivation relates to knowledge that can be gathered while moving through the forest. The trees that are left standing can be mapped and many kind of statistics can be gathered. When the structure of the soil of the specific forest lot is known, this measurement information can be very useful in forest management and planning. The information gathered during the thinning can be valuable when evaluating when the final harvest should be made and what kind of wood material can be expected from it.

The other part of the motivation relates to the human aspect of forest machine operation. Forest machines are difficult to control and operator training is time-consuming and expensive. The working conditions can also be very exhausting to the operator. Long work shifts, repetitive movements, and the relentless concentration can be very wearing in the



FIGURE 1. Ponsse forest harvester.

long run. The burden of the operator could be lightened by increasing the automation level of the forest machine. This could mean, for example, that the control of the crane would be partly automated. The operator could just indicate which tree to cut and the crane would automatically grasp it. This would give the operator the chance to better utilize his greatest strength, which is the expertise about the trees. Increased automation would also make the job easier and the training period could be shortened for new operators.

The Forestrix project studies advanced sensor systems for forest harvesters. One of the sensors used is a scanning laser range finder (henceforth, a laser scanner). As the machine is moving in the forest, the laser scanner can take multiple scans of the trees it passes from different directions.

In the Forestrix project, the laser scanner is used for tree measurement and mapping, and simultaneous localization and mapping (SLAM). In robotics, SLAM is used for solving a chicken and egg problem: When the robot moves in an unknown environment, it has to create a map of the environment on the one hand, and localize itself within the same map on the other.

The standard practice for the height of the tree diameter measurement is equal to 1.3 meters above the ground. When having only one static 2D laser scanner onboard, it is impossible to determine the measurement height without any kind of terrain model. Such a model was not available at the time and therefore we had to settle for an approximation with the measurement height.

In the mapping, a GPS receiver was tried for localizing the measurement platform. However, the thick foliage of the forest prevents a continuous GPS fix and thus pushed us to use some other method for localization. SLAM methods normally require good landmarks in the environment and, in this case, the trees in the forest are ideal for that purpose.

Hardware

Obviously, it would have been too expensive to have a genuine harvester laying in the backyard of our laboratory during the whole project (Figure 1). For that reason, we equipped a Honda ATV (Figure 2) with the necessary sensors and other equipment for the various stages of the project. The requirements for the platform in this particular case were an adequate ability to carry measurement equipment and the capability to maintain a steady and low locomotion velocity since the forest terrain is uneven. Plus, vertical smoothness while moving was also required.

We used a SICK 2D laser range finder as the primary sensor. We mounted the scanner in front of the measurement platform. It is used continuously at a measurement rate of 38 Hz. Additionally, a 3D laser range finder is used, but only for reference measurements and only when the measurement platform is in a stationary state. On the platform, a data acquisition PC is also installed. The measurement data are collected through serial connection and stored in numerical form to files. The measurement platform includes batteries



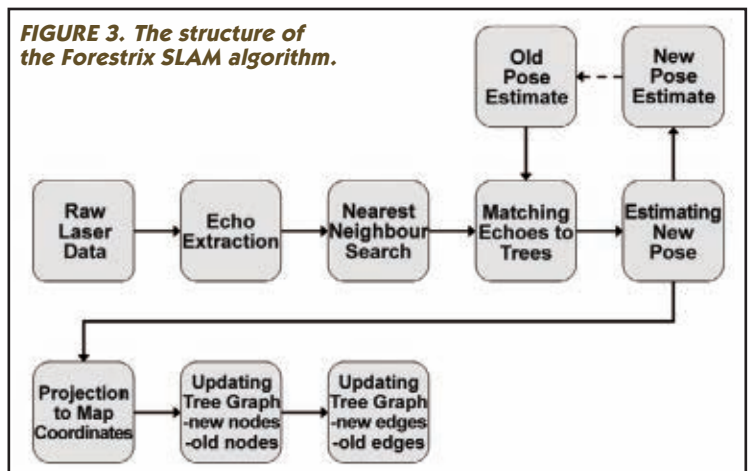
FIGURE 2. Honda ATV used in testing the system.

for a 24V electric system utilized by the measurement devices. An inverter is included for providing 230 VAC for the PC which performs the data acquisition with a conventional Windows XP operating system. The actual data acquisition software is programmed with C++ language. A brute user interface of the program is used to control measurement devices and data storing process.

Software

The tree map is created by using the SLAM algorithm. We'll discuss the operation of the tree measurement algorithm shortly. They work independently but the tree measurement information is added to the map in every cycle of the algorithms. The overall structure of the feature-based SLAM algorithm is shown in Figure 3.

FIGURE 3. The structure of the Forestrix SLAM algorithm.



Navigation and Tree Measurement

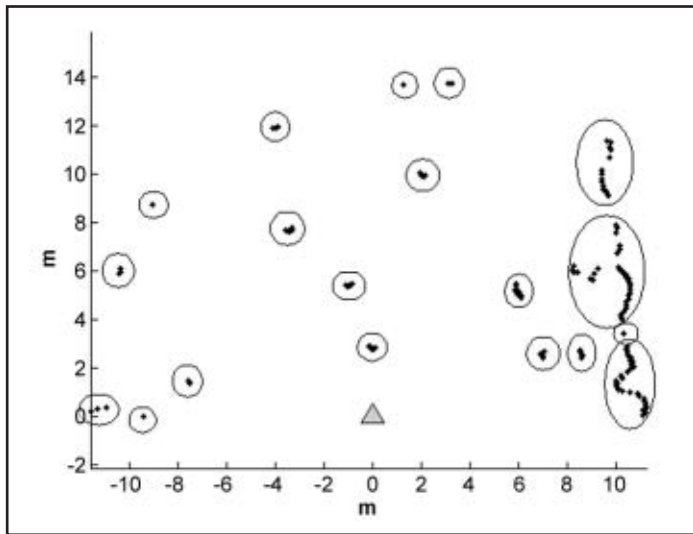


FIGURE 4. The date of one laser scan in a forest for which the tree segmentation is done.

position, not all trees may be visible. The matching of echoes to the tree map is done in the fourth step. After the echoes have been matched to the trees, a simple algorithm is used to estimate the new pose.

First, the “centers of gravities” of both set of points are calculated and the echo set is then translated so that the CoGs coincide. Second, the average angle between all echo-tree pairs is calculated and the echo set is then rotated by this angle. Care should be taken that all the angles are calculated on the correct cycle. This algorithm is based on the realistic assumption that the subsequent scans differ only by a translation and a rotation.

In the final phases, the echoes are projected to map coordinates. At this point, they are classified either as new or old trees. New trees are added to the tree graph and diameter information is accumulated to the old trees.

The closer the scan is taken, the better the diameter information. New edges are also inserted into the tree graph and old edges are updated to reflect this. Currently, only the distances between the trees are recorded but later angles will be used, as well.

One of the problems of this algorithm is that it makes the optimistic assumption that all echoes really are trees. This helps the algorithm to acquire new trees and to keep going, but if echoes of branches are falsely interpreted as trees they will be added permanently to the tree map. To solve this problem, another algorithm was added to remove false positives from the tree map.

The algorithm works by projecting laser scans to map coordinates and then using a collision detection algorithm to find intersections between the “rays” and the trees in the map. If there are more rays passing through a tree than there are valid measurements of that tree, the tree is removed from the map. This second algorithm was added as an afterthought and it reflects the inability of the first algorithm to record what is sometimes referred to as

negative information. Future improvements of the algorithm may use local occupancy grids to keep track of areas that are known to be free of trees.

The calculation of the tree parameters is divided into two consecutive phases. The first part deals with the method used for feature extraction in the measurement data. The second phase contains the determination of the midpoint location and trunk diameter. These methods provide only the relative location between the trees and the measurement device.

The feature extraction can furthermore be divided into two consecutive parts: clustering and validation. A straightforward clustering algorithm is used in this case when the points representing objects are easily distinguishable. The clustering

The SLAM is based on the raw 2D laser scanner data. The first step is the feature extraction. In this application, the features are the surrounding trees. The tree features identified from a single scan are called echoes due to their uncertain nature. In a clean forest with little or no underbrush, it is a relatively simple task to find the tree trunks from the raw laser scans. This is done in the second step.

An example scan and its segmentation are shown in Figure 4. However, in more dense forests, finding the trunks can be extremely difficult, if not impossible. The dense vegetation drastically reduces the effective measurement range. Even in relatively clean forests, the blind areas behind the nearest trunks can be substantial.

Instead of trying to identify individual trees, it is better to identify groups of trees. These groups are identified in the third step. The tree groups offer a variety of features that can be used for identification: distances and angles between adjacent trees, and trunk diameters.

One of the central challenges in this approach is that the tree groups are not constant. Depending on the current

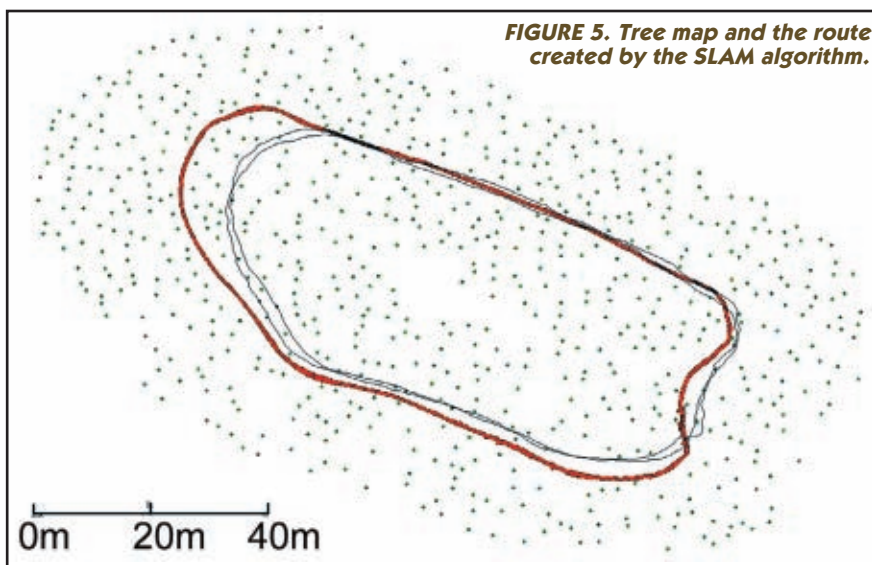


FIGURE 5. Tree map and the route created by the SLAM algorithm.

For Your Info

The forestrix project was funded by the Finnish Funding Agency for Technology and Innovation (TEKES) and participating companies. The description of the project is based on two conference publications authored by the Forestrix team (4), (5).

algorithm handles the distinct tree features in most cases with sufficient certainty.

While performing the scan depicted in Figure 4, the scanner has not been level with the ground. In the right-hand side of the figure, a set of large clusters can be seen. These represent the points where the laser beams have hit the ground. The algorithm is tuned in a way that these kinds of outlier points are discarded. The algorithm is quite picky with the discarding, as was the intention. Only the most definite features are accepted as trees. This way, some actual tree features are dropped but now we can be sure that every passed feature is truly a valid one and its parameters can be calculated.

Results and Conclusions

The current version of the SLAM software is written in the Java programming language. The software can read input data either from data files or from the actual sensors. The data files are essential for testing the system in a laboratory environment. The software can connect to the 2D laser scanner and the DGPS receiver. The SLAM path can be matched to the DGPS path. As a result, the algorithm also gives the tree positions in absolute map coordinates. The measured tree diameters are then stored in the tree map. The resulting absolute accuracy is mostly limited by the DGPS receiver.

A tree map produced by Forestrix SLAM is shown in Figure 5. The DGPS path is marked with a narrow line and the SLAM path is drawn with a bold line. Measured tree positions are also shown. These positions and diameters can be compared to hand-measured tree information provided by METLA (Finnish Forest Research Institute).

During the experimental drive, a total of 277 trees were observed. The testing environment is depicted in Figure 6. At a measurement distance of 1 m, the maximum observed diameter error is 1 mm. The error increases linearly with distance up to 10 m, where it is 6 mm. After that, the error increases more rapidly.

The current algorithms implemented in Java run somewhat slower in modern laptop computers than what is required for real-time applications. Some parts of the presented SLAM algorithm can be easily tuned for better performance. Java is a safe language which makes it easy to program with, but it is not the best programming language performance-wise. Also, interfacing it with exotic hardware such as 2D laser scanners can be difficult. C++ implementation may have to be considered in later phases of the project.

A forest is a very tough environment for precision



FIGURE 6. One of the experimental environments in a pine forest.

instruments. Luckily, there are 2D laser scanners and DGPS receivers that are designed for outdoor use. The forest terrain is often quite rough, which adds additional challenges for the sensor system. It may be necessary to tilt the sensor package when the harvester is working in an inclined position or traversing slopes. More development work is needed to find better solutions to the association problem. The current system works well for small loops (50 m x 100 m). However, the accumulation of errors may be a problem for larger loops. Identifying trees may also be a problem in more dense and cluttered forest environments. Digital forest imaging has turned out to be a very challenging research area, but it is essential for future forest mapping systems which will require features such as tree species recognition. Digital imaging systems have the ability to see textures and other tree qualities that laser scanners cannot perceive. **SV**

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A3950 DC Motor Controller

Tips and Tricks

by Jose I Quinones

DC motors are plentiful. My favorite source for heavyweight robotics projects is electric scooters. A motor capable of transporting a 250 pound human can move a lot of robot. However, DC motors require significant current handling at startup, and during increased load conditions and direction reversal.

Thanks to major advances in H-bridge power FET technology, current handling is no longer problematic. This article examines the Allegro A3950 DC motor controller and details how it can be interfaced to a microcontroller.

The A3950

Allegro Micro's A3950 is a single H-bridge DC motor controller capable of delivering up to 2.8A of current in both directions with up to 36V input voltage. Controlling the motor could not be easier. Two main signals — ENABLE and PHASE — control speed and motor rotation direction, respectively.

An open collector output — the NFAULT pin — offers information on when the H-bridge has become non-operational due to faults such as shorts to ground or power plane, as well as the device becoming too hot (thermal shutdown).

This NFAULT signal is tied to a rather well designed logic which protects the device against most harmful conditions such as the mentioned shorts. Now you can feel safe about mis-wiring your motor. I have tried a good deal of combinations and all of them resulted in the device shutting down and protecting itself until the short was removed.

Other important signals to understand include SLEEP, which considerably limits device total current consumption when the H-bridge is unutilized, to less than 10 uA. This is a priceless feature when dealing with battery saving modes.

The MODE pin selects from different current recirculation modes. This comes in handy when wanting to stop the motor really fast (a.k.a., brake). For braking, MODE must be HI, in which case a slow decay mode is selected. During this configuration, the motor winding back EMF is shorted, causing the braking to occur. Otherwise, during MODE LO or fast decay, the motor coasts down as it is disabled.

How Powerful is the A3950?

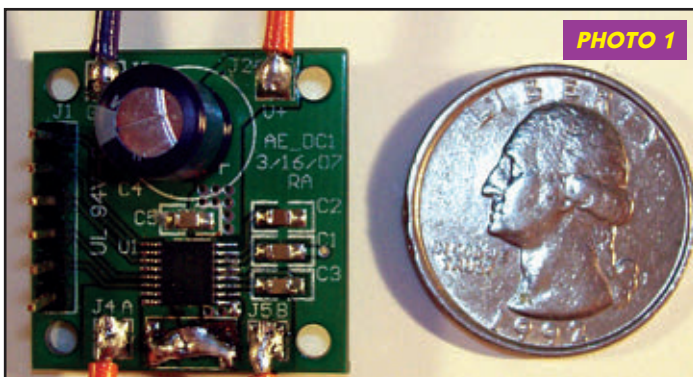
For being a small IC housed on a SOIC16 package with a bottom heatsinking pad, the device is impressively powerful; 2.8A of current at 36V translates to 100.8W of power into the motor load. If that is not enough, stay tuned for a little trick on how to double this.

How can such a small package provide such a large power range? There are two important details. First is the very low RDS_ON parameter on the A3950 FETs. Combined, the path resistance is about 650 megohms at room temperature. Power dissipated on the chip is then around 5.096W ($2.8A * 2.8A * .65\Omega = 5.096W$).

Granted that 5W is a pretty large amount of power to be dissipated as heat on any IC. In order to help against this cumbersome and unwanted heat generation, Allegro Micro has installed the device on a package with a heat slug on the bottom of the package. For this to work, PCB (printed circuit board) design must take this into consideration and offer enough copper space to act as a heatsink. Fortunately, any large copper area spreading into both top and bottom PCB layers seems to be sufficient. A good deal of vias (between five and 10) should be used to connect both copper fills.

The Design

I wanted my DC motor controller to be as small as possible. The idea was to allow for the controller to be used on small robots, as well as large ones. I was able to cram all the external components into a board as small as 1" by 1", and there was still room for improvement. This also makes the bare board



considerably cheaper, which was another requirement as I intended to buy them by the dozens. My hypothesis is that by parallelizing these boards together, a higher amperage rating could be obtained.

As you can see in Photo 1, the controller is not only tiny but the component count is considerably small. In reality, there is no need for much more as the motor driver itself is packaged on the tiny 16-pin SOIC package.

The controller has all the capacitors needed for the charge pump (the H-bridge uses four N-channel power MOSFET transistors and a higher internal voltage is generated to enable the high side drivers), the bulk capacitance for high current transient management — especially when the motor changes direction — and the internal regulator voltage bypass capacitor.

A sense resistor allows for a current max to be set. A zero ohm resistor signals the device to limit current to at least 2.8A. That's a lot of juice! If you want to limit the current to less than 2.8A, place a resistor such that the voltage at the SENSE pin is 500 mV. The equation is:

$$R_{SENSE} = \frac{500 \text{ mV}}{\text{DesiredCurrent(mA)}}$$

Four pads (J2 through J5) allow for battery power and motor leads to be soldered. A six-pin connector (J1) offers the inputs and outputs necessary for the microcontroller application to interface (FAULT, MODE, ENABLE, SLEEP, PHASE, and GROUND).

Table 1 shows the connections to power supply, motor and microcontroller signals.

Figure 1 Shows the DC motor controller Block Diagram and the connections needed to be made. Any microcontroller should be able to tackle on the task of sending these signals and reading the NFAULT output. I decided to leave the NFAULT disconnected for early tests, but will definitely use this feature as a trouble shooting tool on later projects.

Do be warned that I have found the NFAULT to get asserted a few times when the motor starts as the large inrush current is bound to be seen as a short. It is a matter of learning how much time the particular motor needs to start and ignore these NFAULT triggers for that time. Depending on the motor, you may see some of these NFAULT pulses when the motor switches direction, in which case the firmware must take this fact into account, as well.

Artwork used to generate the board can be downloaded from my website at www.avayanelectronics.com or from the SERVO website at www.servomagazine.com. I usually post firmware and other details, as well.

Figure 2 shows the schematic for the entire controller design. Six passive components, a chip, and a bare board offer a low count.

More Current!

For larger or heavily loaded DC motors, 2.8A of current may not be enough. However, what if we could add more

Connector	Name	Description
J1:1	Motor Fault	Connected to the NFAULT pin. Needs to be pulled up to rail (3.3V or 5V, depending on system).
J1:2	MODE	Controls current decay modes (fast vs. slow)
J1:3	PHASE	Controls direction
J1:4	SLEEP	Allows for very little current consumption when on sleep mode
J1:5	ENABLE	Enable/Disable and/or speed control if PWM'd
J1:6	GND	Microcontroller Ground
J2	V+	Solder Motor Battery Positive Terminal (8V to 36V)
J3	GND	Solder Motor Battery Ground Terminal
J4	A	Solder Motor Positive Terminal
J5	B	Solder Motor Negative Terminal

TABLE 1

controllers in parallel and increase current capability? I found the A3950 to allow for this enhanced power potential by plugging two controllers in parallel.

There are various ways of attaching the two controllers in parallel. I tried three different venues and the cleanest is shown in Photo 2. The item to look for is cable length. I am not certain what would happen if cables are grossly different in length, other than possibly inducing some fault on one of the controllers if the signals do get out of phase. However, with cables the same size, I was able to control the motor with two controllers in the same fashion as I had done with one with an added plus, that is a twofold increase in current!

H-bridges are nothing but controlled resistances. It is

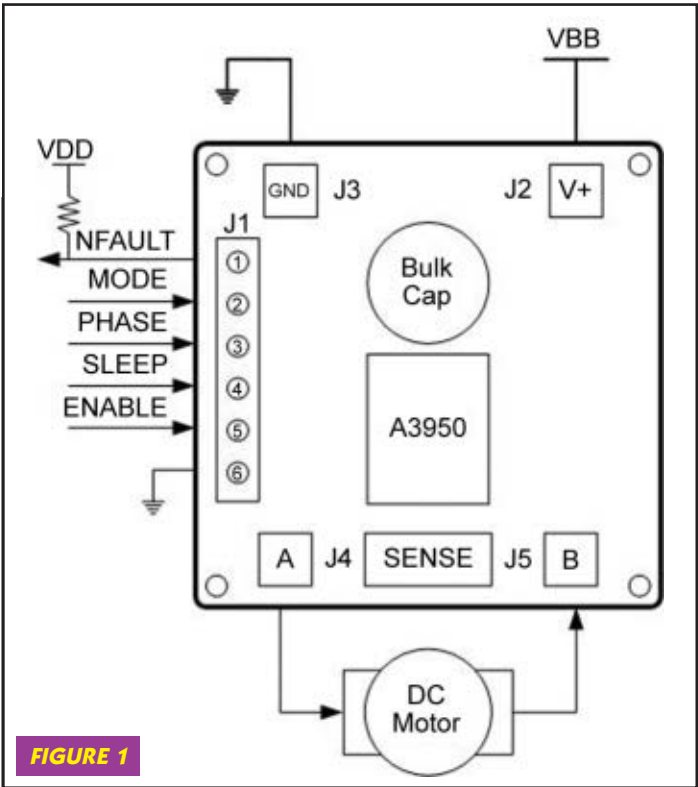
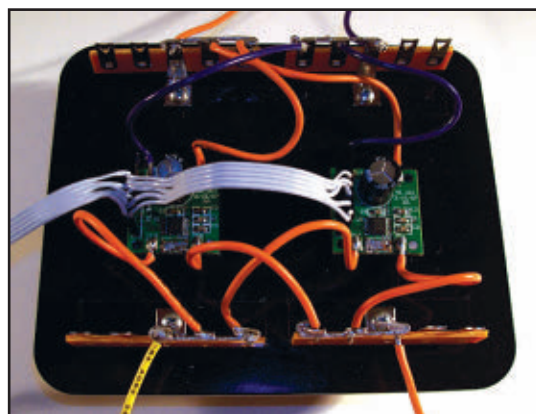


FIGURE 1

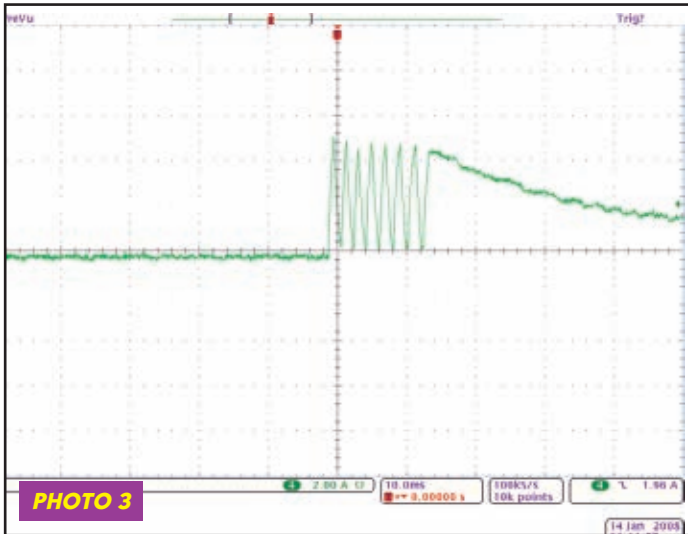
FIGURE 2

The schematic diagram illustrates the A3950 motor driver circuit. The A3950 IC is the central component, with its pins connected to various control signals and power supply components. The control signals are provided by a 5-pin header (Header 6) labeled MA FAULT, MA MODE, MA PHASE, MA SLEEP, and MA ENABLE. The A3950 IC has pins for NFAULT, MODE, PHASE, GND, SLEEP, ENABLE, OUTA, and SENSE. The power supply is connected to VBB and VREG, with decoupling capacitors C1, C2, C3, C4, and C5. The output signals OUTA and OUTB are connected to the motor via SMD TABs (32, 33, 34, 35, 36). The circuit also includes a resistor R2 and a capacitor C1.

PHOTO 2

The second test consisted of applying a power resistor as a load instead of a DC motor. Since the high speed transient is eliminated from the picture, we can better see the current driven being twice on the parallel version when compared to the single controller version. To do this, I used a 5W power resistor. At 24V, this resistor draws a 4.8A current. If my theory is correct, the single controller will not be able to drive this load, while the parallel version will. Photo 4 shows the theory to be correct.

When you first enable the H-bridge and the FETs are cold, the RDS ON is as we explained earlier. However, it is a



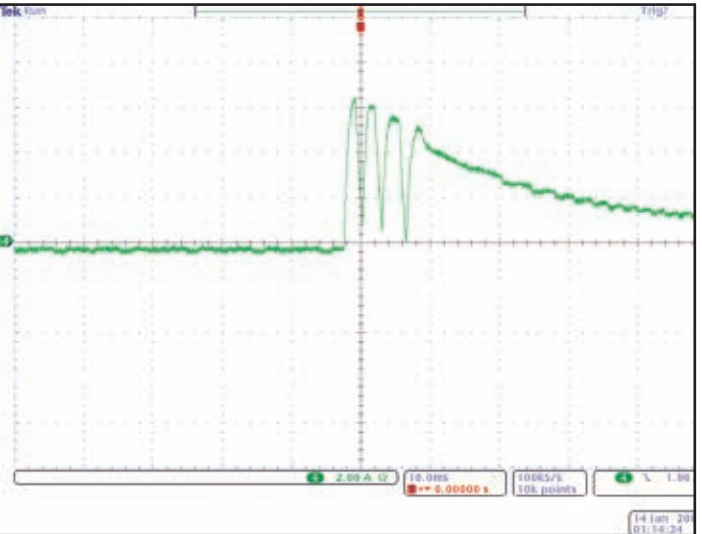
law of physics that material's resistance increases with an increase in temperature. Due to the I^2R power dissipation on these FETs when current is flowing, temperature will increase. This is why you will see current flowing at 4.8A first, but quickly settling at something smaller.

That being said, it is imperative to realize that these H-bridges are rated to a maximum of 2.8A, but are not meant to run at that current level for an undetermined amount of time. You should expect to see thermal shutdown trigger eventually. The solution is to PWM the H-bridge at some duty cycle less than 100%. This will allow larger peaks to be delivered to the load.

In theory, we can add more H-bridges in parallel and increase the current even more, but although possible, I would rather use a different venue. Still, I am glad the A3950 stood up to the parallel test, which means my latest robot will soon be strolling down my kitchen and scaring the cat.

Future Revisions

After designing this little controller, I came to a series of realizations on how it could be improved. First, the size is about right, but there is certainly lots of room to make it even

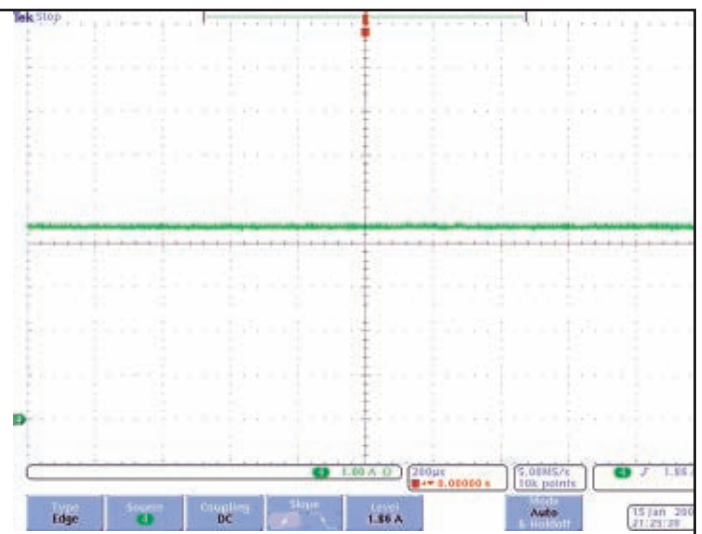
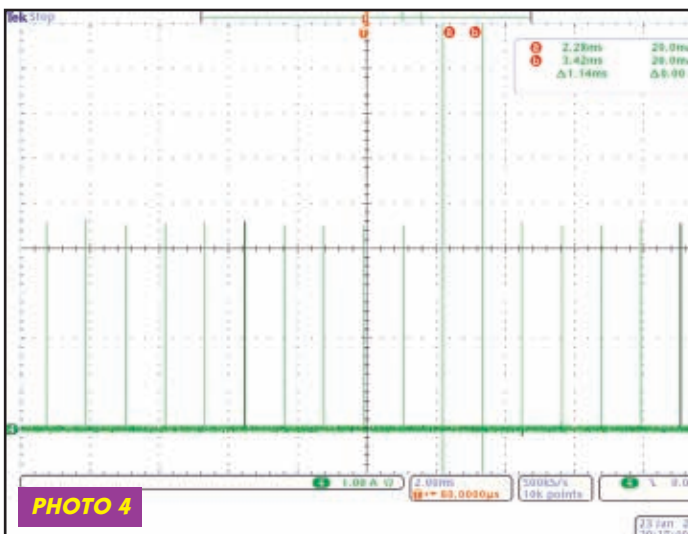


smaller. Currently, I am working on a release which will compress the board to 1" by 0.8" — perfect for tiny robots!

The solder pads are perfect for soldering battery and motor wires. However, when parallelizing controllers, this is not necessarily the best option. I am working on a way for the boards to be assembled together into a larger unit. Getting more current should be a clean solution, not a wire mess which complicates things.

Conclusion

The amount of DC motor controllers on the market is staggering. I have found the A3950 to be so versatile and economical, I believe it should be the solution for a great deal of small to medium sized DC motor control applications. The device could not be easier to use. With very little effort, you should be polarizing DC motors at all speeds and both directions. Thanks to its increased current capability and small size, it is very easy to group these devices together and get more power. I definitely will be using these little devices on most of my future DC motor projects! **SV**





Continued from page 20

- digital servo specifications.
- Requires only one Propeller cog, leaving the other seven for development.
- Fully documented in code comments.

Cost for a single license to run 16 servos on the Parallax Propeller chip, is \$20.

Static RAM Memory Expansion Kit

Another new product from Machine Intelligence Technologies is a memory expansion kit for the Parallax Propeller.

This memory expansion kit supports a five megabyte block transfer rate and can be used to store program overlays for the cogs, overlays for Spin code, and large, fast, bulk data access.

The Memory Expansion Kit expands the Propeller's 32 Kb internal static RAM. It allows the Propeller to natively address from 512 Kb additional static RAM up to 16 Mb additional static RAM.

- There is minimum program overhead.
- It requires only four instructions per byte transferred.

- It boasts a transfer rate of 5 Mb per second.
- It can store anything from data to executable Spin programs to cog programs.
- It uses the Euro Card standard.
- It can store the 16 instruction RISC emulator package under development to run on the Propeller chip.
- It will support up to eight Propeller processors on our proprietary Corpus Collosum Bus (currently under development).
- All external devices such as memory, servos, gigabyte USB memory devices, etc., may be accessed from any Propeller chip in the system.
- Further development in the near future includes adding up to 16 gigabytes of non-volatile RAM accessed via USB 2.0.

For further information on these three items, please contact:

Machine Intelligence Technologies

Website: www.machineinteltech.com

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BIO-FEEDBACK

continued from page 7

material on the next charge cycle.

Even if all cells stay below 100% DOD, at the end of the run the bottom cells will be at a much lower state of charge than the top cells. When they are charged, the lower cells will never get back to the same state of charge as the top cells; that is, they will not reach full charge before the charger turns off. Over time, the bottom cells will develop a substantial voltage (charge) difference from the top cells, and this just exacerbates the problem of driving the bottom cells past 100% DOD. So at the very least, these batteries will have a very short life.

Zac should be using an undervoltage/overvoltage/overcurrent protection circuit, as recommended by the battery manufacturer. Failing that, at the very least he should treat these as two completely separate packs when he charges them. Charge the bottom two cells as a two-cell pack, then charge the top three cells as a 3-cell pack — in both cases, using an appropriate automatic LiFeO₂ charger. At least that way, the cells will all get back to 100% charge before a match.

I strongly suggest you print some sort of clarification to prevent your readers from putting their very expensive batteries — and perhaps more — at risk due to this incorrect application of these cells.

Mark Lewus

Response:

I'm glad you have voiced your concerns. Safety is always the first consideration in any project that deals with this kind of power, and mistreatment of any of a number of parts in these machines can result in expensive or dangerous problems. All of your concerns are very real possibilities if the batteries are undersized for a project, so I'd like to provide some better data on the loads in my system to alleviate some of your worry.

The weapon in Scurrie draws approximately 35 amps when the disk is initially started and it tapers off to somewhere between five and 10 amps while the disk is cruising. Certainly repeated starts and additional resistance in the system caused by bent and worn parts can make these numbers higher through a competition, but they are well below the ratings for the cells.

As you mentioned, the lower

two cells of the battery system are additionally loaded by the drive system, and so will be discharged more than the upper three cells. The drive system pulls approximately another 15 amps when initially starting or pushing against another robot. This load brings the total peak current draw to approximately 50 amps — a number that is considerably lower than the continuous current rating of the cells.

The speed controller that is running the weapon has built-in protection from excessive voltage drop, and the drive speed controllers have both thermal and current limiting built into them. I actually competed with this robot again recently and even in the five minute long "rumble" with multiple other machines in the arena, the lower two batteries were only about half discharged.

In a typical three minute match, I draw between 800 and 1,000 milli-amp hours from the two-cell pack. The cells are not even warm, and I always charge the two- and three-cell packs separately. I always use a balancing connector so that the cell voltages are all as close to each other as possible.

This new battery technology is, of course, not a toy and should always be treated with great care. If treated correctly though, the batteries from A123 Systems can provide a great weight savings over traditional Nickel-based packs while providing added safety over Lithium Polymer packs. I am very glad you read the article and I appreciate your concerns.

Zac O'Donnell

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NABGO 2008 IN A NUTSHELL

NABGO is the North American Big Gun Open — an open competition for 1/144 scale model warships in the Big Gun format, originally founded by the North Texas Battle Group (www.ntxbg.org). It is a multi-day event held in July each year at Star Brand Ranch Executive Retreat (www.starbrandranch.com) in Kaufman, TX, commonly consisting of three major segments and is currently the only national Big Gun Model Warship Combat event in North America. The Big Gun Model Warship Combat World Championships, founded by the Australian Battle Group (www.AusBG.org), were hosted by NABGO for 2008 with permission of the AusBG, resulting in double titles for the medal winners this year.

In general, when and if damage is counted, it is in the form AA/OO/BB PP R, for each sortie made where AA = holes on/above the waterline (in the gray), OO = holes at the waterline (on the "boot" or black line), BB = holes below the waterline (in the red), PP = points scored due to damage, and R = result other than returning to port (S for sunk, L for lost, etc.)

The three major events are: Cargo is King, BYOB, and The Texas Cage Match.

Cargo is King

Generally, the first major battle event of NABGO, Cargo is King is a scenario battle where only the cargo runs completed count toward which side wins or loses — not sinks, not holes (sinks and holes are counted for other awards, like Most Damaged Without Sinking, but not for the scenario score). It was originally conceived as a "low damage" event, but turned out to be high carnage, with skippers putting their ships at great risk to ensure the cargo made it through.





BYOB

Usually the full battle day consisting of multiple sorties, BYOB — or Bring Your Own Battle (Plan) — is a flex-day, where the available skippers decide by consensus what the day's plan will consist of. This year, there were two sorties by normal NTXBG standards and a third under Western Warships Combat Club custom (much shorter sorties, different rules as to ships counted lost, etc.)

The Texas Cage Match

Many rules are reversed for this Last-Man Standing event. It is every ship for itself, as the rule that there is no firing in port becomes all ships sequestered in port and all combat takes place in the port basin. Battle reverse is allowed, as there is not room to turn. After the first reload, the rate of fire restrictions are often lifted. It is a wild and woolly shoot-out in a small corral.

We did have some of the NAMBA Thunderboats racers from the Cedar Creek area come by to watch.

A few highlights of NABGO 2008:

- Our first hosting of the AusBG's Big Gun Model Warship Combat World Championships at NABGO.
- Our youngest armed captain ever: Charlie Webster, age 10.
- Our oldest battler ever: Dr. Bob Fristrom, age 87.
- Our farthest traveled battler ever: over 1,600 miles.
- Most captains and ships attending.
- Smallest ship in the Texas Cage match (USS Reluctant, also the first sunk, too).
- Youngest ship in the Texas Cage Match (built that week, the Lutzow, was "commissioned" on Friday, battled all weekend, and still finished second in the cage match on Sunday against several much larger, more heavily armed ships).
- First time in the history of NABGO that Axis won all (or, for that matter, any) team competitions and this in spite of having two rookie captains and one untested battleship built on-site the week of the event.



Twin Tweaks

THIS MONTH:

Bug Sport



FIGURE 1. THE OLLIO BUG KIT.

This month, we have the pleasure of presenting another robot kit from esteemed Korean company Robotis, who brought us the incredible Dynamixel and Bioloid. The kit is the Ollio Bug — a kit targeted at that elusive demographic of young budding roboticists. Inspiring youngsters to become interested in science and technology is an admirable goal, but it is a competitive niche already dominated by the LEGO

Mindstorms and NXT kits. Does the bug have what it takes to carve out a segment in this competitive market? To test the mettle of the Ollio Bugs, we've set up a scenario worthy of a Dick Francis novel — our little bugs will compete in an adrenaline-pumping, line-following race against another tiny opponent, our Dad's OOPIC Mark III robot. Will the bugs be ready for race day, or will malevolent forces crush their chances for a successful run in the United States? Will the kid friendly

robots hold their own against the sophisticated veteran? Put away the bug spray and read on to find out!

Twice Fly

The Ollio Bugs are part of a larger Ollio system, with the interesting moniker supposedly a portmanteau of the words "all" and "robot" because the Ollio kit purports to allow tinkers to build any robot that can spring from their imagination.

FIGURE 2. OLLIO BUG PARTS.



FIGURE 3. OLLIO RIVETS AND SLEEVES.



They immediately jump out at you from the moment you see the box because of their bright colors and charming cartoonish appearance. Opening the box reveals that the parts are divided into plastic bags for sorting. The kit also contains an instruction manual and a mix and match line-following track. A fold-out pamphlet shows how to get started with the bugs right away, and it details some of the exciting add-ons not present in the original kit.

Our first order of business was to pop open the parts bags to see what we were dealing with. The target audience of the Ollo Bugs invites a comparison to the LEGO Mindstorms kits, and as we sifted through the parts we remembered the anticipation and excitement of our initial introduction to the world of kid friendly robots.

The kit itself is comprised of two classes of parts. One contains the main structural pieces, which are plastic plates choc full of holes that come in a variety of shapes and sizes. The other type is the unique kind of fastener, referred to in the kit literature as a sleeve and rivet system.

LEGOs have become famous for their ingeniously simple mechanism that is literally the building block of the brand – the classic dimples. Would the Ollo kit have a similarly innovative mechanism to make kit building easy and fun? The folks at Robotis have

FIGURE 4. THE OLLO BUG MOTOR.



certainly made an admirable effort with the unusual rivet and sleeve system. The color coding on the rivets refer to different lengths and sizes of rivets, and the bright colors make the right parts easy to find quickly.

The Ollo kit comes with a special tool for the removal of the rivets. The rivets can usually be inserted by hand easily, but removal is a bit more difficult. The removal tool – reminiscent of the sharp end of a claw hammer, does actually make the process relatively painless.

The instruction manual for the bugs is mercifully devoid of written instructions, and instead relies on detailed three dimensional models to communicate the process of constructing the robot. The instructions alert the builder to the type and number of each part required for each step beforehand, and the colorful models are fairly easy to read. There can be some difficulty when dealing with parts that demand

FIGURE 5. THE OLLO BUG INSTRUCTION MANUAL.



a large number of rivets because it can be confusing to keep track of where they are supposed to go, but a careful accounting of the holes is a surefire strategy for success.

The instruction manual gives directions on how to build four different creepy crawlies: a ladybug, a grasshopper, an Atlas beetle, and a mysterious weevil looking creature. All of the bugs except for the grasshopper start from a common base, so we decided to build that first. We soon discovered that reaching into the parts bags for a specific piece was a tedious test of dexterity, so we emptied out the bags into the open Ollo Bug box for easy retrieval.

For Ticks

The Ollo Bugs' box declares that the kits are suitable for ages 10 and

FIGURE 6. THE OLLO BUG BRAIN.

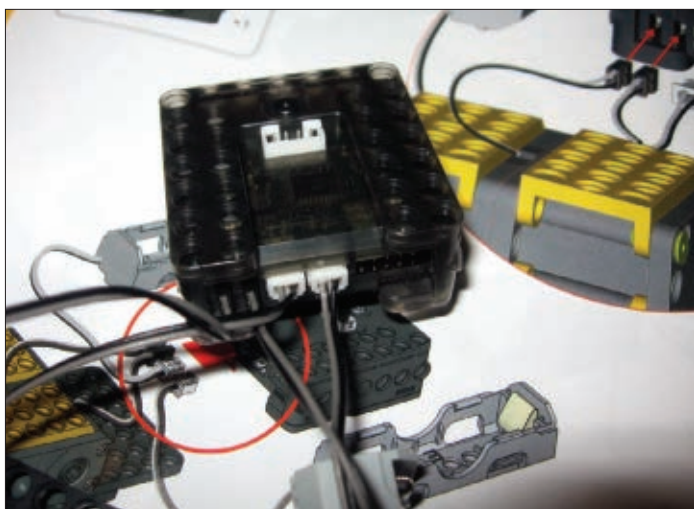


FIGURE 7. THE OLLO BUG BASE.



FIGURE 8. AN OLLO LADYBUG IN THE WILD.



up, and the engaging construction of the bugs is sure to capture the attention of even the most distracted youngsters. The basic shapes of the bugs are simple, and they begin to look like creepy crawlies in no time. The Olloladybugs also galvanize the imagination by their originality – one detail that immediately sets the Olloladybugs apart from other kits is the fact that they are walking robots. Kid friendly walking robot kits are certainly a rarity, and we think is likely because walking robots are typically thought of as necessitating difficult and esoteric designs.

The folks at Robotis deserve some serious accolades for coming up with

a walking robot design that is simple and accessible. Each set of legs requires only one motor, and the resulting motion is entertainingly bug-like. Unfortunately, the Olloladybug kits don't come with the fancy Dynamixel servo featured prominently in the

Bioid kit, but the motors in the kit are suitably tiny and come in a transparent casing sure to arouse the curiosity of a builder of any age.

The wiring of the Olloladybugs occurs in the middle of the building process as the motors and brain are being mounted, and we think the folks at Robotis have made it simple and accessible. The main electronic components that need to be wired are the motors, battery packs, brain, and infrared receiver. Everything can easily plug into the brain, with the wires from the motors and battery packs being the perfect length to allow for out-of-the-way routing without being a long mess. The Olloladybug

Bug battery packs are two holders for AA batteries, and the batteries seem to last the little critters for a good long while.

In addition to being delightfully painless, the wiring also teaches an important life lesson of carefully checking the colors of the wires to ascertain the correct connection orientation. In this, all that is demanded is a careful differentiation of gray from black, but the lesson is still a good one.

Despite their structural simplicity, the Olloladybugs feature a number of aesthetic embellishments that really enhance the appeal and overall bugginess of the little critters. The long antenna of the grasshopper, the imposing horn of the Atlas beetle, and the wings of the ladybug all add a colorful and imaginative flair to the creations. The bright coloration might be a poor choice for camouflage, but the visual pizzazz is sure to entice the youngsters that the bugs target.

After we built our first bug – a ladybug – we were eager to test it out. The Olloladybugs come with a sleek controller that strongly resembles something intended for a video game console, so the target audience should feel right at home controlling them.

There are four main buttons to command the critter to move forward, backward, left, and right. Other buttons can also switch the channel of the controller so that two Olloladybugs can be controlled simultaneously with two different controllers. This feature is gleefully advertised in the quick start guide as a way to have a bug battle.

Before we descended into a beetle battle that would be the envy of Dr. Seuss, we decided to try our hand at controlling a solitary bug. The movement of the bug is delightfully squirmy, and the critter is surprisingly fast on its six legs. The controller is responsive, and the clever positioning of the infrared receiver on the bug eliminates any line-of-sight difficulties that can be the downfall of some infrared controllers. The bug also has a surprisingly good turn radius – a skill that can be tested by arranging

FIGURE 9. THE OLLO LADYBUG AND GRASSHOPPER ENJOY THE GRASS.



hairpin turns with the mix and match line-following track. While one bug is tons of fun, we thought a second bug would be a great addition to the family. We decided to build the grasshopper out of a second kit (a single kit contains two motors, which is enough for only one bug at a time). We were drawn to the grasshopper because of the antenna that seemed to defy gravity. The construction of the grasshopper is quite similar to that of the ladybug, even though the base is slightly modified.

The charming antenna are given their shape by an innovative little assembly that uses interlocking serrations to provide an adjustable joint that stays in place fairly nicely. These joints are the secret to many of the insect-like details on the Ollo Bugs, and while they certainly seem robust at first blush it remains to be seen if they will be able to hold up after prolonged usage. We think that they would, but one has to remember that these bugs are going to face some serious wear and tear at the hands of the intended audience.

Splattered

After building the ladybug, the grasshopper went together quickly, and we were ready to test the new specimen. We were excited to see the grasshopper work because the leg design is slightly different than that of the other bugs, with longer back legs to more accurately represent its insect inspiration.

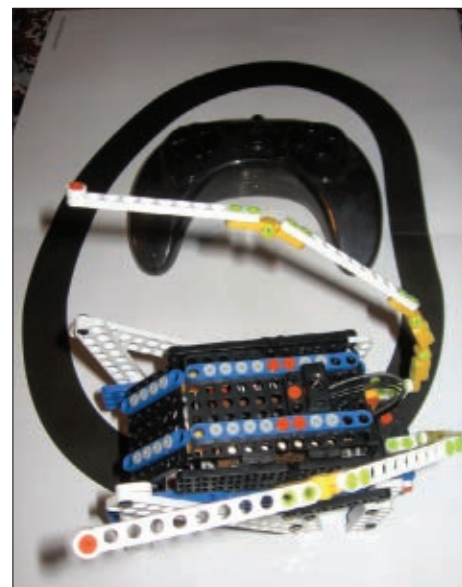
You can imagine our disappointment when we first activated the grasshopper robot only to discover that it sorely lacked the agility that would earn it props from a martial arts master. The basic problem was easy to see – one of the motors was not functional, and the intrepid insect could only move in a wide circle despite its most sincere efforts.

We resolved to try out some insect surgery. The first step would be to disassemble the grasshopper to the point where we could remove the motor. This, unfortunately, was almost to the point of complete disassembly,

but we readied our Ollo Tool and set to work. That is one disadvantage of the Ollo kit – disassembly of the bugs is not as quick as one might hope. The parts themselves can snap apart fairly easily, but removing the rivets and sleeves properly can require individual attention that can become somewhat time-consuming.

Once the motor was freed, we were able to give it a thorough inspection. We began our investigation with the classic technique of jiggling the wires to see if there was a loose connection on the end with the connector or at the end that connected to the motor itself. When our jiggling provided no answers, we were forced to look elsewhere for the cause of the problem. We even went so far as to take the motor apart to see if anything was jamming the gears, but we found nothing. After doing our best to clean out the gears, the motor would actually move, but it would not start without a little helping hand. When reattached in the leg assembly, the motor would not start, and we were forced to bid farewell to our dreams of an epic bug battle. We must make it clear, though, that problems with motor modules are not phenomena confined to the Ollo Bugs. We've had similar experiences

FIGURE 11. THE OLLO GRASSHOPPER, CONTROLLER, AND TRACK.



with motors from the LEGO Mindstorms kits, and the companies were more than happy to speedily replace defective units.

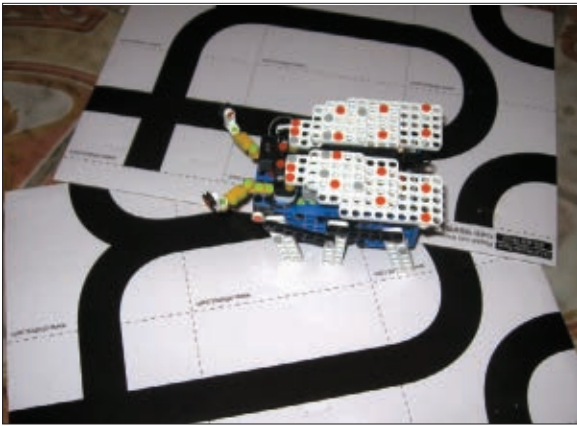
Gnat Race

The Ollo Bugs come with a preprogrammed line-following mode, easily accessed by a few clicks of the power button. After activating the

FIGURE 10. NOT THE BEST CAMOUFLAGE.



FIGURE 12. MIX AND MATCH TRACKS.



line-following mode, the bug pauses to allow a proper positioning at the starting line before sounding a charming charge jingle and setting off.

The Olo Bug kit comes with two line-following tracks of its own. One is a simple oval on the back of the quick start pamphlet, and the other is made out of cardboard squares that can be cut up, mixed, and matched. The small oval is a nice testing ground, and the mix and match track is tons of fun.

One caveat about the mix and match track is that the cardboard squares tend to get pushed around by the scurrying legs of the bugs. Setting up the track on a high friction surface mitigates this problem, and we think something between carpeting and tile would be ideal. Come to think of it, the short, dense carpet that covers the floors of many school classrooms

would work perfectly on a number of levels for the educational bots.

Due to the tragic and unforeseen injury suffered by our grasshopper, the ladybug was left to roam the tracks alone. A true test of the bug's line-following ability would demand a proper race, and a proper race would demand a proper rival. Our dad's Mark III OOPIC robot was eager for a challenge, and perhaps a bit

overconfident due to the advantage it possessed in the form of wheels.

The small oval and the mix and match track were a bit on the small side for a race between the ladybug and the Mark III, so we turned to a larger line-following track. The LEGO Mindstorms starter kit comes with an excellent large scale line-following track, and it would be a perfect proving ground for our bug. We allowed it the luxury of a couple of test laps, and we were very pleased with what we saw.

The walking robot navigates corners surprisingly well, and even when it seems to be veering off course it always corrects itself. If the bug does happen to stray from the course, it has an automatic shutoff feature to keep it from wandering off into oblivion. Hopefully, it wouldn't take advantage of that feature during

the showdown with the Mark III.

Speaking of the Mark III, things were not as they should have been with regards to the programming. The Mark III is equipped with three phototransistors for line-following (the same exact type that we used on our safe-cracking robot, incidentally), and it did at one time have a killer line-following program on it. But now, in a twist of sabotage that would feel right at home in another Francis novel, the Mark III was experiencing some serious issues downloading programs. The only logical culprit that we could identify was the serial cable that connected the robot to the programming computer, but the problem was not resolved in time for race day.

With the Mark III lacking a program, the Olo ladybug ran circles around it in an unexpected upset, enough to arouse the suspicion of any amateur detective. The Olo Bug was triumphant, but the serial cable debacle left us a bit unsatisfied.

Flying Finish

The Olo Bug line-following program is quite effective, but intrepid tinkerers that think they can do better have the opportunity to find out. The kits that we were given did not have the necessary components to allow the bugs to be programmed, but Robotis does make a downloader module to allow for custom programming.

The greatest thing about the Olo downloader is that it takes the form of a simple USB plug-in that uses infrared communication to connect with the Olo Bug. We are positively ecstatic that a robotics kit has finally seen the light and chosen a USB connection over the tragically outdated and anachronistic serial connection that seems inexplicably preferred in almost all of the other robot kits that have meandered through our column.

We were especially bitter about serial cables this time around since it contributed to the

FIGURE 13. HEADING FOR THE HOME STRETCH.

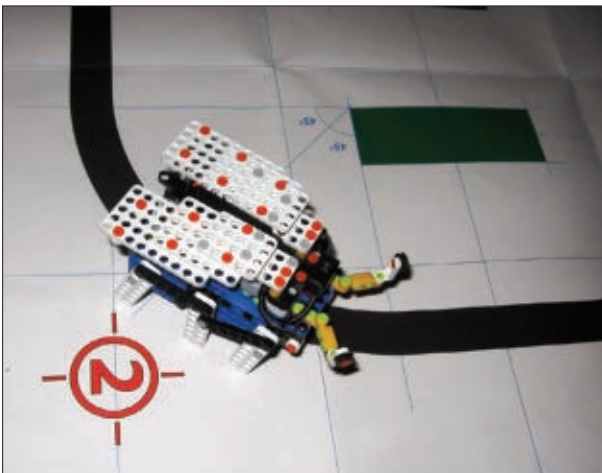
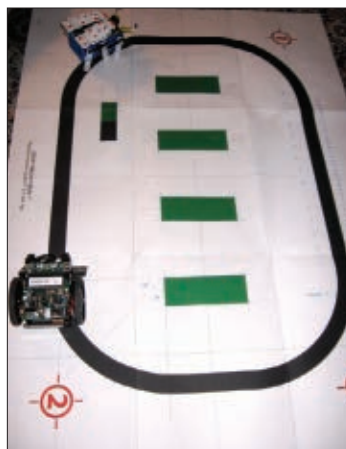


FIGURE 14. MAKING DICK FRANCIS PROUD.



sabotage of the Mark III. The morbid fascination that so many robot kits seem to have on serial ports is positively baffling to us, and we cannot think of a reason as to why the Ollo downloader is in the minority with its USB connection.

Is the motive for the serial connector driven by an egalitarian instinct to not exclude those roboticists with outdated computers that are without USB ports? This seems hardly likely, because in our experience most roboticists tend to be people at the cutting edge of technology, and something like an outdated computer would be considered nothing short of anathema. The folks at Robotis have shown an admirable understanding of their target demographic, because

unless they are inheriting a family heirloom most kids will have computers that sport USB ports only.

Could it be a cost issue? It seems unlikely to us that a USB connector would be that much more expensive than a serial connector to put onto a PCB, and the generally smaller size of USB connectors would save valuable circuit board real estate. Companies that distribute kits with serial cables must also realize that they are saddling the buyer with the extra cost of acquiring a serial to USB adapter, and the extra hassle of pulling out an extra cable whenever anyone wants to download a program.

Could it be performance? We don't think so.

Beyond the obvious issues of applications and cost — both of which USB connectors have a clear advantage — we cannot fathom any other reason that would justify the maddening prevalence of serial connections in robot kits.

Now that we've blown off some steam, we'd like to cool down by looking at some of the other great features of the Ollo kit in general. Additional Ollo kits like the Ollo Figure and Ollo Action kits can be bought for \$20 and \$30 (the Ollo Bugs will run you about \$100), and while the Ollo Figure kit comes only with structural pieces, the Ollo Action kit comes with motors, as well. This indicates potential for easy expansion, allowing young roboticists to practice using the most important engineering tool of all — their imagination.

Once again, the stellar folks at Robotis have earned our highest praise for putting out a high quality kit with enough innovative features sure to carve out a niche in a competitive market. The Ollo Bugs are hugely entertaining kits poised to be both powerful educational tools and an exciting introduction to the world of robotics for many budding roboticists. **SV**

RECOMMENDED WEBSITES

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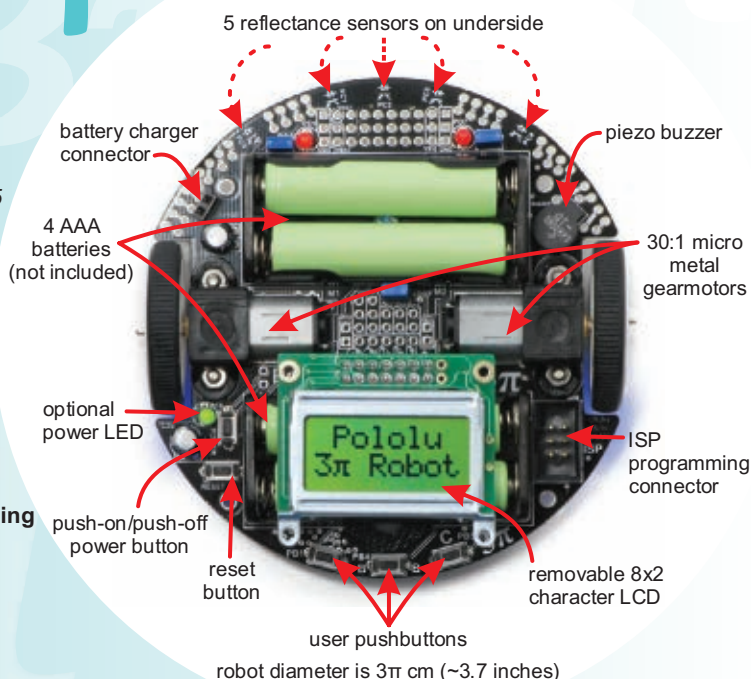
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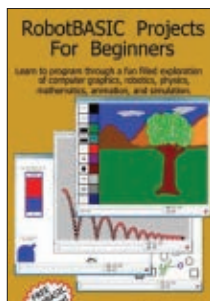
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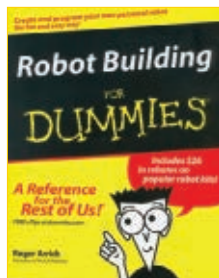


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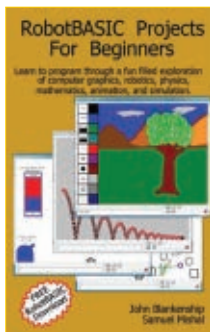
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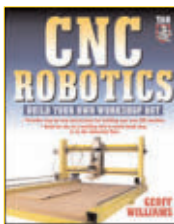
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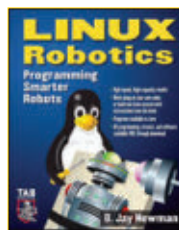
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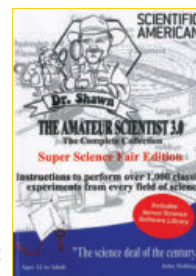
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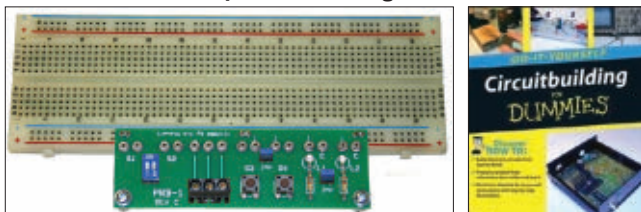
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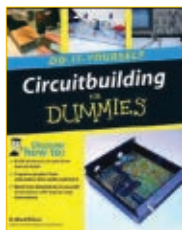
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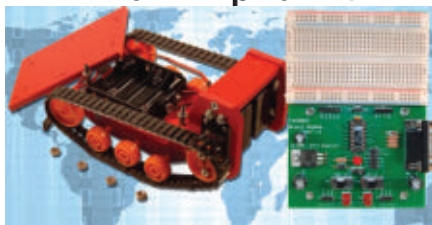
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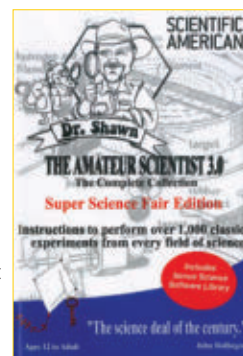
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BY GORDON MCCOMB



Organizing Your Robotics Workbench

In last month's Robotics Resources, we looked at the major electronics tools and supplies used in building robots, such as volt ohmmeters and soldering irons. Like all workbenches, how tidy you keep your robot building home goes a long way to how much you'll enjoy the process.

There are a number of solutions for organizing the bits and pieces of your robotics hobby, including all sorts of toolboxes, chests, cabinets, drawers, boxes, bins, bags, and more. We'll cover many of these in this installment, including handy online sources if you don't happen to have a "Storage Shelves R Us" nearby!

A Place for Everything, and Everything in Its Place

Over his 20 years in the military, one of the main jobs my father had was taking charge of supplies. He had the knack for organizing; the Marine Corps had to keep tabs on everything from huge trucks and airplanes, to miniscule ball bearings. He's long since retired, but he has numerous jobs and hobbies where his organizational skills still come into play.

When he got into model airplanes, his portable "shop box" was

neatly filled with everything he'd need while out in the field. While other members of the team would just toss things into a plastic fishing tackle box (and often had trouble finding things, or even knowing if it was in the box), my dad devised special compartments, and he knew where everything was.

If apple trees are organizational traits, this apple (me) fell far from the tree, because I inherited virtually none of the mindset to keep things in their place. All while growing up — and even now as an adult — I usually forgot to put things back where they belonged. I lost a bunch of stuff, and ended up buying the same thing two and even three times because I lost track of it all in the garage.

So, knowing my shortcomings, I have to make a conscious effort to maintain order in the shop, or else things get rapidly messy and increasingly uncomfortable. When I skip the step of organizing my stuff, I enjoy the robot building process far less. Finding parts becomes a half-hour job. No fun.

While my shop is far from Robotics Eden, it's more or less well organized with the help of some small parts cabinets, a few plastic parts bins, zipper lock-style plastic bags, and a few other odds and ends. While my system works for me, you'll undoubtedly want to improvise your own. Everyone has slightly different requirements; for example, because I usually build small robots, my storage needs tend to be for small parts. If you build large and heavy combat

One of the many Amazon storage-related offerings. Do a search under the sections Home & Garden and Office Products & Supplies for more.

Amazon.com: Rubbermaid 9A6100BLA Mini storage drawers, 11-1/4 high, black with 4 clear drawers - Windows Internet Explorer

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robots, you'll need larger and more heavy duty places to keep your stuff.

Small Parts Drawer Cabinets

Plastic parts drawer cabinets are really the mainstay for any activity that deals with many small parts. You can get them in all types of styles and sizes, from little units with just six or nine 1" x 2" drawers (these fit on bookshelves quite well), to much larger parts chests with 20, 30, even 40 drawers of different shapes and sizes. These won't fit on a bookshelf — unless your shelves are very large — but are ideal for work benches and assembly tables.

I prefer cabinets someplace in the middle of the size range, with about 25 to 35 drawers. You can get these where all the drawers are the same; a common size (for the larger cabinets) is about 2" wide, 1.5" high, and 4" deep. Each drawer can use a plastic divider so you can keep different sizes or styles of something organized; for example, three different sizes of 4-40 machine screws: 3/8", 1/2", and 3/4".

Another option is a cabinet with several drawer sizes to accommodate parts of different bulks. For example, the cabinet might sport 24 or 30 small drawers of about 1.5" by 2", and six, nine, or 12 large drawers of about 2" by 4" (the depth may vary from 4" to 5.5", depending on the model). You might, for instance, place individual values of resistors in the smaller drawers, and large electrolytic capacitors or power resistors in the large drawers.

Drawer cabinets vary greatly in quality, and I prefer to check them out in person when possible. You can find them at any home improvement store, Wal-Mart, Target, even some supermarkets. Look for drawers with thick enough plastic that won't easily break when you use them to store something heavy — like a couple hundred metal fasteners. The drawers should slide in and out without excessive binding. Avoid units where the drawers easily fall out. On better

cabinets, the drawers will have a lip at the back so they can be removed only if you lift them up and out. Otherwise, you may end up with hundreds of little pieces scattered all over your workbench, floor, and lap.

Heavy Duty Storage Drawers

Going up a step on the size and cost ladder is the plastic, heavy duty storage bin. These units are one to three feet high and sport between three and six heavy duty plastic drawers. The drawer sizes may be different or all the same; find a model that suits your needs. Drawer dimensions are typically large enough to hold household items like a ream of paper or other craft goods. The most robust models can readily hold heavy motors, metal gears, and other construction parts. Popular brands are Sterilite and Rubbermaid.

Larger still (and somewhat more flexible) are stacking drawers, with sizes large enough to hold bulky sweaters and pants. They're made to stack on top of one another, so you can get as few or as many as you want. (You want to avoid overdoing the height of your Tower of Babel, or

else things will topple over if you open a heavy drawer near the top of your stack.) Department stores, craft stores, and "get organized" specialty stores are the best sources for stacking drawers. For example, The Container Store (www.containerstore.com) offers many sizes and styles of stacking drawers. Most of these are intended for clothing and are large enough to hold the bigger robot parts in your inventory, but they may not be strong enough to support the weight. So, make sure you get the heavy duty models. Drawers for use in the garage rather than the bedroom are preferred.

Tool Boxes, Tackle Boxes, and Totes

You may not always build your robots in your own shop. Sometimes you need to go where the action is, and you'll need to bring your tools and supplies with you. For the really heavy stuff like electric drill motors, saws, and other tools a standard metal or heavy duty plastic toolbox is the best choice. I still have my Craftsman 24" steel toolbox I bought over 30 years ago, and use it regularly.

For lighter jobs, a plastic fishing

The Randmh.com site is geared toward heavy-duty industrial storage solutions, but their products are available in small quantities and at reasonable prices.



tackle box or tote makes lugging around your supplies a lot easier. The typical tackle box has a storage drawer on the top for small parts. When you open the top, the drawer slides up and over, and you can reach into the bottom of the box for larger tools and supplies. (If you happen to also fish, it's a good idea to invest in a separate tackle box for your robot building endeavors, so you're not mixing worms and old fish guts with solder and stainless steel fasteners.)

Plastic totes are available in a variety of shapes and styles, and are ideal if you work with larger tools and supplies. Most totes are listed in gallon (or liter) capacity. For example, a 10 gallon tote measures about 24" x 16" x 9". I prefer totes with an integrated handle (test the handle to make sure it won't easily break or come off) and reliable secure plastic locking clamps. Some totes come with an extra hasp for a padlock. If you need several totes, you'll want one made of clear (or semi-clear) plastic, so that the contents are visible.

There are, of course, many other sizes, styles, and types of storage bins, boxes, or drawers. There's something for every occasion. Whatever you chose, you'll want to consider the

following general guidelines:

The construction of the storage container should be adequate for the weight, not just the size. Cheaper containers are often made of thinner plastic which can break under load.

Opaque containers hide what's in them. Sometimes this is a good thing, but most often you'll probably want to keep the innards visible so you can quickly find what you need without opening up the container each time. Keep your containers indoors and away from heat sources or even windows. Heat and sunlight can degrade the plastic, causing it to become brittle. Containers with hinges, hasps, clamps, and other mechanical parts should be tested out before you buy them (or at least buy from a known brand name). The hardware on the cheaper units can fail after even modest use.

Keeping Track of Your Inventory

Unless you have perfect memory, you'll need a system to keep track of what has gone where. On the low end of the scale is "the old magic marker on the side of the parts bin trick." (You can use any type of felt tip

marker; a Sharpie is my favorite.) The problem with this approach is that the marking tends to be permanent, and you may want to change what goes inside a container or drawer.

Instead of writing directly on the plastic, you could tape an index card to the container and write on it instead. If you change the contents, just peel the card off and start over.

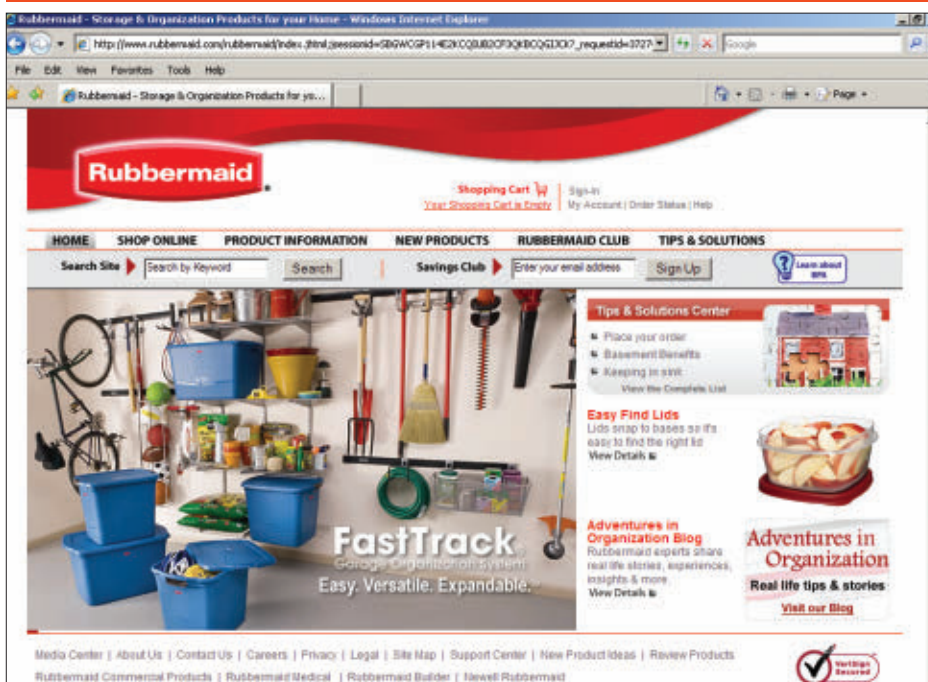
Or, if you have a spool of wide packing tape (the kind used to wrap packages for mailing and shipping), just apply a couple of inches of tape to the container, and write directly on it. To replace the writing, yank the tape off and affix a new length. This usually works well, depending on the plastic used on the container. Some plastic resists the adhesive on the tape, so it never sticks. In other cases, the tape adheres so well it can be difficult to take it off. Try a small piece in a corner of the container and leave it on for a week or so. If the packing tape idea doesn't work, there's also duct tape, but that can leave a sticky residue.

For smaller parts drawers, an electronic labeler is the absolute best way to keep track of parts. (Think of the character Vanessa Kensington and her suitcase of labeled bags in the first Austin Powers movie.) There are low-cost labelers that start around \$30, though I prefer the larger and more featured models, such as the Brother P-Touch PT-65 (\$60 to \$90). You can get different types of labels, with a choice of backing and ink color. Black lettering on a white background is a good choice for parts drawers. Larger machines can accommodate labels of different widths. Common sizes are 6 mm to 12 mm, or about 1/4" to 1/2". The labels come in handy cartridges that can be readily exchanged from one size to another, in case you have different label requirements.

Ad-hoc Storage Solutions

Not everything needs a fancy plastic storage bin. Sometimes simple — and free — is better. Here are some

Rubbermaid's website, where you can browse their offerings for purchase.





storage ideas for when you don't need a fancy solution.

Shoeboxes still make great storage containers. Keep the lid so you can stack the boxes, or just use it open if you need quick access.

Zipper-locking food storage bags — particularly the heavy duty ones for freezer use — make ideal containers for odd-sized items. Mark the contents with a Sharpie or other wide felt pen, and toss into a box (a shoebox) for safe keeping. Tip: The wide bottom bags stand up.

Baby food jars — plastic or glass — continue to be an excellent storage solution for very small parts such as 2-56 size hardware or even surface-mount components. (Use glass jars for electronics parts as it doesn't generate static.) For fast access to the contents of baby food jars, nail or screw the lid into the underside of a shelf near your workbench. Then, just twist the jar on and off. Be sure to leave room between the jars for your hands to grab them.

Empty egg crates and egg boxes are also useful for holding small parts, but take care not to overturn the crate or box, as the lid doesn't close over the hollow for the egg. If you're not careful, your parts may spill out or get mixed up.

assortment of totes, drawers, bins, and other heavy-duty items.

Organize It Online

www.organize-it-online.com

Online specialty store catering to home storage solutions. Click the Storage link for totes, drawers, and bins.

Rand Material Handling

www.randmh.com

Heavy-duty plastic storage bins for shelving. Open front; may be stacked several high to save space. Up to 18" deep. Dividers let you separate parts in a single bin.

Quantum Storage

www.quantumstorage.com

Industrial strength storage, including heavy-duty plastic bins for shelves or workbench.

Rubbermaid

www.rubbermaid.com

Storage solutions from small boxes to large bins. Shop online or sold at department, discount, and home improvement stores. Visit their site to see what's available.

Spacesavers

www.spacesavers.com

Free-standing and stackable totes, bins, boxes, drawers, and other storage solutions. Plastic, wood, and other materials.

Stacks and Stacks

www.stacksandstacks.com

Specializing in storage solutions; products sorted by application.

Staples

www.staples.com

Online and local stores; cardboard file boxes (cheap way to store bulky items), plastic storage drawers.

Sterilite

www.sterilite.com

Manufacturer of plastic storage drawers, bins, and other products. Sold in stores such as Target and home improvement outlets.

Uline

www.uline.com

Sells plastic and cardboard storage bins and boxes. Most must be purchased in quantity, so share your order if you don't need dozens of fold-together cardboard shelf parts bins. **SV**

Gordon McComb can be reached via email at robots@robotoid.com

Sources

Here are some online retailers that specialize in storage solutions. Be sure to also check out local retailers such as Home Depot, Lowe's, Wal-Mart, Target, Bed, Bath, & Beyond, Ikea, and others.

Amazon

www.amazon.com

Yes, Amazon even sells storage containers. Look under "storage drawer" in the Home & Garden and Office Products & Supplies sections. Good selection of storage drawers, in various drawer sizes.

The Container Store

www.containerstore.com

Online store specializing in storage products and solutions. Good

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Beginner Electronics



Basic Atom & Robotics

by William Smith

I introduced the new Atom Nano chips from **BasicMicro.com** in my last column, and now there are more new development tools to help the beginner. There is also a great robotics platform based on the Atom that is a great platform for the beginner so I'll give it a mention later since it's built around the 28 pin Atom interpreter chip. Let's start with the Atom Nano chips.

Atom Nano Chips

The Atom Nano chips come in 28-pin and 40-pin packages just like the original interpreter chips, but now there is an 18-pin package based on the PIC16F88. You can now debug right in your circuit and we'll talk about this a little later. Table 1 shows the three new chips and feature details.

The original Atoms cost more, but offered higher speed than the Nanos because they relied on an external 20 MHz resonator for the clock. This is an area that Nanos improve on as far as simplicity. The Nano chips use an internal 8 MHz oscillator which means you give up about 2.5 times the speed, but all you need is power and ground to run a Nano, so it's actually much easier to hook up. Being 2.5 times slower may sound like a lot, but in many applications speed isn't a major factor plus, the Atom Nano has more

program memory space. The Nano chips also use the exact same pin-out as the original interpreter chips, so if you have a design that doesn't need the higher speed, a Nano chip can be plugged right in.

Atom Debugger

I mentioned the debugger in my last column, but I wanted to show anybody new to the Basic Atom the features this debugger offers. I borrowed some of this information (with permission from the author) from Chuck Hellebuyck's book *Programming the Basic Atom Microcontroller* that can be purchased from the *SERVO Magazine* bookstore (<http://store.servomagazine.com>). The Atom debugger is a special tool that makes your software run in slow motion so you can step through the code. Once you use it, you won't ever want to develop without a debugger again. The Atom debugger is completely controlled by software on the PC that sends command codes to the Atom Nano chip. You only need the serial programming connection to use the debugger. No other hardware is required.

When you finish writing your program, you would normally press the "Program" button to compile and download your program into the Nano chip. To use the debugger, you simply press the "Debug" button instead. The difference between the Program button and the Debug button is hidden. They both compile and then program the Nano but the Debug adds another hidden step: It adds a block of code to your program that is used to communicate with the debugger software running on your PC.

When your program is running, the added debugger block of code sends variable values, internal register values,

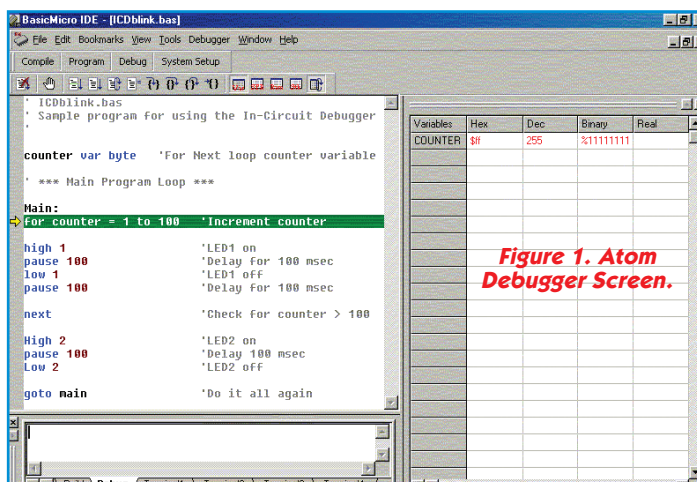


Figure 1. Atom Debugger Screen.

and other details to the PC through the programming cable. The debugger software on the PC will display that data in the way you choose by clicking on the different setup buttons in the debugger tool bar.

The debugger controls allow you to run the program continuously or in animate mode. Animate mode automatically steps command by command in slow motion through the program. You can also manually step through your code command by command by using the PC mouse to advance the program via the various single step buttons. This gives you total control of how the program advances. Figure 1 shows the Atom IDE with the debugger enabled.

When the debugger is in run mode, it will run close to but not exactly actual speed. Because of the communication back to the PC, an added delay will occur between commands which will slow things down a bit. You must take this into account when running time critical code. Each command will run in full runtime mode (SERIN and SEROUT will function normally) but added time will appear between commands. Also, the programming cable must be connected to the PC or the debugger will not operate and neither will the Nano – it will wait for commands from the PC.

Debugger Controls

The debugger controls can be found under the Debugger selection in the top menu line or via the debugger toolbar line that appears when you press “Debug” to compile your program. The debugger toolbar line can be switched on and off under the View main menu selection. The total tool bar is shown in Figure 2. A summary of the debugger control features follows.



Connect/Disconnect

The Connect/Disconnect button is used to establish communication between the debugger and the Nano. When the “Debug” button is pressed and the program is downloaded, the debugger will automatically connect to the Nano. A green bar will highlight the first line of the program indicating the debugger is successfully connected. You can stop the debugger connection at any time by just clicking on this icon.



Toggle Breakpoint

The Toggle Breakpoint button allows you to turn a breakpoint on or off at any point in the program. A breakpoint is a highlighted line that will stop execution of the program when it gets to that command line. This is handy if you want to see what the variables and I/O pins look like when a specific command is encountered in the program without having to step through each command. To use it, just position the cursor to the line you want the program to stop at. If a break point is not set on that line, click on this icon or right-click on your mouse and select “Toggle Breakpoint.” This will highlight that command line in red and enable the breakpoint action. To turn it off, just click on the icon again

Figure 2.



 18 Pin Atom Nano	The 18-pin Atom Nano chip sells for \$7.95 and is based on the PIC16F88 which has the following features: 7K Program Flash 368 Bytes RAM 256 Bytes EEPROM 16 I/O External Interrupt Change on PORTB Interrupt Two Eight-bit Timers One 16-bit Timer Capture/Compare/PWM Port 10-bit, Seven Channel A/D Converter SPI / I ² C Hardware Peripheral Two Analog Comparators Hardware USART MCLR Feature can be internal
 28 Pin Atom Nano	The 28-pin Atom Nano chip sells for \$8.95 and is based on the 16F886 which has the following features: 14K Program Flash 368 Bytes RAM 256 Bytes EEPROM 24 I/O External Interrupt Change on PORT B Interrupt Two Eight-bit Timers One 16-bit Timer Two Capture/Compare/PWM Ports 10-bit, 11 Channel A/D Converter SPI / I ² C Hardware Peripheral Two Analog Comparators Hardware USART MCLR Feature can be internal
 40 Pin Atom Nano	The 40-pin Atom Nano chip sells for \$10.95 and is based on the 16F887 which has the following features: 14K Program Flash 368 Bytes RAM 256 Bytes EEPROM 35 I/O External Interrupt Change on PORT B Interrupt Two Eight-bit Timers One 16-bit Timer Two Capture/Compare/PWM Ports 10-bit, 14 Channel A/D Converter SPI / I ² C Hardware Peripheral Two Analog Comparators Hardware USART MCLR Feature can be internal

Table 1. Atom Nano Chips.

or right-click to turn it off.



Animate

This is a nice feature of the debugger. The Animate function will automatically step through your program command by command in slow motion. Each command being executed is highlighted in green. When the command is completed, the next line is highlighted. This allows you to watch and verify the program is flowing where you expect it to go. If the “Auto Update” feature is selected (described below), then variables and internal information will be updated after each command. To view those values though, it is often best to stop the program as the Animate mode can sometimes run too fast to allow you to read the data.

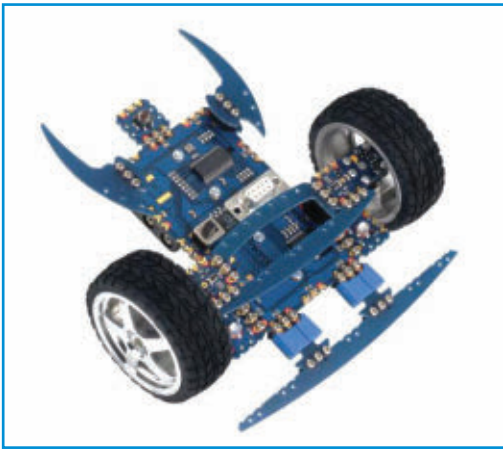


Figure 3. Microbric Viper Robot.

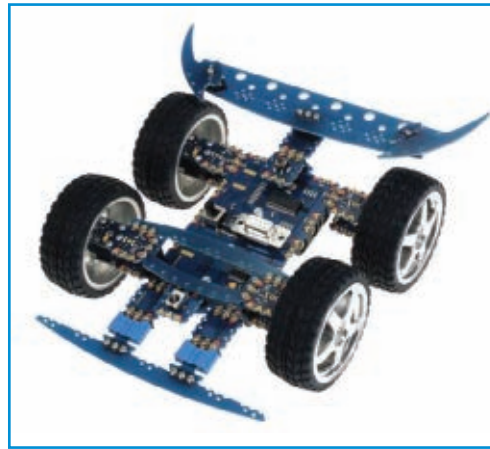


Figure 5. Four Wheel Viper Robot.



Run

This option allows you to run the program in the Nano at full speed (minus a minor delay for the debugger block of code) without stopping to check for variables or other data. The green command line indicator will not step through each command. It will just stay at the last line executed before pressing the Run icon.



Reset

Reset is used to start the program at the beginning. Any information stored in the variables is not erased. This is a simple way to start at the beginning or to see how your program will react if a hardware reset were to occur.



Pause

The Pause button will halt the program at the current command line. To resume execution, the Run or Animate button is pressed. The Pause button is handy to stop the Run or Animate mode so variables and other data can be viewed.



Step Into

This is the button you press to step through your program command by command line using your PC mouse.



Step Over

This button is a special step button that allows you to jump over a part of the program such as a gosub or for-next routine. Sometimes a gosub or for-next routine will take many clicks of the mouse to get through the routine using Step Into. This allows you to jump over it and move on to the command lines after them.



Step Out

This is another special step button that allows you to leave a gosub routine. It's handy for looking at part of a gosub routine and then lets you leave when you have seen enough. Clicking this will jump you to the command line after the end of the gosub routine.



Run To Cursor

Clicking on any command line in the program will produce a blinking cursor. If you then click on the "Run To Cursor" button, the program will execute in Run mode until the cursor line is encountered. The program execution will stop at that command line.



Show Variables

This control button will toggle the Variables window open or closed. When it's selected, a separate window will open and the variables defined in your program will automatically be listed. The values of those variables will be displayed in HEX, Decimal, and Binary formats. (Make sure Auto Update is selected so these are updated after every command).



Show SFRs

SFR stands for Special Function Registers. These are special internal locations within the Nano chip that indicate how the internal program is responding to modifying the internal registers. This is really a function for the advanced user but can be handy for understanding how the Nano program controls the internal features.



Show RAM

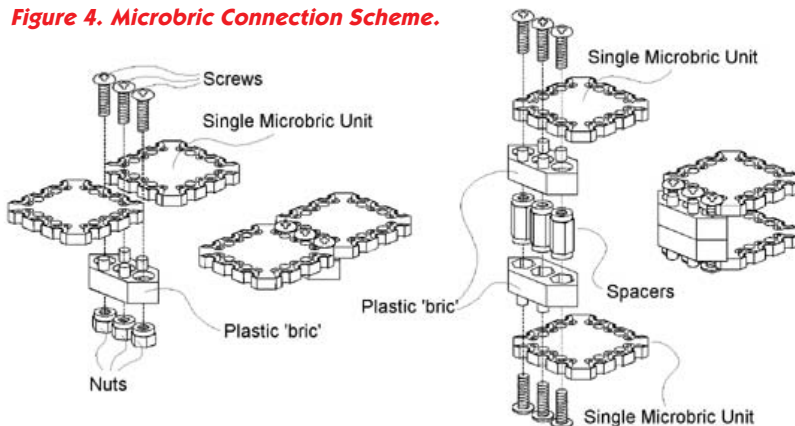
This feature shows all the Random Access Memory in the Atom chip/module, not just the variables. Again, this is handy for the advanced user to see the inner workings of the microcontroller.



Show Gosub Stack

This displays the Gosub Stack. The Gosub Stack is the list of location pointers

Figure 4. Microbric Connection Scheme.



within the microcontroller that directs where to jump to when a gosub command is encountered. By monitoring this, you can make sure multiple gosubs are not somehow getting lost. This is really an advanced user function.

Set Auto Update

This should always be selected. It tells the debugger to update the variable, RAM, SFRs, and Stack after every command is executed. You should select this when the debugger is first connected, but it can be turned on or off at anytime.

Atom Robot

One of the most interesting robot platforms I've found that is based on the Basic Atom chips is the Microbric Viper robot shown in Figure 3. I did a search for locations that offered this kit but the only place I could find that still had some in stock was www.electronickits.com. The Microbric website www.microbric.com shows a few other resellers but www.robotshop.us was the only one that had add-on parts for the kit but didn't have full kits in stock. I assume this kit may be phasing out or a new version is in the works. Microbric has other robot options as well, but they don't appear to be Atom based.

What makes this robot kit so interesting to me is that it's designed for the beginner. The robot is assembled with just a screwdriver and this includes the electrical connections. The boards have color-coded connections and plastic locks that hold the boards together. The screws serve two purposes: to both hold the boards together and to make the electrical connection. Figure 4 shows the secret to the Microbric connections.

The two-wheel version looks a little weird at first but it can make a great robot. If you can find the add-on kit for an extra set of wheels, then a four wheel robot can easily be made. Figure 5 shows that type of bot.

Atom Nano Development Boards

Last time, I also mentioned and showed pictures of some Nano development boards, including an early version of the little breadboard programming adapter (Figure 6). This board contains the programming interface circuitry. The board is designed to plug into a breadboard and then connect to the PC's USB port. I've since been told by Basic Micro that they are working on an improved version that powers itself off the USB port of the PC. The early prototype I have requires you to supply power. I look forward to that new design.

Conclusion

Watch for more Nano information in future articles. I'm told there are several new development boards on the way that will make it easy to use the Nano chips. I hope to use these to demonstrate some basic applications for the Basic Micro Nano. In the meantime, check out what they have at

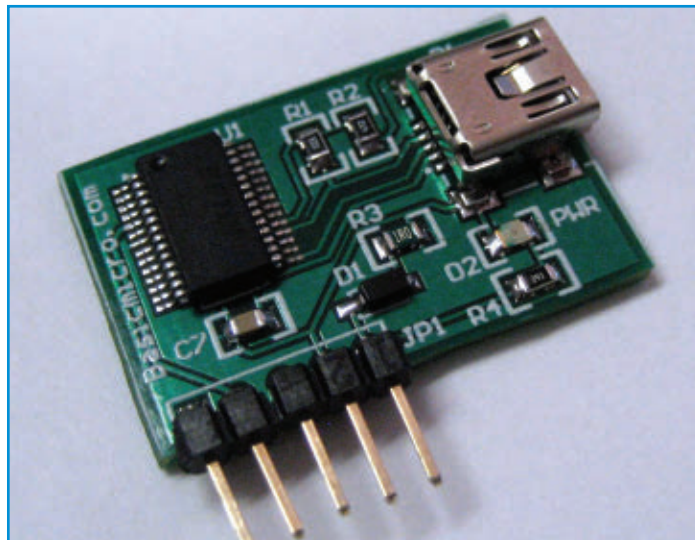


Figure 6. USB Programming Adapter Board.

BasicMicro.com and also look for it at our site at www.BeginnerElectronics.com **SV**



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Then and NOW

EUROPEAN ROBOTS

by Tom Carroll

Rather than delve into 18th century 'clock-work' automatons that were famous across Europe (and ones that I've written about before), I'd like to center on the past few decades of more modern robotics. We have to admit that our neighbor across the Pacific — Japan — inarguably has implemented more robots into their industries than any other nation. We seem to forget that the European Union across the Atlantic has long been in the forefront of robot progress.

Nations in Europe have designed and implemented some of the world's best industrial robots, many of which are installed in factories across the US and have been in use since the '70s. Delving a bit deeper, we also find that some very unique mobile and service robots have been created in this part of the world.

Swedish Robotics

Sweden has one of the most automated economies in the world with 107 industrial robots installed in their factories for every 10,000 people employed in manufacturing. BAE Systems and the Saab Group are two large forces in the development of defense robot systems. ABB Manufacturing Automation is also known around the world for factory automation and robotics. Previously known only as Asea, Brown Boveri joined forces with Asea in 1987 to form ABB. Their bright orange robots (as seen in Figure 1) are some of the most popular industrial robots installed in factories around the world.

Vacuum cleaner and appliance manufacturer, Electrolux (based in Stockholm), is also a world leader in consumer robotics with their famous

Trilobite 2.0 robot vacuum cleaner (Figure 2). The newest version supposedly has over 200 improvements but still carries a hefty price tag of \$1,800. The robot is also heavier (5 kg) and higher (13 cm) than the US-made Roomba and may not be able to clean under the lowest furniture.

Electrolux literature speaks of eight ultrasound 'radar' furniture avoidance sensors to prevent the robot from actually touching delicate antique furniture legs by sending radar beams to allow it to stop within 1/4" from objects. Electrolux is also very proud of their efficient 'mapping technology' that allows the Trilobite to intelligently cover areas of a carpet only once, thus spending less time per room. At 90 watts, it's a powerful machine for a battery-powered robot vacuum cleaner and automatically

Tom Carroll can be reached at TWCarroll@aol.com.



FIGURE 1.



FIGURE 2.

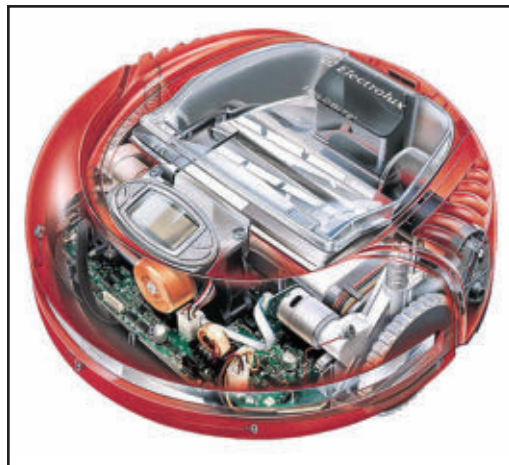


FIGURE 3.

returns to its charger base just like the Roomba.

Unlike the iRobot Roomba, however, it uses magnetic strips instead of virtual walls/lighthouses to define its working area. Figure 3 shows an interior view of the Trilobite. This is just one of many types of robots being developed in Swedish universities and then produced by their world-class industries.

An interesting Electrolux robot that was developed more as an 'ecological green' demonstration device is the HVF robot vacuum cleaner. Looking more like a roving flower pot than a useful household device, it supposedly runs on hydrogen fuel cells and produces water as a byproduct for the plants. Figure 4 shows the robot in all its glory.

News of the device made headlines in the blogosphere about a year ago when it was touted "for the home of 2020." I have yet to read anything mentioning the technical aspects and actual cost to purchase it here in the US.

British Robotics

One particular British robotic device that I've heard about the past few years is the MAS Mower 01 and 05 — large scale robotic grass mowers intended for stadiums and soccer pitches. The Gloucestershire-based company, McMurtry Limited, has developed a mower that can act on its own and mow an entire sports field without human intervention. The MAS Mowers are substantially larger and more expensive (£40,000+) than smaller residential types (see Figure 5). The mower utilizes a 36 inch wide, six blade reel cutting

system and the robot mower is guided by a rotating laser shown in Figure 6. The on-board computer can be pre-programmed to follow any pattern desired, even to actually cut a sports team's logo in the grass. When the 70 liter catcher is full, the mower automatically goes to a specified dump point to dispose of the clippings and then returns to the place where it left off to continue cutting.

The rotating laser scanner senses reflective strips placed around the field to accurately locate and orientate itself on the area being mowed. A hand controller can be used to program or control the mower.

Considering the robot's size and potential danger inherent in an autonomous lawnmower, the MAS Mowers have ultrasonic obstacle detectors, soft front and rear bumpers, a warning horn, a flashing strobe light, and manual override capability with three emergency kill switch buttons. The massive mowers can cut up to two acres of grass before needing recharging (by manual means).

Dyson DC06 Robot Vacuum Cleaner

Dyson is known as a manufacturer of quality and fairly expensive vacuum cleaners. James Dyson started the company after examining a sawdust collector that used cyclonic action to collect the dust without replacing



FIGURE 4.

bags. The collector was too expensive for him to buy, so he designed and built his own that was superior and cheaper. His uniquely colored vacuums are popular around the world and he decided to follow suit with the other robot vacuums that were beginning to appear in the late '90s and developed the 9.2 kg DC06 that was unveiled in 2001 (see Figure 7).

The project was placed on hold in 2004 after many beta models were placed into various households for testing. It was expected to cost over \$4,000.

"Dyson engineers are researching robotics, but it takes time. We could have launched DC06 and heralded it as the first robotic vacuum cleaner. It has three onboard computers, 2,000 electronic components, 27 separate circuit boards, and 70 sensory devices. As robots go, it's highly advanced, more so in fact than robot vacuums available today. But, we want one that cleans properly and guides itself more logically than a human would, a



FIGURE 7.



FIGURE 6.



FIGURE 5.



FIGURE 8.

truly autonomous machine, rather than a glorified carpet sweeper. Propulsion and battery life will be key, and we're getting closer," Dyson commented. He intends to bring out a superior robot vacuum cleaner very soon.

His reasoning for scrapping the original model was its high price, poor battery performance, and suction not up to his company's high standards. Dyson was satisfied with the machine's sensor/microcontroller system and the required programming, but his 'Dual Cyclone' vacuum system was difficult to implement in such a small device. "An automatic vacuum cleaner must clean as well as the best mains powered vacuum cleaner and with the sort of methodical coverage a human being

couldn't possibly achieve," Dyson explained.

German Robotics

KUKA Roboter, based in Augsburg, Germany is another robot company that has installations in most of the industrialized countries of the world. With a line of over 100 models,

mostly painted orange (is this a European robot color tradition?), this German company began operations in 1989 and entered the US market in 1995.

One of their most unique products is the Robocoaster — a large robot arm with one or two seats attached, acting as an amusement park ride. Figure 8 shows two people experiencing the ride at a robot exhibition.

Germany has not been left out of the consumer robot market. The German designed and built Kärcher RC3000 Robocleaner is a pretty good robot vacuum. It basically operates much the same as the other popular brands. It does, however, have one unique feature: It automatically takes its collected dirt and dust back to the

combination base station/charger unit as seen in Figure 9 and deposits it, recharges itself, and heads back out to clean another room. Unlike some of the other top end robot vacuums, the Kärcher does not use room mapping, relying instead on the random cleaning approach. Kärcher has been around for years and has built the popular KMR 1200 commercial road sweeper that they are now looking to make entirely autonomous. Priced at an average of \$1,500, the RC3000 is quite a bit more expensive than the Roomba but does offer some attractive features. It has not been as popular as the Trilobite here in the US but seems to be selling quite well in the UK and the rest of Europe.

Robotics Development is Shared in Most European Countries

Europe has long been a leader in robotics technology. Reis Robotics in Obernburg, Germany, the Fraunhofer Institute for Manufacturing Engineering and Automation in Stuttgart, Germany, the European Robotics Association based in Brussels, Belgium, the LIRA-Lab at the University of Genova, Italy, and the TKK Automation Technology Laboratory at the University of Finland in Helsinki are just a few of the many world-class companies, organizations, and universities in Europe that have made world news lately in the field of robotics.

NASA is not the only space agency developing lunar and planetary rover technology. The European Space Agency (ESA) has been interested in studying the lunar craters at the Moon's poles, long thought to hold deposits of frozen water. Not only is this necessary for lunar-based astronauts to drink, it is also an important source of hydrogen and oxygen for fuel and breathing. Lunar orbiters have detected water vapor emissions but rovers are the key to determining just how much ice is available.

This European agency established

FIGURE 9.



the ESA Lunar Robotics Challenge that was first conducted on the island of Tenerife in the Canary Islands off the coast of Africa. The pumice landscape of Minas de San Jose within Tenerife's Teide National Park was an excellent stand-in for a stark lunar landscape.

Operating from a trailer situated 2 km up the side of the mountain, various teams controlled their rovers to descend a steep 40° slope down a crater to grab a 100 gram sample of specially selected soil and return it back up the slope — in the dark. All of the competing rovers had to be a specific size, weight, and could not exceed a set power consumption. Considering that this was actually Earth, the contest was further complicated by rain and clouds.

The German rover CESAR won the competition in much the same manner as many robotic competitions — it was the only one that actually completed the course. The three-wheeled rover shown in Figure 10 gained the acronym from Crater Exploration and SAmple Return. Though situated 2,000 km from the mainland, the trailer — and thus the competing rovers — were in constant communication with the ESA Telecom Directorate through satellite ground stations and data links. As with most robotic contests operating in adverse conditions, many of the eight competing teams faced mechanical, software, and other problems.

ERA: The European Robotic Arm

The ESA has worked on a design for a space robotic servicing system for many years, and typical of very expensive projects, it has been delayed and mission uses have changed. Called the ERA and built by Fokker Space, the European Robotic Arm will be used in the assembly and servicing of the Russian segment of the International Space Station called the Russian Multipurpose Laboratory Module. It is expected to be launched in late 2011 on a Russian Proton booster from the Baikonur Cosmodrome.

Figure 11 shows the ERA robotic

manipulator, remarkably similar to the original Canadarm used on the Space Shuttle. It is designed to work with the new Russian airlock to transfer small payloads into and out of the International Space Station, thus reducing the astronaut's EVA activities. Another prime use will be the positioning of a cosmonaut (astronaut) for operations on the external locations of the Space Station.

The ERA manipulator system consists of two end effectors (to hold tools and payloads), two wrist joints, two carbon fiber cylindrical 'limbs,'



FIGURE 10.

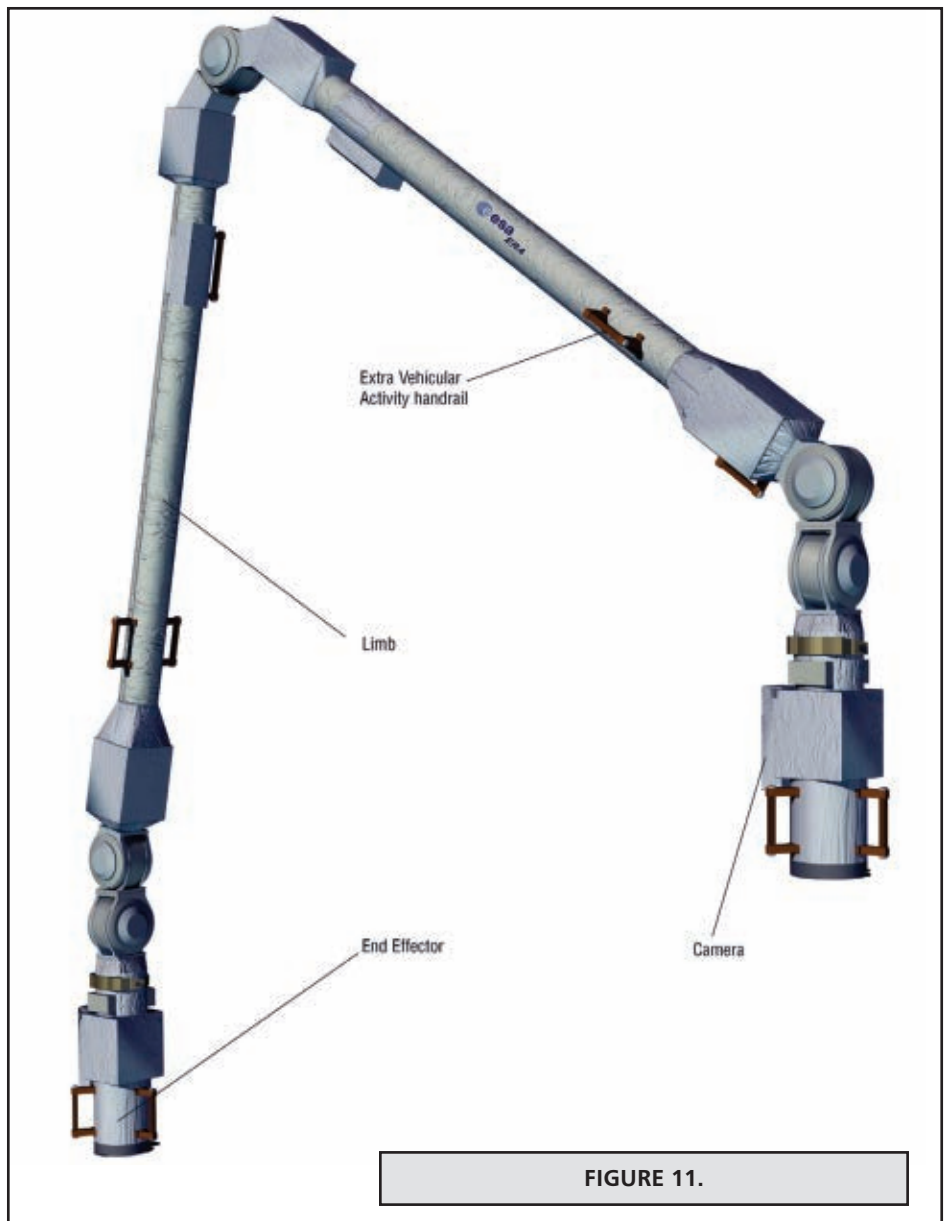


FIGURE 11.



FIGURE 12.

and one elbow joint. Both of the end effectors can function either as a gripper or an attached joint for a base. The end effectors have high resolution cameras attached for video

control by the operators.

The total length of the whole assembly is 11.3 meters and it has a reach of 9.7 meters. Weighing in at 630 kg, it can handle payloads up to 8,000 kg. There are no plans at present to have it operate on a mobile transporter system that could place it in all locations of the Space Station.

Robot Lucy

Europe is not just about building expensive robot vacuum cleaners, world-class industrial robots, and space-borne rovers. The European Union has funded robotic rovers to delve into the depths of erupting Mount Etna in Italy. EURON, the European Robotics Network has interconnected the robotics labs across the many European nations to foster transfer of technology and cooperation amongst the many universities and industrial labs. This cooperation

was instrumental in the development of the iDroid, a fully-functioning humanoid robot that consumers could build for themselves. One of the most interesting humanoid robots that I saw in my research for this article was the bipedal robot, LUCY.

The Robotics and Multibody Mechanics Research Group of the Department of Mechanical Engineering at the Vrije Universiteit, Brussels, Belgium (VUB) started research on computer-aided analysis of rigid and flexible mechanical systems back in 1990 and the evolution of this research culminated in the creation of LUCY. Now, we know of many bipedal robots developed around the world, but few — if any — use the unique pneumatic muscles shown in Figure 12.

Lucy is a lightweight anthropomorphic biped at 30 kg and stands 150 cm tall (about 5 feet). See Figure 13. The design team bypassed complex gear systems in favor of these lightweight pairs of antagonistic pneumatic actuators that act like our own muscles operating in pairs, opposing each other in motion. The 'McKibben Muscle' has been used by experimenters here in the US and these researchers used the Air Muscle from the Shadow Robot Company. Seven 16-bit MC68HC916Y3 microcontrollers operate the complete system built on an AlSiMg1 aluminum alloy body structure.

Wrap-Up

As always, a trip through the many Internet sites on these and many other unique sites will introduce you to some cutting edge robots being designed and built. Virtually every country in Europe has robotic products or on-going dedicated robot research. **SV**

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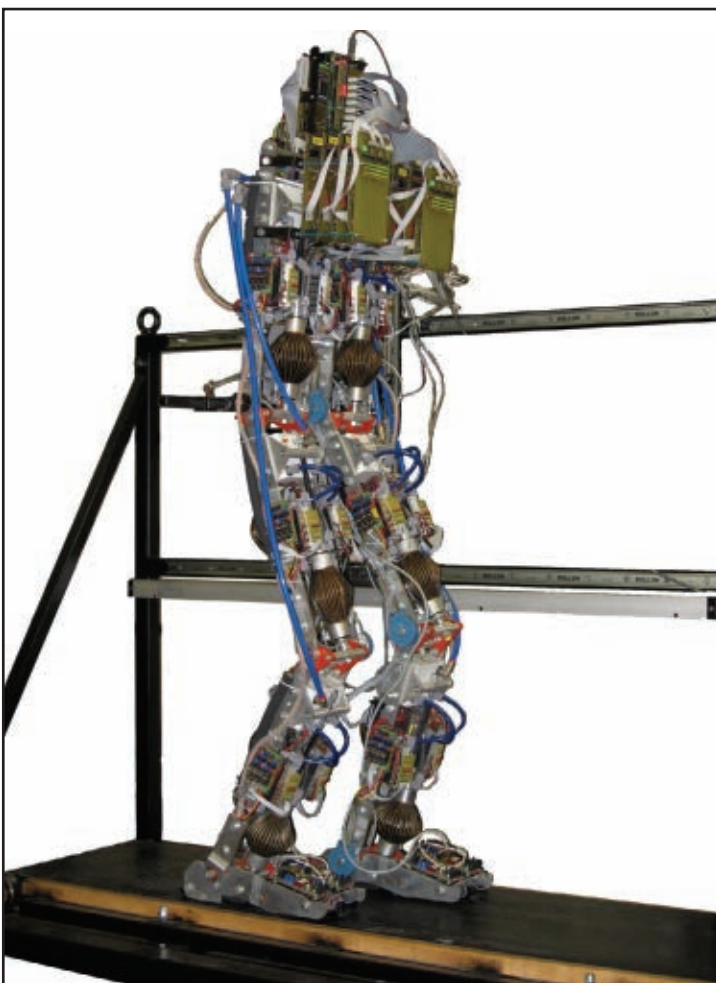


FIGURE 13.

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Basic Micro75	Lynxmotion, Inc.2	RobotShop, Inc.81, 83
Budget Robotics17	Maker Faire82	Solarbotics/HVW15
CrustCrawlerBack Cover	Maxbotix81	superbrightleds81
CipherLinx Technologies81	Parallax71	Technological Arts81
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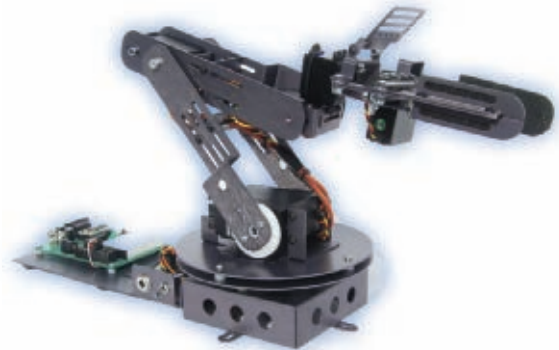
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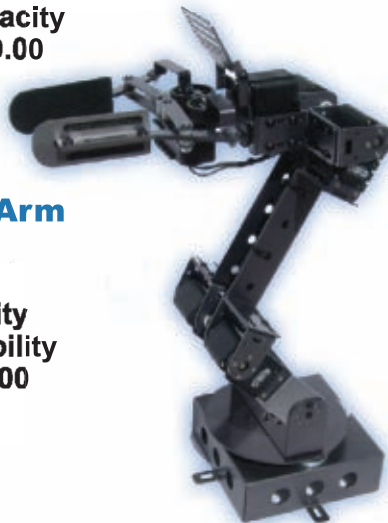


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